

Measuring discrimination using natural experiments *

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August 25, 2025

Abstract

We often see disparities between social groups (e.g., gender, race) at crucial decision points in many contexts, from hiring to policing. These disparities are difficult to interpret as discrimination if the groups are unobservably different. We show how to use a binary instrumental variable (IV) strategy to measure discrimination adjusted for group differences in unobserved potential outcomes. Our approach does not require random assignment to decision-makers, a prerequisite for existing discrimination measurement techniques. We implement our approach to study racial discrimination in misdemeanor prosecution, the most common form of contact with the criminal court system in the United States. We use a difference-in-difference IV strategy generated by a budget cut that reduced prosecution rates in King County, Washington, but did not affect adjacent counties. Before the cut, we find no evidence of discrimination in prosecution conditional on unobserved potential recidivism. After the cut, white defendants became more likely to be prosecuted than minority defendants with similar unobserved potential recidivism. We find suggestive evidence that cases involving minority defendants were more likely to be lower quality, potentially due to disparities created in previous stages of the criminal legal system (e.g., arrests), and that prosecutors responded to the budget cut by focusing on high quality, less resource-intensive cases.

*We thank Ashley Craig, Sara Heller, Sarah Miller, Michael Mueller-Smith, Benjamin Scuderi, Kevin Stange, Mel Stephens, and Basit Zafar for their comments and guidance. We also thank Lea Bart, Micah Y. Baum, Jordy Berne, Charlie Brown, Elisa Facchetti, Jamie Fogel, Benjamin Hansen, Christopher Hollrah, Emily Horton, Brian Jacob, Amanda Kowalski, Steven Mello, Emir Murathanoglu, Tyler Radler, Katherine Richard, Roman Rivera, Brock Rowberry, Damián Vergara, Iris Vrioni, and numerous seminar participants for helpful suggestions. We are grateful to Kevin Cottingham at the Washington Administrative Office of the Courts for assistance with the court data and to Jordy Berne, Brian Jacob, and Christina Weiland for their help in working with the ‘Read by Grade 3’ project data. Rao gratefully acknowledges financial support from the Poverty Solutions Doctoral GSRA award and the Rackham One Term Dissertation Fellowship.

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1 Introduction

Disparities between social groups (e.g., race, gender, socio-economic status) are common in many contexts, e.g., employment, housing, and the criminal legal system. To understand whether such disparities are the result of discrimination, researchers often seek to compare how similar individuals from two groups are being treated.¹ This comparison is difficult to make in practice if there are important but unobserved group differences (Aigner and Cain, 1977; Charles and Guryan, 2011).² For example, in the canonical context of hiring discrimination, it is challenging to measure the hiring gap between equally productive people from two groups, because productivity is only observed for hired workers.

An extensive literature has focused on overcoming similar challenges by operationalizing the Becker (1957) ‘outcome’ test, using theoretical restrictions and quasi-experimental tools (Knowles, Persico, and Todd, 2001; Anwar and Fang, 2006; Arnold, Dobbie, and Yang, 2018; Canay, Mogstad, and Mountjoy, 2024; Bharadwaj, Deb, and Renou, 2024). These tests detect taste-based discrimination (Arrow, 1973) at the margin of a decision, but do not quantify accurate statistical discrimination (Phelps, 1972) or discrimination away from the margin, regardless of its source (Hull, 2021). Recent work addresses these concerns by using random assignment to decision-makers to extrapolate unobserved outcomes (e.g., productivity of workers not hired) and directly adjust for these unobservables when measuring discrimination (Arnold, Dobbie, and Hull, 2022). While this approach captures discrimination whether it is taste-based, statistical, or due to incorrect beliefs, it cannot be applied when individuals are not randomly assigned to decision-makers.

In this paper, we greatly expand where researchers can measure discrimination while adjusting for key unobserved differences. We show how to use natural experiments that yield a binary instrumental variable (IV) framework (Angrist, Imbens, and Rubin, 1996) to measure average discrimination in a treatment decision. We map commonly used quasi-experimental techniques, e.g., regression discontinuity (RD) or difference-in-difference (DiD), to the binary IV framework and use mild assumptions on selection behavior to recover the distribution of unobserved outcomes for each group. We directly condition on these unobservables to either bound or point identify discrimination that arises from multiple sources.

We implement our approach to provide novel evidence on racial discrimination in the context of misdemeanor adjudication, the most common form of contact with the criminal court system. Roughly 4,000 misdemeanor cases are filed annually per 100,000 people in the United States, a rate that is triple the felony filing rate (Stevenson and Mayson, 2018). These large caseload sizes can create incentives to process cases quickly with limited oversight, potentially exacerbating disparities and discrimination (Gershowitz and Killinger, 2011). We study racial discrimination in misdemeanor prosecution in King County, Washington (Seattle metropolitan and suburban areas).

¹Prior work documents disparities and discrimination in hiring (e.g., Hellerstein, Neumark, and Troske, 1999, 2002; Goldin and Rouse, 2000; Bertrand and Mullainathan, 2004), housing (e.g., Ross, 2005; Ross et al., 2008), and the criminal legal system (e.g., Abrams, Bertrand, and Mullainathan, 2012; Anwar, Bayer, and Hjalmarsen, 2012).

²Such differences might be due to multiple factors, including preferences or discrimination prior to the decision of interest, i.e. ‘systemic discrimination’ (Bohren, Hull, and Imas, 2022).

Cases are not randomly assigned to prosecutors in this setting, rendering existing discrimination measurement approaches infeasible. Instead, we construct a binary IV using a DiD strategy generated by an unexpected budget cut that affected prosecutors in King County but not in adjacent counties. Our data consist of all cases that prosecutors accept from law enforcement and we examine the decision to continue to pursue an accepted case. Throughout the paper, we use the term ‘prosecuted’ if a case was pursued and all charges were not dismissed.

To outline the details of our approach, consider a potential outcomes (Rubin, 1974) framing, where we define treatment as $D_i = 1$ if an individual is prosecuted and $D_i = 0$ if dismissed. Let the treated potential outcome, $Y_i(1)$, be a measure of re-offending if prosecuted, and the untreated potential outcome, $Y_i(0)$, be a measure of re-offending if dismissed. We start by defining racial discrimination as the racial gap in prosecution rates between individuals who would have the same re-offence outcome if prosecuted, $Y_i(1)$, and later discuss extensions to conditioning on the i) re-offence outcome if dismissed ($Y_i(0)$) or ii) treatment effect of prosecution ($Y_i(1) - Y_i(0)$).³ Focusing on the treated potential outcome, if the distribution of $Y_i(1)$ varies by race, then the raw disparity in prosecution rates could be due to differences in $Y_i(1)$, and may not align with our definition of discrimination. Let $Y_i(1)$ be an indicator for whether a defendant would re-offend if prosecuted. Since $Y_i(1)$ is binary, the distribution of potential recidivism if prosecuted is the share of individuals who would re-offend if everyone were prosecuted—the “average prosecuted outcome”.⁴

Assume for now that we can observe the average prosecuted outcomes for each racial group. If the average prosecuted outcomes differ by race, we quantify discrimination conditional on the prosecuted outcome in three steps. First, we rescale observed race-specific prosecution rates using the ratio between the observed outcome and the average prosecuted outcome for a given racial group, e.g., the ratio between the prosecuted defendants’ recidivism rate and the overall recidivism rate (Arnold, Dobbie, and Hull, 2022). This rescaling yields race-specific prosecution rates that are conditional on the prosecuted outcome. Second, we use the resulting prosecution rates to construct racial gaps in prosecution for each value of the prosecuted outcome. That is, we construct racial gaps in prosecution for people who: a) would re-offend if prosecuted ($Y_i(1) = 1$) and b) would not re-offend if prosecuted ($Y_i(1) = 0$). Third, the weighted average of these two racial gaps yields average discrimination in prosecution conditional on the prosecuted outcome.

Since we typically cannot observe the average prosecuted outcome for each racial group, we estimate it using a binary IV that shifts treatment rates—prosecution rates in this example. First, we use the IV to estimate how potential outcomes (e.g., recidivism if prosecuted) vary across defendants who would always be prosecuted (‘always takers’) and marginal defendants prosecuted due to the IV (‘compliers’) within each racial group (Imbens and Rubin, 1997). Since we do not observe the outcomes dismissed defendants (‘never takers’) would experience if they were prosecuted

³Recent work critiques discrimination definitions that condition on potential outcomes and suggests alternative definitions that may be more or less appropriate depending on the context (Grossman, Nyarko, and Goel, 2024; Bushway et al., 2025). While the focus of this paper is expanding where researchers can measure discrimination conditional on potential outcomes, we discuss these critiques in Section 2.1.

⁴We discuss modifications for multi-valued potential outcomes in Section 2.2.

instead, we extrapolate their outcomes using behavioral assumptions from the marginal treatment effects literature (Brinch, Mogstad, and Wiswall, 2017; Mogstad, Santos, and Torgovitsky, 2018; Kowalski, 2023a,b). Depending on the strength of the assumptions, such extrapolations recover either bounds or point estimates of the unobserved average prosecuted outcomes for each racial group, which we use to measure average discrimination conditional on the prosecuted outcome.⁵

Our approach is also applicable when a natural experiment generates a conditionally-random binary IV. In particular, we discuss implementation with a DiD, a setup where time might independently influence treatment status and potential outcomes, making it difficult to identify the proportions and outcomes of always takers, compliers, and never takers. We show how restrictions on the effect of time and an assumption of parallel trends in potential outcomes permit identification of these key quantities that are crucial for our approach.⁶

Taking our approach to the data, we use administrative court records from Washington State to bound racial gaps in misdemeanor prosecution among individuals who are equally likely to commit a new offence if prosecuted.⁷ Using the shifts in prosecution rates and re-offence outcomes generated by the budget cut, we find racial differences in the unobserved prosecuted outcome.⁸ Before the budget cut, 25–29% of white defendants would commit a new offence if prosecuted, while the same is true for 32–37% of minority defendants. These stark differences imply that discrimination estimates that do not account for differences in unobserved prosecuted outcomes, e.g., covariate-adjusted prosecution gaps, will be biased.

We use the estimates of the average outcomes if everyone were prosecuted to bound the racial gap in prosecution before and after the budget cut, conditional on the prosecuted outcome. Before the budget cut, we cannot reject the null of no racial gap in prosecution rates. This pattern changes after the cut, which sharply reduced prosecution rates—white defendants became 1.3–4 p.p. (1.8–5.6%) more likely than minority defendants to be prosecuted. The gap implied by our bounds is lower than and excludes the covariate-adjusted disparity (4.4 p.p.).

Next, we investigate potential reasons for the relatively higher white prosecution rate after the budget cut. One explanation might be that: i) resource constraints after the budget cut made prosecutors less likely to pursue low quality cases, and ii) minority cases were supported by weaker evidence due to discrimination in pre-prosecution decisions, e.g., policing (Goncalves and Mello, 2021; Owens and Ba, 2021; Agan, 2024). If this is the case, we should expect more positive white–minority gaps after the budget cut among a subset of low quality cases. We define low quality cases as those filed for offences that were relatively less likely to be successfully sentenced before the

⁵While we estimate bounds in our empirical application, Appendix B.3 illustrates the point-identification approach using a brief empirical example studying racial discrimination in incarceration decisions in Texas.

⁶We also discuss various approaches to use local variation generated by an RD to measure discrimination conditional on potential outcomes at an RD cut-off as well as away from it. Appendix B.4 illustrates using an RD design with our approach through an empirical example studying socio-economic discrimination in the decision to allow Michigan public school students to progress to the next grade.

⁷Since racial differences in sentencing might complicate the interpretation of discrimination conditional on re-offence outcomes if prosecuted, we also present exercises conditioning on the re-offence outcome if dismissed, which compare defendants who are similar in the baseline likelihood of re-offending.

⁸We probe the assumptions needed to use a DiD with the IV approach and do not find evidence of violations here.

budget cut. When splitting the sample along this margin, we find that the post-cut racial gap is driven by these low quality cases. We interpret this pattern as suggestive evidence that prosecutors shifted their focus to high quality cases after the budget cut, offsetting disparities generated in prior stages of the criminal legal system. These patterns mirror recent work on prosecutorial discretion (Harrington and Shaffer, 2023; Jordan, 2024), and highlight the importance of considering the criminal legal system as a multi-stage institution (Harrington and Shaffer, 2024).

This paper makes several contributions. First, we expand where researchers can measure discrimination conditional on key unobserved factors. While existing approaches to do so require random assignment to decision-makers (Arnold, Dobbie, and Hull, 2022), our approach only requires a binary IV and assumptions from the marginal treatment effects literature (Brinch, Mogstad, and Wiswall, 2017; Mogstad, Santos, and Torgovitsky, 2018; Kowalski, 2023a,b). Our approach is especially useful in contexts where individuals are not randomly assigned to decision-makers (as in our empirical application) or if researchers do not even observe decision-makers in the data, conditions that render existing approaches inapplicable. For example, since students or employees are rarely randomly assigned to teachers or managers, researchers could use our approach to study discrimination in student suspensions or worker promotions. As a final methodological point, we add to the literature mapping DiD to IV (De Chaisemartin and D’Haultfoeulle, 2018) by showing how to use DiD variation to estimate average potential outcomes.

Second, our empirical analysis presents the first evidence of racial discrimination in misdemeanor prosecution that documents and directly accounts for unobservable racial differences. Although recent work has examined disparities in misdemeanor prosecution (Kutateladze and Andiloro, 2014; Sloan, 2022; Jordan, 2025), most of the prior literature on disparities in prosecutorial decisions has focused on felony prosecution (e.g., Spohn, Gruhl, and Welch, 1987; Davis, 2014; Rehavi and Starr, 2014; Yuan and Cooper, 2022; Harrington and Shaffer, 2023; Tuttle, 2023; Agan, 2024; Jordan, 2024). In either case, much of this evidence is either descriptive or from contexts where unobservable differences between the comparison groups are minimal.

2 Estimating discrimination using a binary instrument

In this section we show how to use a binary instrumental variable (IV) to measure discrimination conditional on potential outcomes. First, we formalize the definition of discrimination that we are interested in measuring. Second, we highlight which components of the definition are observed in the data and which are not. Third, we show how to use insights from the IV and marginal treatment effects literatures to estimate the unobserved components, and discuss practical details related to implementing our approach.

2.1 Discrimination estimands of interest

We begin with a general potential outcomes framework in which individuals are chosen for a binary treatment, D_i , as a function of the unobservable potential outcomes, where $Y_i(D_i = 1)$ is the

treated outcome and $Y_i(D_i = 0)$ is the untreated outcome (Imbens and Angrist, 1994). Each potential outcome is only observed if the associated treatment state is realized. These potential outcomes may be continuous, discrete, or binary. Individuals belong to one of two groups, denoted by $R_i \in \{r_1, r_2\}$ and the distribution of potential outcomes may differ across groups.⁹

Our goal is to quantify group differences in a treatment decision for individuals with the same potential outcome, a definition of discrimination that encompasses multiple sources of discriminatory behavior (statistical discrimination, animus, and biased beliefs) and is consistent with the legal interpretation of ‘disparate impact’ (Becker, 1957; Phelps, 1972; Arrow, 1973; Bordalo et al., 2016; Bohren et al., 2019; Arnold, Dobbie, and Hull, 2022).^{10,11} The most appropriate potential outcome to condition on should be such that any group differences in treatment that remain after accounting for differences in the potential outcome can be interpreted as an unwarranted disparity.¹² For example, in the canonical context of hiring discrimination, any group differences in hiring rates that remain after accounting for differences in worker productivity could be interpreted as unwarranted. The exact choice of potential outcome will typically depend on a combination of the empirical, theoretical, and normative features of the particular context.

To simplify exposition, we focus our discussion on measuring differential treatment between individuals who would realize the same outcome **if they were treated**, formally described in Definition 1. Appendix B.1 discusses conditioning on the untreated outcome and the treatment effect. These extensions may be relevant for contexts where the most appropriate potential outcome to condition on is not clear ex-ante. For example, consider the decision to nominate students for advanced educational programs. Researchers studying discrimination in such a context may want to quantify the unwarranted disparities in educational program nominations between students who would: i) do equally well in the program, ii) do equally well without the program, or iii) have equal gains from the program.

Definition 1. Differential treatment conditional on treated potential outcome $Y_i(1)$

$$E[D_i | R_i = r_1, Y_i(1) = y] - E[D_i | R_i = r_2, Y_i(1) = y]$$

⁹This framework is a general version of the framework described in Arnold, Dobbie, and Hull (2022). In their setting of bail reform, individuals vary in a latent unobservable Y^* , which is only observed among treated (released) individuals, while no outcome is observed for untreated (detained) individuals. We consider a framework that applies to settings where untreated outcomes are also selectively observed among those not treated.

¹⁰This definition does not require individuals to be identical in terms of all non-race characteristics, as in Canay, Mogstad, and Mountjoy (2024). We think of differences in observed covariates, even conditional on potential outcomes, as possible drivers of discrimination, and investigate them in our empirical application.

¹¹Using the pretrial detention context as a motivating example, Grossman, Nyarko, and Goel (2024) and Bushway et al. (2025) argue that comparisons like Definition 1 are unsuitable for measuring discrimination, preferring definitions that compare individuals with similar predicted misconduct. While such definitions may be attractive from a legal standpoint in certain contexts, they may not always compare unobservably similar individuals, especially if prediction quality varies by group (e.g., race).

¹²This approach requires the researcher to specify the potential outcomes to condition on, $Y_i(D_i)$. It is possible that the underlying decision-makers also value other factors that are not captured by the chosen $Y_i(D_i)$ (Kleinberg et al., 2018). If the relationship between such omitted factors and our chosen $Y_i(D_i)$ varies by group, this definition would not quantify differential treatment conditional on all unobservable factors. However, such gaps can still be interpreted as unwarranted disparities if conditioning on our chosen $Y_i(D_i)$ maps to a well-defined notion of fairness.

The discrimination estimand in Definition 1 is generally difficult to estimate empirically. Since treated outcomes are only observed among treated individuals, it is not feasible to directly condition on $Y_i(1)$. The simple approach of computing the raw gap in treatment rates by group will differ from Definition 1 if treatment decisions are a function of $Y_i(1)$ and if the distribution of $Y_i(1)$ varies by group. An alternative approach of computing the treatment gap conditional on a set of observed covariates can also introduce bias if the covariates themselves are generated due to some discriminatory behavior, while also altering the interpretation of the test. Instead of measuring how differently people with the same potential outcome are treated, conditioning on covariates yields a narrower test measuring how much two individuals with identical **observables** are treated differently, generating ‘included variables bias’ (Ayres, 2010). Even if such bias is small, controlling for covariates will typically not recover the quantity in Definition 1 unless the covariate used is perfectly correlated with $Y_i(1)$.

In practice, groups often differ in terms of unobserved potential outcomes in many settings where measuring discrimination is of interest. For example, discrimination by police could generate cross-race differences in the underlying likelihood of recidivism among arrested individuals, in turn making it challenging to study racial discrimination in prosecution. As another example, differential access to educational inputs by socio-economic status might generate group differences in student skills, making it difficult to study socio-economic discrimination in educational decisions such as student promotion or retention.

We next demonstrate how to use a natural experiment to identify the estimand in Definition 1. Consider an intervention that generates a binary instrument, Z , where $Z \in \{0, 1\}$ denotes periods before and after an intervention. Assume Z satisfies the usual instrumental variables (IV) assumptions of relevance, independence, exclusion, and monotonicity. For ease of exposition, assume that these assumptions are unconditionally satisfied for the binary IV, Z . Section 2.3 below discusses implementing our approach with common quasi-experimental designs that generate conditionally-random binary IVs, such as regression discontinuity (RD) and difference-in-difference (DiD) designs.

Definition 2 describes a time period-specific version of Definition 1, where the periods are delineated by the values of Z : Δ_{zy} is a group treatment gap that is specific to a given period and value of the treated outcome. Each such gap is composed of treatment rates that are conditional on time period, group and potential outcome (π_{zry}).

Definition 2. Differential treatment within a given period, conditional on $Y_i(1)$

$$\begin{aligned}\Delta_{zy} &= (E[D_i|Z = z, R_i = r_1, Y_i(1) = y] - E[D_i|Z = z, R_i = r_2, Y_i(1) = y]) \\ &= (\pi_{zr_1y} - \pi_{zr_2y}) \\ \Delta_z &= \sum_{y \in \text{supp}(Y_i(1))} Pr(Y_i(1) = y) \Delta_{zy}\end{aligned}$$

Our objects of interest are the period-specific estimates of discrimination that are conditional on having the same outcome if treated (Δ_z). These are averages of the period- and treated outcome-specific gaps, weighted by the population prevalence of each value of the treated potential outcome

($Pr(Y_i(1) = y)$). These period-specific discrimination objects can also be differenced to measure changes in discrimination **due to the intervention**, $\Delta_{z=1} - \Delta_{z=0}$.¹³

To understand how to estimate our main object of interest, Δ_z , note that its building blocks are treatment rates that are conditional on time period, group, and treated outcome (π_{zry}). As discussed earlier, since the treated outcome is selectively observed, we cannot directly condition on it to compute each π_{zry} . However, following Arnold, Dobbie, and Hull (2022), we re-write π_{zry} in Equation 1 using: 1) the definition of conditional expectations, and 2) the IV assumptions. The second line follows from the definition of conditional expectations, while the third line follows from the fact that $Y_i(1) \perp Z$.

$$\begin{aligned}
\pi_{zry} &\equiv E[D_i|Z = z, R_i = r, Y_i(1) = y] \\
&= \frac{E[Y_i(1) = y|Z = z, R_i = r, D_i = 1] \times E[D_i|Z = z, R_i = r]}{E[Y_i(1) = y|Z = z, R_i = r]} \\
&\quad \text{Observed in data} \\
&= \frac{\overbrace{E[Y_i(1) = y|Z = z, R_i = r, D_i = 1] \times E[D_i|Z = z, R_i = r]}^{\text{Observed in data}}}{\underbrace{E[Y_i(1) = y|R_i = r]}_{\text{Unobserved}}}
\end{aligned} \tag{1}$$

Equation 1 highlights how to quantify the time period-, group-, and potential outcome specific treatment rates (π_{zry}) used to estimate discrimination conditional on having the same outcome if treated. We need the following objects:

1. $E[D_i|Z = z, R_i = r]$:
Share of individuals of each group r treated in each period z
2. $E[Y_i(1) = y|Z = z, R_i = r, D_i = 1]$:
Share of treated individuals with treated outcome y , for each group r and period z
3. $E[Y_i(1) = y|R_i = r]$:
Prevalence of treated potential outcome y in each group's population

Objects 1 and 2 in the list above are observed in data—we see the share of individuals of each group who are treated and the outcomes realized for treated individuals. Object 3, the share of individuals of group r who would experience a given value of the treated potential outcome, would only be observed in a counterfactual where everyone of that group was treated. This share is an especially crucial element because it also provides a test of whether the distribution of potential outcomes varies by group. If the distributions differ by race, we cannot interpret the raw observed

¹³Even though Z represents quasi-experimental variation, the cross-group gap in the impact of Z on D will not generally recover $\Delta_{z=1} - \Delta_{z=0}$ unless 1) potential outcomes are similar across groups or 2) the impact of Z on D is uncorrelated with potential outcomes (see Appendix B.2). Such cross-group comparisons can also suffer from the pitfalls of conducting marginal outcome tests with discrete instruments (Canay, Mogstad, and Mountjoy, 2024).

group differences in treatment as discrimination. ¹⁴

Object 3 is typically not directly observable in the data unless unique institutional features generate a subsample where everyone is treated (and there is no discrimination in that sample by definition). In certain contexts, random assignment to supremely lenient decision-makers who treat almost everyone permit statistical extrapolations of $E[Y_i(1) = y | R_i = r]$ (Arnold, Dobbie, and Hull, 2022; Baron et al., 2023; Reeves, 2023). However, there are many settings where measuring discrimination is of interest, but the empirical conditions needed to apply these existing approaches are not satisfied. For example, current extrapolation techniques are difficult to apply if decision-makers are not randomly assigned or they are unobserved in the data, both of which might be common in empirical work. In our main empirical application, criminal misdemeanor defendants in King County are not randomly assigned to prosecutors. Thus, using existing approaches in such a setting to assess if the distribution of potential outcomes differs by group and measure discrimination would be difficult.

Below, we provide an approach to measuring discrimination that greatly expands the settings where researchers can measure discrimination conditional on potential outcomes. We incorporate insights from the IV and marginal treatment effects literatures to estimate $E[Y_i(1) = y | R_i = r]$, which we need to understand if the groups to be compared differ in terms of the potential outcomes if treated. We then measure discrimination accounting for any differences in these distributions following [Equation 1](#) and [Definition 2](#).

2.2 Implementation with a binary instrument

We describe the framework in the context of our empirical application studying racial discrimination in misdemeanor prosecution in Washington State. Individuals belong to either ‘White’ or ‘Minority’ groups, denoted by $R_i \in \{w, m\}$, and the treatment decision is prosecution.¹⁵ We continue our discussion of estimating racial differences in prosecution rates for individuals who would have the same outcome if treated, i.e., prosecuted (see [Appendix B.1](#) for discussions of measuring discrimination conditional on the untreated potential outcome or the treatment effect).

Individuals, indexed by i , are chosen for treatment, D_i . In the context of prosecution, this treatment decision is made by multiple agents who influence case outcomes, including prosecuting attorneys and judges, rather than by the individual i . Let $D_i = 1$ if an individual’s case is prosecuted, and $D_i = 0$ if an individual’s case is dismissed. Let the potential outcomes be binary indicators for whether an individual commits a new offence in the future, after prosecution or dismissal, i.e., $Y_i(D_i) \in \{0, 1\}$. While we use binary potential outcomes in this section for simplicity, the expressions in [Definition 2](#) and [Equation 1](#) accommodate multi-valued potential outcomes. Finally, let Z be a binary instrument that satisfies the standard IV assumptions listed in [Section 2.1](#)

¹⁴We also use Object 3, along with each group’s shares of the population (p_r), to compute the underlying prevalence of treated potential outcome y in the population: $E[Y_i(1) = y] = p_{r_1} E[Y_i(1) = y | R_i = r_1] + p_{r_2} E[Y_i(1) = y | R_i = r_2]$, which we use to aggregate each Δ_{zy} to estimate discrimination in a given period.

¹⁵This definition of racial groups, rather than a White–Black comparison, is motivated by the context of Washington State, which we describe in detail in [Section 3](#).

and shifts the rate of prosecution. In the context of our application in Section 3, let Z represent periods before and after an unanticipated budget cut that sharply **reduced** prosecution rates.

Recall that we need to estimate each time period-, group-, and potential outcome-specific treatment rate (π_{zry}) in Equation 1 to quantify the discrimination estimands in Definition 2. The key challenge is that the denominator, $E[Y_i(1) = y | R_i = r]$, is unobserved. Overcoming this challenge requires estimating the proportion of each group r that would realize treated outcome $Y_i(1) = y$ if everyone in that group were treated. Since $Y_i(1)$ is binary, this proportion coincides with the average outcome that would be realized if everyone in group r were treated. We next show how to use the binary instrument Z to estimate bounds and point estimates of $E[Y_i(1) = 1 | R_i = r]$.

Under the standard IV assumptions discussed earlier, the variation from the binary instrument Z partitions the population into three “compliance groups” (always takers, compliers, and never takers) and identifies their population shares as well as certain average potential outcomes (Angrist, Imbens, and Rubin, 1996). If Z shifts treatment for both racial groups, these quantities are identified separately by race group. For each race, we directly observe the proportions of always takers (p_A) and compliers (p_C) in the data by examining the share of the population that would receive treatment regardless of the reform and would only receive treatment because of the reform, respectively. In the example here, since the reform decreases prosecution rates (the treatment), p_A is the share of people prosecuted after the budget reform and p_C is the change in prosecution rates due to the reform. Since always takers, compliers, and never takers partition the population of group r , the share of never takers is $p_N = 1 - p_A - p_C$.

The variation from the binary IV also provides estimates of average potential outcomes for a subset of the “compliance groups”. Focusing on treated potential outcomes again, first note that in the prosecution application, the average treated outcome for always takers is the average outcome of people treated (i.e., prosecuted) after the budget cut. Second, note that the group of people treated before the budget cut consists only of compliers and always takers. Hence, the average outcome of people treated before the reform is a weighted average of treated outcomes for compliers and always takers. Using these two averages, along with the population shares of always takers and compliers, we estimate the average treated outcomes of compliers (Imbens and Rubin, 1997). This information recovers average treated outcomes for two of the three “compliance groups” that partition the population of a given racial group: always takers and compliers. However, we do not observe the average treated outcomes of never takers, since they are never treated by definition. That is the final piece to estimate the average outcome that would be realized if everyone of each racial group were treated, $E[Y_i(1) = 1 | R_i = r]$.

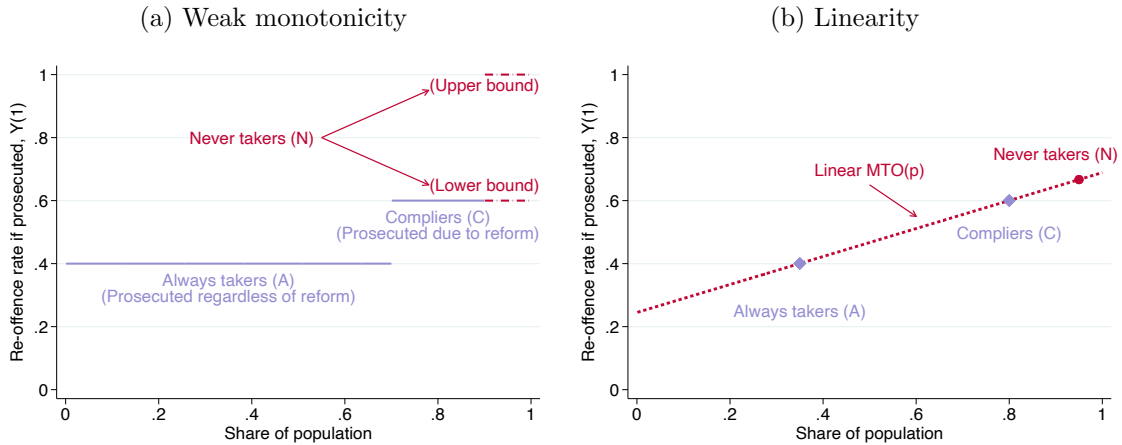
We estimate bounds (or point estimates) for the average treated outcomes of never takers by placing restrictions on the relationship between treatment propensity and average treated outcomes.¹⁶ Each “compliance group” is defined by its propensity to be treated: always takers are more likely to be treated than compliers, who are in turn more likely to be treated than never takers.

¹⁶When both a natural experiment and random assignment to decision-makers are present, the approach that identifies outcomes using information for a larger proportion of treated (or untreated, if conditioning on the untreated potential outcome) individuals will require less extrapolation and involve less extrapolation error.

Figure 1 depicts a hypothetical example where always takers and compliers are roughly 70% and 20% of the population respectively. In this example, compliers have greater treated outcomes than always takers. In the context of prosecution, this implies that on average, marginally prosecuted individuals are more likely to commit a new offence if prosecuted than those who are very likely to be prosecuted.¹⁷ We use this estimated relationship to infer the treated outcomes of never takers.

In Panel a) we assume that **average** treated outcomes are weakly monotonic in the treatment propensity of “compliance groups”, following Mogstad, Santos, and Torgovitsky (2018) and Kowalski (2023a). This assumption extends the relationship between always takers’ and compliers’ average treated outcomes to never takers’ average treated outcomes. In this example, the assumption implies that the average treated outcomes for never takers must be weakly greater than the average treated outcomes for compliers, pinning down the lower bound for never takers’ average outcomes.¹⁸ Since $Y_i(1) \in \{0, 1\}$, the average treated outcomes for never takers is bounded above by 1. Next, we compute a weighted average of 1) the bounds on the treated outcomes for never takers and 2) the point estimates for the average treated outcomes for compliers and always takers, where the weights are the shares of each compliance group. Together, these yield bounds on the average treated outcomes for each racial group, $E[Y_i(1) = 1 | R_i = r]$.¹⁹

Figure 1: Identifying the average treated outcome with a binary IV



Note: This figure uses simulated data. Lower values of the x-axis denote individuals who are more likely to be treated. $Y(1)$ denotes the treated potential outcome. The diamonds and dots in Panel b) reflect outcomes of the median individual in that group. Solid lines and diamonds represent moments observed in the data, and dashed lines and circles represent objects that are extrapolated.

¹⁷Such a pattern might arise if prosecution was focused on individuals who would reduce their future offending if prosecuted.

¹⁸This restriction is mathematically similar to the “Performance Bound” in Jordan (2024), who studies racial discrimination in felony review in a context where prosecutors are randomly assigned cases. However, those bounds arise from modelling the underlying objectives of prosecutors, while ours place restrictions on the behavior induced by the policy reform.

¹⁹We can still use this approach if potential outcomes are multi-valued. In such a case, we require an estimate of the prevalence of each possible value of the potential outcome, $E[Y_i(1) = y | R_i = r]$, to identify the discrimination estimand in Definition 2. The logic underlying the bounding approach will still hold, since each expectation $E[Y_i(1) = y | R_i = r]$ represents a proportion and is hence also bounded between $[0, 1]$.

Note that the restrictions we place to bound average treated outcomes do not require assuming the underlying decision-makers focus on a single narrow objective. Rather, this approach allows decision-makers to adopt a range of multi-dimensional models as long as their decisions generate patterns that imply that average potential outcomes for each of the “compliance groups” is weakly monotonic in their likelihood of being treated. In the context of prosecution, the extrapolation in Panel a) of [Figure 1](#) assumes that never takers, who are least likely to be prosecuted, are at least as likely as compliers to re-offend if prosecuted. This assumption might be violated if other inputs into prosecution decisions generate contradictory patterns. For example, say that prosecutors are also less likely to pursue cases with poor quality evidence, such that all the never takers’ cases are poor quality. If individuals whose cases are poor quality never re-offend if prosecuted, the weak monotonicity assumption could be violated. In such situations, an alternative approach is to bound never takers’ treated outcomes between 0 and 1—the widest logically possible bounds (Manski, 1989).

Instead of bounds, we can also obtain point estimates of discrimination if we restrict the relationship between the underlying treatment propensity and treated outcomes to be linear. Panel b) of [Figure 1](#) demonstrates this, where the diamonds plot the treated outcomes for the median always taker and complier against their respective treatment propensities.²⁰ Assuming this relationship is linear lets us extrapolate the treated outcomes across the support of the treatment propensity and point identify the treated outcomes of never takers. This restriction in turn identifies the marginal treated outcome function which we would integrate to estimate $E[Y_i(1) = 1 | R_i = r]$ (Heckman and Vytlacil, 2000; Brinch, Mogstad, and Wiswall, 2017; Mogstad, Santos, and Torgovitsky, 2018; Kowalski, 2023b).²¹ Since this approach involves assuming that the relevant marginal treatment response function is linear, it places stronger restrictions on behavior than the previous partial identification approach. We illustrate this point-identification approach in the context of racial discrimination in incarceration decisions in [Appendix B.3](#), which sketches a simple model of selection, describes the required assumptions, and implements a brief empirical example.

Implementing this procedure separately by race group r provides either bounds or point estimates for the average outcome that would be realized if everyone of each group were treated, $E[Y_i(1) = 1 | R_i = r]$. As mentioned earlier, this is the final object that we need to estimate discrimination that is conditional on treated potential outcomes, which is reproduced for the case of binary treated outcomes in [Equation 2](#).

²⁰Linearity and the uniformity of the underlying latent index determining treatment implies that the median outcome of each compliance group has the average treated outcome of that compliance group (Kowalski, 2023b).

²¹Linearity assumptions can identify all the marginal treatment response functions. We would identify the marginal untreated outcome function by using the average untreated outcomes for compliers & never takers to extrapolate the untreated outcome for always takers. Along with the marginal treated outcome function, this identifies the marginal treatment effect function.

$$\begin{aligned}
\pi_{zr1} &= \frac{\overbrace{E[Y_i|Z = z, R_i = r, D_i = 1] \times E[D_i|Z = z, R_i = r]}^{\text{Observed in data}}}{\underbrace{E[Y_i(1) = 1|R_i = r]}_{\text{Extrapolated}}} \\
\pi_{zr0} &= \frac{\overbrace{E[(1 - Y_i)|Z = z, R_i = r, D_i = 1] \times E[D_i|Z = z, R_i = r]}^{\text{Observed in data}}}{\underbrace{1 - E[Y_i(1) = 1|R_i = r]}_{\text{Extrapolated}}} \tag{2}
\end{aligned}$$

$$\begin{aligned}
\Delta_{zy} &= \pi_{zwy} - \pi_{zby} \\
\Delta_z &= \sum_{y \in \{0,1\}} Pr(Y_i(1) = y) \Delta_{zy}
\end{aligned}$$

We observe the re-offence rate among prosecuted individuals in each period and race in the data: $E[Y_i|Z = z, R_i = r, D_i = 1]$. We also observe the prosecution rate for each period and race: $E[D_i|Z = z, R_i = r]$. Plugging the bounds/point estimates for $E[Y_i(1) = 1|R_i = r]$ into the first two lines of [Equation 2](#) generates bounds/point estimates for each period-, race-, and potential outcome-specific treatment rate, π_{zry} . Using π_{zry} in the third line yields bounds/point estimates for the period- and potential outcome-specific discrimination Δ_{zy} . We then construct the period-specific discrimination estimates, Δ_z , as a weighted average of the period- and potential outcome-specific discrimination, where the weights are defined by the prevalence of the treated outcome in the population, $Pr(Y_i(1) = 1)$ (fourth line of [Equation 2](#)).

2.3 Implementation with common quasi-experimental variation

So far, we have discussed using an unconditionally random binary IV to assess if treated outcomes differ by group (e.g., race) and then estimate discrimination, conditional on treated outcomes. Here, we discuss the practical details of implementing our approach with common quasi-experimental designs that generate conditionally random binary IVs, e.g., regression discontinuity (RD) or difference-in-difference (DiD).

Regression discontinuity

As is well known, RDs are well-suited to identify treatment effects and average potential outcomes at a particular cut-off. The simplest way to implement our approach with an RD is in the spirit of the local randomization approach, which typically involves only using information in a small region just below and just above a cut-off (Cattaneo, Idrobo, and Titiunik, 2024). In such a setup, the binary instrument will be an unconditionally random indicator for being above or below the cut-off, letting us follow our approach as discussed above. However, the sample size reduction from limiting to a very small window around a cut-off will typically result in power issues.

When using our approach with an RD_i we can use data from a larger window around the cut-off with some minor adjustments. To see how, consider the following expression of an RD design: $X_i = \alpha + \beta \mathbb{I}(S_i > c) + \delta_1 f(S_i) + \delta_2 \mathbb{I}(S_i > c) \times g(S_i) + \varepsilon_i$. Here S represents the running variable, $f(\cdot)$ and $g(\cdot)$ govern the relationship between S and the outcome, c represents the cut-off, and the binary instrument Z is given by $\mathbb{I}(S_i > c)$. When estimating this equation with X_i as an indicator for treatment, α estimates the share of individuals at the cut-off receiving treatment despite having $Z = 0$ —in other words, the share of always takers. β represents the share of individuals at the cut-off receiving treatment due to the instrument, pinning down the share of compliers. Next, consider estimating a version of the expression above on the set of treated individuals, this time with X_i representing the outcome of interest. In this case, α pins down the average treated outcome for always takers at the cut-off. $\alpha + \beta$ pins down a weighted average of treated outcomes for always takers and compliers at the cut-off.

As in the case with a simple binary IV, we have all the information we need to 1) back out the average treated outcomes for compliers, 2) obtain bounds or point estimates for the average treated outcomes in the population, and 3) use that to measure discrimination at the RD cut-off conditional on potential outcomes. Appendix B.4 illustrates using an RD design with our approach, through an empirical example studying socio-economic discrimination in the decision to allow public school students to progress to the next grade.

While this implementation uses observations away from the cut-off, the resulting estimates of discrimination conditional on potential outcomes are still informative only for observations at the RD cut-off rather than away from it. Extending the approach to measure discrimination in a small window around the cut-off requires two additional assumptions (described in detail in Appendix B.5). First, the running variable alone cannot influence treatment. Second, the average outcome if everyone at the cut-off were treated would need to be the same as the average outcome if everyone above or below the cut-off were treated.²² Since the running variable often has a tight relationship with both treatment and potential outcomes in many settings, these assumptions are generally strong and may be less likely to be satisfied.

Difference-in-difference

Consider a setting, as in our misdemeanor prosecution application, where the key variation comes from the timing of a budget reform adopted in one county but not others. Mapping this variation to the IV framework that is the core of our approach, individuals rather than counties select into treatment (i.e., prosecution) and time might independently influence potential outcomes and treatment. With this time variation, IV designs in the spirit of two-way fixed effects that condition on indicators for county and time period will not identify the proportions and average potential outcomes for each of the “compliance groups”, information that we need to measure discrimination.

²²These assumptions are stronger than those in typical approaches to extrapolate away from RD cut-offs (e.g., Angrist and Rokkanen, 2015; Cattaneo et al., 2021; Ricks, 2022) because our focus is on extrapolating average potential outcomes rather than treatment effects.

Under additional assumptions, we can account for the time trends in the county that *adopted the reform*, which we do in our application studying discrimination in misdemeanor prosecution. Focusing on the treated potential outcome and the bounding approach, we account for time trends by using the change in treated outcomes in counties that *did not adopt a reform* as an estimate of the change in treated outcomes the county that *adopted the reform* would have experienced if it had not adopted the reform. This adjustment assumes that for individuals in each racial group: 1) time alone does not influence treatment status, and 2) time trends in average treated outcomes are the same for always takers and compliers, and are independent of county (in a window around policy adoption).^{23,24,25} Appendix B.6 formally discusses the assumptions and adjustment, and also discusses extensions to condition on untreated potential outcomes. This adjustment essentially provides a way to map DiD designs to existing IV methods of estimating average potential outcomes for each “compliance group” (Imbens and Rubin, 1997).

After correcting for the effects of time in the county that adopted the reform, we only use information from that county and define the instrument as we would in the simple binary case, where Z represents periods before and after the reform. Note that since there are time trends in potential outcomes, the average treated outcomes for each compliance group and the average outcomes if everyone were treated will vary by time period.

The logic of this approach also extends to DiD with staggered policy adoption. Following prior work on estimating policy impacts in settings with staggered adoption, we can pair each county that adopts a policy (‘adopter’) with a set of counties that never adopted or have not yet adopted the policy (‘not-yet-adopter’) (Cengiz et al., 2019). Then, the second assumption described above needs to hold between each ‘adopter’ and its associated set of ‘not-yet-adopters’. Combining the IV extrapolation approach described earlier with these DiD adjustments for each ‘adopter’ provides estimates of discrimination for each ‘adopter’. These estimates can be aggregated across ‘adopters’ to construct an average measure of discrimination across all ‘adopters’.

2.4 Recap: Step-by-step guide

In this section we have discussed how to use a natural experiment to estimate discrimination between two groups, accounting for any underlying differences in the distribution of potential outcomes.²⁶

²³The second assumption is stronger when using our approach to obtain point estimates of discrimination, which requires assuming that the outcomes of always takers, compliers, and never takers are linearly related. Hence, the assumption of equal of time trends across always takers and compliers must extend to never takers as well.

²⁴While this second assumption is in the spirit of parallel trends, it contrasts with the usual DiD implementations in which everyone in counties where a policy occurs are considered ‘treated’ by the policy, while the rest are considered ‘untreated’. There, the standard parallel trends assumption identifies impacts of the policy by assuming parallel trends in the average untreated potential outcomes between the two counties.

²⁵These assumptions are similar to those underlying the “time-corrected” Wald estimand in De Chaisemartin and D’Haultfœuille (2018), which allow identification of the local average treatment effect but not of the average potential outcomes of each “compliance group” separately.

²⁶So far we have discussed estimating aggregate measures of discrimination. One can also use our approach to quantify decision-maker-specific discrimination estimates with large enough samples and first stages within the subsample for each decision-maker, j . Repeating our approach within each j ’s subsample yields estimates of the average potential outcomes in each j ’s subsample, $E[Y_i(D_i)|j]$. Decision-makers with similar $E[Y_i(D_i)|j]$ observe

Implementing this requires the following:

1. A natural experiment that generates a binary IV
2. A potential outcome, $Y_i(D_i)$, that corresponds to a notion of fairness. That is, the choice of $Y_i(D_i)$ should be such that group differences in treatment between people with the same $Y_i(D_i)$ can be interpreted as an unwarranted disparity.
3. Use the natural experiment to estimate whether the underlying distributions of potential outcomes differ between the groups. For example, if $Y_i(D_i)$ is binary and we want to condition on the treated potential outcome, this involves comparing the average outcomes if everyone of each group were treated.
4. If the potential outcomes differ by group, estimate average discrimination conditional on potential outcomes following [Equation 1](#).

In the next section we apply this approach to study racial discrimination in prosecution. We adopt the partial identification approach since the data are inconsistent with the linearity restriction required for point identification. We first bound the average potential outcomes separately by race, and then plug these bounds directly into the expression for the average period-specific discrimination (Δ_z).²⁷ We compute the bounds for Δ_z by searching over a grid defined by the combinations of the average potential outcome bounds for each racial group.

For inference, we generate 95% bootstrapped confidence intervals for these bounds using a Bayesian bootstrapping procedure (Rubin, 1981). We use weights randomly drawn from $\Gamma(1, 1)$ to compute all the moments in the estimation procedure, enforcing the weak monotonicity restriction within each re-weighted bootstrap sample. Finally, we report confidence intervals for the true underlying parameter (rather than each bound) using the resulting bootstrap distribution (Imbens and Manski, 2004).

3 Racial discrimination in misdemeanor prosecution

In this section we use quasi-experimental variation from a cut to the King County prosecutors' budget in Washington State to measure racial discrimination in the decision to prosecute individuals arrested for misdemeanors. Misdemeanors represent the most common form of contact with the criminal court system (Stevenson and Mayson, 2018) and the decision to prosecute cases like

subsamples with similar potential outcomes. j -specific discrimination estimates are comparable, in a 'selection-on-unobservables' strategy, among such a subset of decision-makers.

²⁷Note that one could also bound average period-specific discrimination by 1) first constructing gaps using the bounds on the group- and potential outcome-specific treatment rates, 2) computing the average, and then 3) taking the minimum and maximum. This will differ from our approach of plugging in the bounds on average potential outcomes directly into the equation for average period-specific discrimination because minimum/maximum are not linear functions. We prefer directly plugging bounds on average potential outcomes into the expression for Δ_z in [Equation 2](#) since our main goal is estimating discrimination conditional on potential outcomes.

these can have long-lasting adverse impacts on individuals’ lives (Leasure, 2019; Mueller-Smith and Schnepel, 2021; Agan, Doleac, and Harvey, 2023).

We first find that in this setting, potential re-offence outcomes if prosecuted differ by race. Accounting for these racial differences in potential outcomes, we find no evidence of discrimination in prosecution before the budget cut. After the budget cut, which sharply reduced prosecution rates, we find that white defendants were more likely to be prosecuted than minority defendants. We find suggestive evidence that this gap is driven by prosecutors dropping cases that were likely low quality, which were more prevalent among minority defendants. Crucially, misdemeanor cases are typically not randomly assigned to prosecutors in King County, ruling out existing random assignment-based tools to measure discrimination here.²⁸

3.1 Natural experiment: King County budget reform

We study racial discrimination in misdemeanor prosecution in King County, Washington (Seattle metropolitan and suburban areas) using administrative records on all criminal cases from 2002–2014 from the Washington Administrative Office of the Courts.²⁹ We consider an individual as having their case prosecuted if their case **did not** meet the following condition: dismissed without requiring any punishments.³⁰ Our primary definition of punishment excludes fines, but we assess robustness to including them.

We focus on differences in prosecution between white (non-Hispanic) and ‘minority’ defendants. This comparison is motivated by the fact that the population that has contact with the Washington criminal legal system is quite diverse. A large proportion of non-white defendants are of Native Hawaiian and Pacific Islander descent—these groups often face disadvantage of various forms and are over-represented in the criminal legal system in the Western United States (Hu and Esthappan, 2017; Malott, 2024). Nevertheless, we later demonstrate that our results are robust to comparing prosecution rates between white (non-Hispanic) and Black & Hispanic defendants.

As discussed in Section 2.2, we need a natural experiment that shifts prosecution rates to assess if unobserved potential outcomes vary by race, and then estimate discrimination conditional on the unobserved potential outcomes. We use the fact that King County, facing a \$60 million budget shortfall in September 2010, cut the Prosecutors’ Office budget by approximately \$3.9 million, the equivalent of 33 full-time employees (Constantine, 2010). In response, the Prosecutors’ Office warned that the unanticipated reduction in resources would reduce their ability to prosecute challenging and time-consuming cases, and that they would have to focus resources on high-priority

²⁸Correspondence with the King County Prosecuting Attorney’s Office suggests most cases submitted by police are assigned to an attorney (or a team of attorneys) based on offence type, experience, and workload, who decide whether to file the case. The filing decision is then approved by supervising attorneys. While some subsequent proceedings for filed cases may be quasi-randomly assigned (e.g., arraignment based on a stipulated calendar), prosecutors for many steps are often assigned at the discretion of supervising attorneys.

²⁹Since these data consist of individuals in the court records our estimates are representative of individuals who have been arrested and whose cases have been sent by police to the prosecutors’ office.

³⁰Our data do not allow us to accurately distinguish between situations where prosecution was pursued and: individuals were convicted, prosecution failed, and charges were dismissed upon successful completion of a sentence. Our definition considers all of the above scenarios as ‘prosecution’.

offences (Ervin, 2010). The County tried to mitigate the budget cuts’ impact on the criminal legal system by holding a referendum to raise funding via a sales tax increase. However, the referendum failed in November 2010, consigning the King County Prosecutors’ Office to the new budget realities (Ballotpedia, 2010).³¹

We use the budget reform with our discrimination estimation approach to study racial discrimination in prosecution. The sharp contraction to prosecutorial resources should result in many cases being dropped, especially given the Prosecutors’ Office’s public statements. A shift in prosecution rates would let us partition each racial group into always takers, compliers, and never takers for prosecution. We would then examine how the average prosecuted outcomes vary across racial groups, and then account for any racial differences in average prosecuted outcomes.

We isolate the quasi-experimental variation using a difference-in-difference strategy that compares changes in prosecution and recidivism rates around the budget reform in King County, relative to changes in the adjacent counties unaffected by the reform (Chelan, Kitsap, Kittitas, Pierce, and Snohomish). We construct our analysis sample using criminal cases disposed in the District Courts of these counties. County prosecutors who work in District Courts typically hear criminal misdemeanor cases of varying severity. Given the messaging from the Prosecutors’ Office regarding the types of cases they will find difficult to pursue, cases in District Courts are most likely to be affected by the prosecutorial budget cuts. We limit to misdemeanor cases disposed in the relevant District Courts between October 2008 and September 2012, a two-year span on either side of the budget cut announcement.³²

We construct re-offence outcomes and measure criminal history using information on cases filed by law enforcement in Washington State, including felony offences and offences filed outside of the October 2008 and September 2012 interval. We measure re-offending by tracking whether a defendant appears again in these data after disposition. This measure of re-offending will not capture undetected criminal activity or arrests that police did not forward to prosecutors. However, it will capture new arrests that police forward to prosecutors, including cases that prosecutors choose not to pursue.

The final sample, described in Table 1, consists of around 120,000 unique cases. 30% of the sample consists of non-white defendants, who broadly represent the diverse population in this context—43% are Black, 32% are Hispanic and 18% are Asian American or Pacific Islander (AAPI). We refer to the group of non-white defendants as ‘minority’ defendants. Women make up almost a quarter of the sample, and the sample consists of a wide range of individuals in terms of age and

³¹We rule out that the change in the Seattle City Attorney, who pledged to reduce racial disparities and prosecution of minor offences, on January 1, 2010 poses a confounding threat. The City Attorney has jurisdiction over Seattle’s local municipal courts (and county prosecutors do not), which we exclude from our sample of case dispositions. Hence, direct impacts of the City Attorney change are unlikely to be present in our analysis. Since this change occurred before the county budget cut, any indirect effects of the City Attorney’s reforms on the broader courts in our sample should show up as differential pre-trends in the year leading up to the budget reform. We do not find evidence of such differential pre-trends during this period in our event studies examining prosecution, recidivism, or caseload composition (discussed below).

³²If an individual has cases filed on multiple dates within this time frame, we only include the first case to ensure that the probability of multiple appearances in our sample is not a function of prosecution decisions.

criminal background. The average defendant is almost 35 years old, and 47% of individuals have had at least one conviction in the past (averaging 3.5 prior convictions conditional on having any). Prosecution rates are quite high and individuals are unlikely to face jail time if they prosecuted. Only 9% of individuals prosecuted in King County (and 6% overall) in our sample were sentenced to any jail time. Among those who serve any time in jail, the average sentence is around 40 days long. By statute, the longest jail sentence for misdemeanors in Washington is one year, a sentence which only occurs for 0.5% of prosecuted cases in our sample.

Table 1: Characteristics of the Washington District Court sample

	Overall	King County	Adjacent Counties
N	122,156	51,242	70,914
Demographics			
White (Non-Hispanic)	0.72	0.65	0.78
Black	0.12	0.16	0.09
Hispanic	0.09	0.09	0.09
AAPI	0.05	0.08	0.03
Age at disposition	34.6	34.8	34.4
Male	0.74	0.73	0.74
Criminal history			
Any prior convictions	0.47	0.42	0.50
# prior Any	3.8	3.5	3.9
Case outcomes			
Case prosecuted	0.86	0.82	0.88
<i>White</i>	0.86	0.83	0.87
<i>Minority</i>	0.85	0.81	0.89
Jail sentence Prosecuted	0.06	0.09	0.05
Sentence length (Days) Jail sentence	40.4	39.6	41.3

Note: Sample includes all criminal cases disposed in the District Courts in Chelan, King, Kitsap, Kittitas, Pierce, and Snohomish counties in Washington State between October 2008 and September 2012. For defendants with multiple dispositions in this time frame, we include only the first case. “AAPI” stands for Asian American or Pacific Islander.

Estimating the first stage & shifts in recidivism outcomes

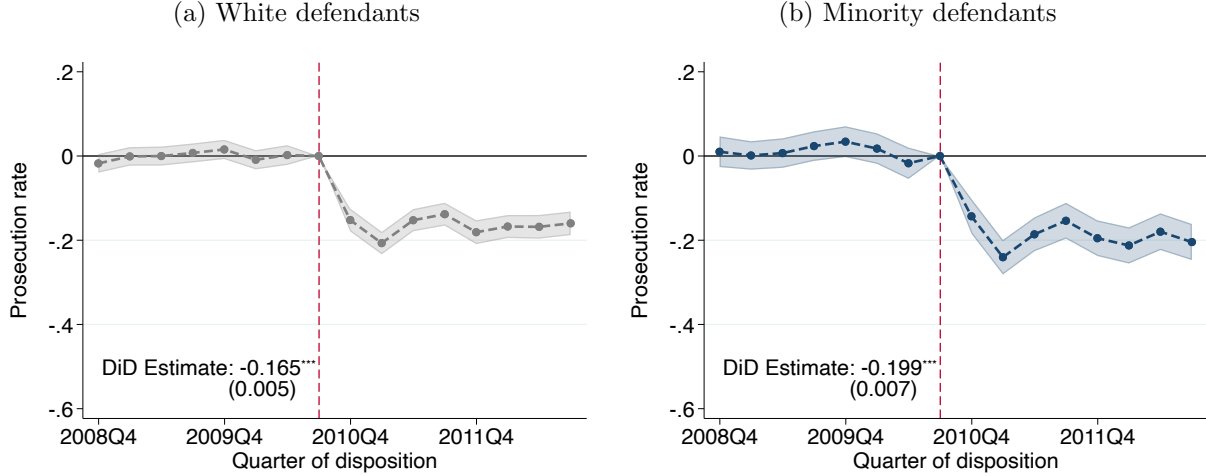
We first compare the changes in prosecution rates before and after the King County budget reform to changes in prosecution rates in adjacent counties that were unaffected by the budget reform. This involves estimating the specification in Equation 3, where D_{itg} denotes whether defendant i was prosecuted in quarter t and g denotes whether the case was disposed in King County or the adjacent counties. We investigate the changes in prosecution rates separately for white and minority defendants to ensure that we have sufficient variation in prosecution rates in each racial subgroup.

$$D_{itg} = \theta_t + \delta \mathbb{I}[\text{King County}] + \sum_{k \neq -1} \beta_k (\mathbb{I}[t - \text{Budget Reform} = k] \times \mathbb{I}[\text{King County}]) + \varepsilon_{itg} \quad (3)$$

Figure 2 documents large drops in prosecution rates due to the King County budget reforms. Prosecution rates fall by 16.5 p.p. and 19.9 p.p. for white and minority defendants respectively

(17.8% and 21.7% of pre-reform average prosecution rates in King County). Reassuringly, we do not see any evidence of differential pre-trends in prosecution rates, which provides initial evidence in support of a credible natural experiment (we discuss additional validity tests below). The magnitude of these shifts in prosecution rates pin down the complier share of white and minority defendants. Given the overall high rate of prosecution, these large shifts in prosecution suggest that we will have to bound outcomes for a relatively small share of the population.

Figure 2: Impact of King County budget reform on prosecution rates



Note: Each Panel presents event study estimates investigating the impact of the King County budget reform. Sample includes all misdemeanor defendants as described in Table 1. ‘DiD Estimate’ pools the coefficients on relative time indicators and estimates $D_{igt} = \alpha + \delta_1 \mathbb{I}[\text{King County}] + \delta_2 Post_i + \beta^{DD} \mathbb{I}[\text{King County}] \times Post_i + \epsilon_{igt}$, where $Post_i = 1$ if the case is disposed on or after September 28, 2010, when the budget reform was announced. 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

We next examine whether these shifts in prosecution rates influenced individuals’ re-offence outcomes, which will reduce the width of the bounds for never takers’ prosecuted outcomes. Figure 3 repeats the event study approach, assessing the impact of the budget reform on the probability that a defendant re-offends one year after disposition. We see that not being prosecuted reduces one year re-offence rates by 3.1 p.p. for white defendants and 4.7 p.p. for minority defendants (13.1% and 15% of pre-reform average re-offence rates in King County).^{33,34,35} Again, we see no evidence

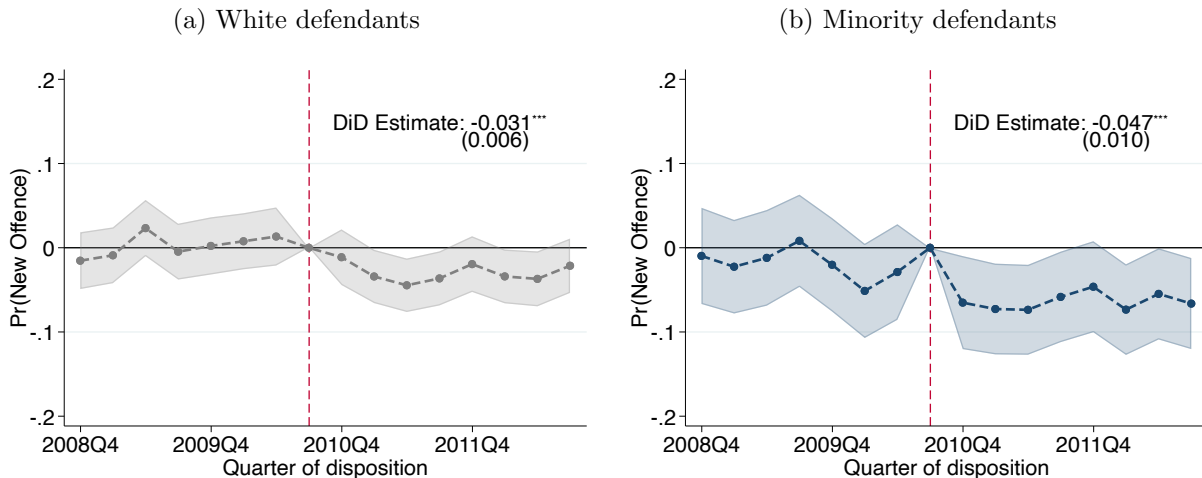
³³These reductions are unlikely to be driven by incapacitation since only 9% of sentences for prosecuted defendants in King County involved jail. Among those with jail sentences, the average length was 40 days, much shorter than the one year horizon used to compute the outcome (see Table 1). As a result, potential outcomes are not mechanically censored by jail spells.

³⁴These reduced form estimates are not driven by prosecutors systematically changing what cases they accept from law enforcement, e.g., by refusing all low priority cases. Figure A1 presents results excluding charges for offences commonly dropped right after the budget reform announcement (resisting arrest, criminal trespass, driving with a suspended licence, minor marijuana possession, reckless driving and DUI). Assuming these commonly-dropped charges are those that the prosecutor’s office considers low-priority, we designate other charges as ‘high-priority’. We see reductions in the probability of ‘high-priority’ re-offending that are similar to our baseline estimates in terms of their proportion of the relevant pre-reform means.

³⁵The direction and magnitude of these estimates are consistent with recent work finding that avoiding prosecution for minor offences reduces future interactions with the criminal legal system (e.g., Mueller-Smith and Schnepel, 2021; Agan, Doleac, and Harvey, 2023).

of pre-trends in re-offence rates. Figure A2 presents event study estimates showing how the reform impacts prosecuted outcomes—we later use the shift in prosecuted outcomes to estimate average prosecuted outcomes for always takers and compliers.

Figure 3: Impact of King County budget reform on re-offence within one year



Note: Each Panel presents event study estimates investigating the impact of the King County budget reform. The outcome includes any new misdemeanor or felony offences committed anywhere in Washington State. Sample includes all misdemeanor defendants as described in Table 1. ‘DiD Estimate’ pools the coefficients on relative time indicators and estimates $Y_{igt} = \alpha + \delta_1 \mathbb{I}[\text{King County}] + \delta_2 \text{Post}_i + \beta^{DD} \mathbb{I}[\text{King County}] \times \text{Post}_i + \epsilon_{igt}$, where $\text{Post}_i = 1$ if the case is disposed on or after September 28, 2010, when the budget reform was announced. 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

Assessing identifying assumptions

Next, we rule out various threats to the natural experiment’s validity. A key concern is the presence of other concurrent policy or behavioral changes that could influence the determinants of crime and confound our estimates of the budget reform’s effects. These might occur if other aspects of King County institutions, e.g., police or social services, were affected by the budget reform, or if prosecutors altered their behavior in ways other than prosecuting fewer cases. We assess how likely these concerns are through multiple exercises.

First, we test whether the composition of cases that prosecutors choose to accept from law enforcement changes due to the budget reform. We estimate a series of regressions where we compare the change in the share of individuals with a given baseline covariate before versus after the reform in King County to that same change in the adjacent counties. Figure A3 presents each of these coefficients and shows limited evidence of compositional shifts across multiple observable characteristics (Figure A4 presents the underlying event study patterns for each covariate). We see some evidence that individuals whose cases were accepted after the reform were 1.8 p.p. more likely to have been previously charged with an offence but this is a small shift relative to the pre-period mean in King County (3.5%).³⁶

³⁶This minor compositional shift might be due to the end of the federal investigation of Seattle Police Department (SPD) which resulted in lower SPD stops and arrests (Campbell, 2023). This compositional shift may arise if police

A related concern is whether law enforcement agents in King County were laid off or reduced their arrest effort (perhaps anticipating that prosecutors would stop prosecuting certain cases) due to the reform, and whether the reform affected other economic determinants of crime. In [Figure A6](#) & [Figure A7](#) we show that there are no significant reductions in either officer employment or different categories of arrests. [Figure A8](#) similarly finds no differential changes in house prices, unemployment rates, or population, suggesting that the reform did not meaningfully impact other economic factors that might influence criminal behavior.

The results of these exercises suggest that the drop in prosecution rates and resulting reduction in recidivism are driven by the unanticipated budget reform that affected the operations of the Prosecutors’ Office, and not by concurrent policy factors or behavioral changes. Together with the stable pre-trends, these results provide strong evidence that the King County budget reform is a valid natural experiment. Next, we use this natural experiment to estimate racial differences in prosecution among individuals who would have the same re-offence outcome if prosecuted and then estimate the analog, conditioning on the re-offence outcome if dismissed.

3.2 Estimating discrimination in prosecution using the budget reform

As a refresher, we begin by mapping the objects in the empirical discussion to the potential outcomes framework discussed in [Section 2.2](#). Individuals are considered ‘treated’ if they are prosecuted ($D_i = 1$) and considered ‘untreated’ if dismissed. Potential outcomes in each treatment state are binary indicators for whether an individual assigned to a given treatment state would re-offend within one year of disposition. We start by considering the re-offence outcome if prosecuted.³⁷ Z indicates periods around the King County budget reform, which reduced prosecution rates. As described in [Section 2.2](#), the first step is to use the quasi-experimental variation from the budget reform to estimate the average re-offence outcome that would be realized if all defendants of each racial group were prosecuted. These objects would help us understand whether the unobserved potential outcomes vary by race, in which case we adjust the discrimination estimates for such differences.

Given the time-varying nature of potential outcomes in a DiD setting, we first need to adjust for any trends in potential outcomes due to the passage of time. Following the DiD adjustments discussed above, we use the trend in re-offence outcomes among prosecuted individuals in adjacent counties as an estimate of the time trend in outcomes that prosecuted individuals in King County would have experienced had the budget reform not occurred. As described in [Section 2.3](#) and [Appendix B.6](#), this adjustment is valid under two additional assumptions. In this context, we need

focused on more serious offences. Consistent with this mechanism, Campbell (2023) finds the largest reductions in SPD activity after the investigation ended (December, 2011), which corresponds to when we see changes in defendants with a prior charge (see Panel c) of [Figure A4](#)). Our first stage and reduced form estimates are nearly identical if we exclude this time period (see [Figure A5](#)), and so we proceed with the full sample for our baseline analysis.

³⁷Any racial differences in re-offence outcomes if prosecuted might be partially driven by differences in sentencing, complicating the interpretation of discrimination conditional on re-offence outcomes if prosecuted. Below, we also present exercises comparing defendants with similar baseline likelihood of re-offending by conditioning on the re-offence outcome if dismissed.

the following conditions to hold within each race group:

A1 Without the reform, time does not shift individuals’ prosecution status.

A2 Without the reform, re-offence outcomes if prosecuted, $Y_{it}(1)$, would trend similarly for always takers & compliers and is independent of county.

While these assumptions are fundamentally untestable, Appendix A.2 provides suggestive evidence that they are not violated in this setting. To examine **A1**, we use pre-period data and find negligible shifts in prosecution rates over time across covariate subgroup (gender, criminal history, age)-by-race cells (see Figures A22 and A23). These results build confidence that individuals are not shifting treatment status over time. To examine **A2**, which requires parallel trends to hold between always takers and compliers, we test for differential pre-trends by covariate subgroup, since baseline characteristics might be correlated with being an always taker or complier. Using pre-period data, we find limited evidence of differential trends in $Y_{it}(1)$ across counties within the various subgroups (Figure A24) and within counties but across subgroups (Figure A25). While these patterns are not definitive evidence that these assumptions are satisfied, they build credibility that they are reasonable in this setting.

3.2.1 Baseline estimates of discrimination

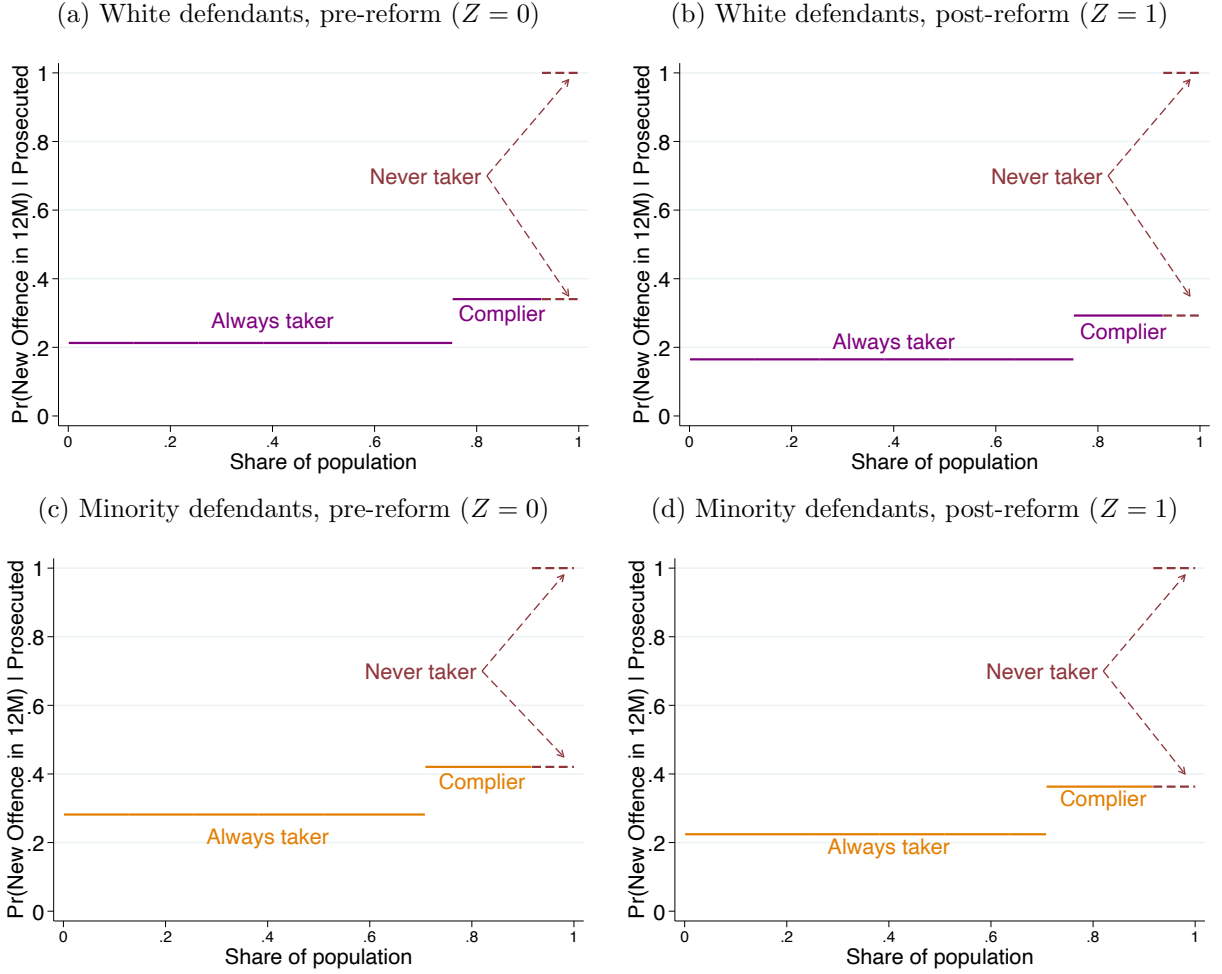
We now use the variation from the budget reform DiD to estimate if there are racial differences in the average re-offence outcome that we would see if all defendants of each race were prosecuted. Since Y_{it} varies with time, we estimate average prosecuted outcomes in each period t : $E[Y_{it}(1)|R_i = r]$. We then measure racial gaps in prosecution that account for such differences, following Section 2.

Bounding the race-specific average outcomes if prosecuted

Figure 4 displays how average re-offence outcomes if prosecuted vary across always takers, compliers, and never takers for white and minority defendants.³⁸ Panels a) and c) present estimates from before the reform, and Panels b) and d) present estimates from after the reform. The average outcomes for always takers and compliers for white defendants are uniformly lower than for minority defendants. This pattern suggests that there likely are racial differences in the average re-offence outcomes that we would see if all defendants were prosecuted. Additionally, never takers comprise 7-8% of the population of each racial group, implying that we have to bound outcomes for a relatively small portion of the population. As a result, the estimated bounds on the average outcomes if everyone were prosecuted will be relatively tight. Since these bounds are inputs into estimating the discrimination estimands (see Equation 2), we should obtain relatively tight bounds on discrimination as well.

³⁸As discussed in Section 2, estimating average potential outcomes by ‘compliance group’ is valid under IV monotonicity, which rules out the existence of ‘defiers’. Here, ‘defiers’ would be individuals who would not be prosecuted before the reform but would be after the reform. We assess the likelihood of ‘defiers’ in this context by re-estimating the first-stage separately by race and baseline covariate (age bins, gender, criminal history). We find consistently large and significant reductions in prosecution similar to the estimates in Figure 2 across all race $\times$ covariate cells, suggesting that ‘defiers’ are unlikely to be present here (Figure A9).

Figure 4: Average re-offence outcomes if prosecuted ($Y_i(1)$) by compliance group



Note: This figure shows the average treated outcomes for always takers, compliers, and never takers for each time period. The treatment is prosecution and the treated outcome, $Y_i(1)$, is whether an individual re-offends one year after disposition, if prosecuted. The bounds for the treated outcomes for never takers come from the assumption of weak monotonicity of average treated outcomes across compliance groups, and that $Y_i(1) \in \{0, 1\}$.

As discussed earlier, the next step assumes that the average potential outcome if prosecuted is weakly monotonic across always takers, compliers, and never takers, which implicitly places assumptions on the underlying decision-maker behavior. Here, the assumption implies that the defendants who prosecutors are least likely to prosecute (never takers) are at least as likely to re-offend if prosecuted as marginal defendants are.³⁹

Figure 5 estimates bounds for the average prosecuted outcome by computing a weighted average of the average outcomes for always takers, compliers, and never takers from Figure 4. We find

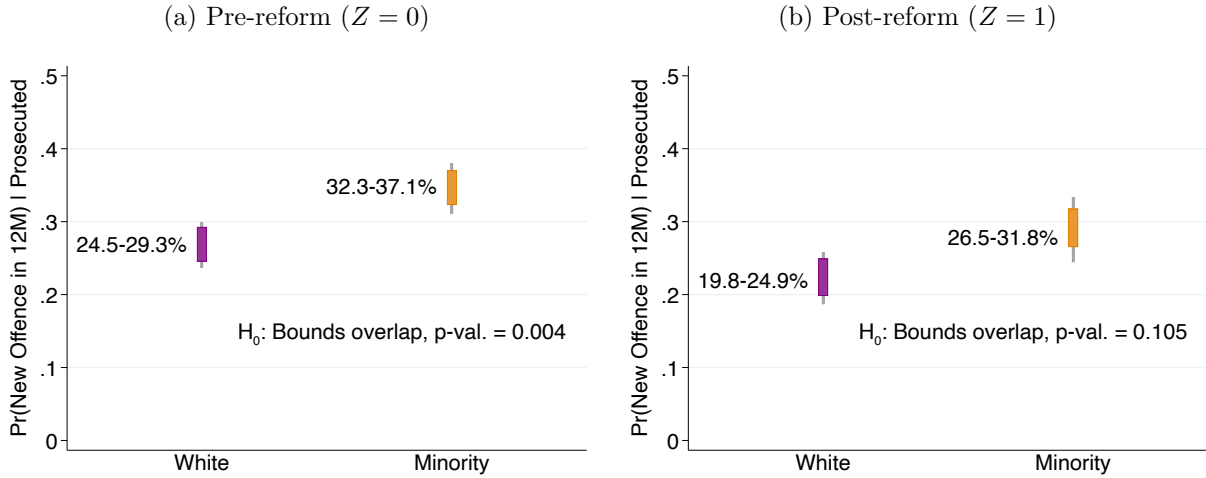
³⁹This assumption could be violated if other inputs into prosecution decisions are correlated with re-offence outcomes in specific ways. E.g., if i) never takers are not prosecuted because their cases are backed up by low quality evidence, and ii) individuals whose cases are low quality are systematically unlikely to re-offend if prosecuted. While we do not have evidence that such violations occur in our setting, we test robustness to not assuming weak monotonicity and find similar results.

meaningful and significant cross-race differences in the average outcomes that would be realized if all defendants were prosecuted. Before the budget reform, approximately 25–29% of white defendants would have re-offended if prosecuted, while approximately 32–37% of minority defendants would have done so. Using a bootstrapped inference procedure described in Appendix B.7, we reject the null that these bounds overlap ($p = 0.004$). After the reform, the bounds on average prosecuted outcomes are still meaningfully different (approximately 20–25% vs 27–32%), although testing the probability that they overlap is less precise ($p = 0.105$).⁴⁰ Given that re-offence outcomes if prosecuted differ meaningfully by racial group, not accounting for unobservable cross-race differences would yield incorrect estimates of discrimination.

Estimating the racial gap in prosecution

Next, we estimate bounds for the racial differences in prosecution rates that condition on re-offence outcomes if prosecuted. Following Equation 2, we need, for each racial group and time period, 1) the average re-offence outcomes among prosecuted defendants, 2) the prosecution rate, and 3) the average re-offence outcomes if everyone were prosecuted. 1) and 2) are directly observed in the data, and we use bounds for 3) from Figure 5.

Figure 5: Average re-offence outcomes if prosecuted, $E[Y_{it}(1)|R_i = r]$

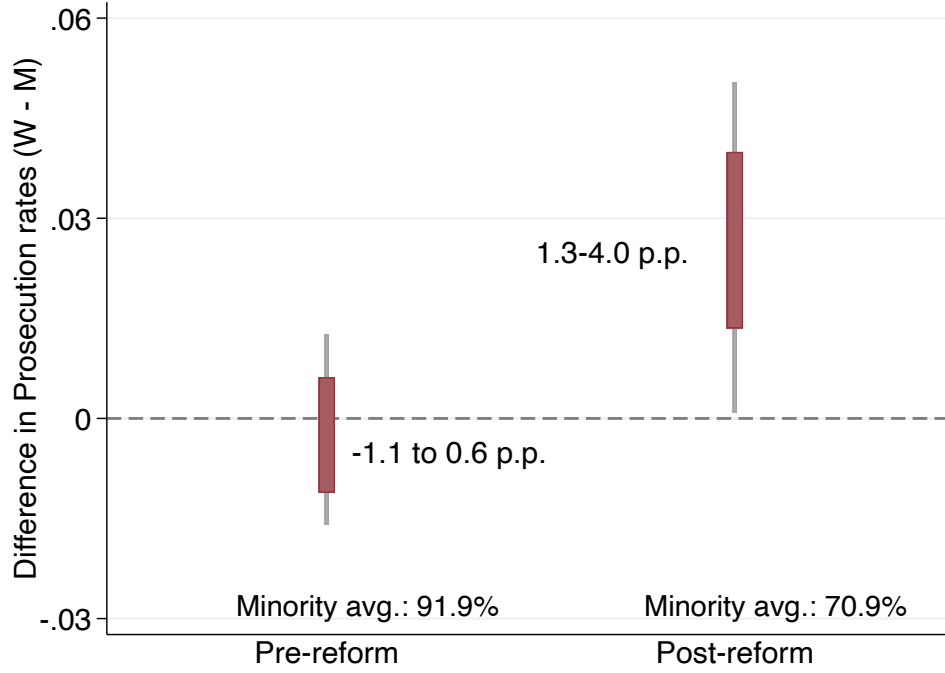


Note: This figure presents bounds on the average treated outcome obtained using the approach described in Section 2.2, separately by race and time period. The treatment is prosecution and the treated outcome, $Y_i(1)$, is whether an individual re-offends one year after disposition, if prosecuted. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap. The p-value is from a formal bootstrapped test of whether the identified sets overlap, described in Appendix B.7.

Figure 6 displays bounds on the average white–minority gap in prosecution rates conditional on re-offence outcomes if prosecuted. After accounting for racial differences in the outcomes if prosecuted, we cannot reject that white and minority defendants were prosecuted at similar rates

⁴⁰These patterns are not due to our decision to define a broad minority subsample. Figure A10 disaggregates the average re-offence outcomes if prosecuted for minority defendants separately by the largest race/ethnicity subcategories. While there is variation across these subcategories, the average re-offence rates for non-Black and non-Hispanic defendants are not large outliers.

Figure 6: Racial prosecution gap conditional on prosecuted outcome



Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on prosecuted outcomes, $Y_i(1)$, using the approach described in Section 2.2. $Y_i(1)$ is whether an individual re-offends one year after disposition, if prosecuted. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

before the budget reform. Our bounds suggest that white defendants before the reform were between 0.6 p.p. **more likely** to 1.1 p.p. **less likely** to be prosecuted compared to their minority counterparts. While these bounds suggest that prosecution in this context may not have been discriminatory before the reform, this does not speak to discrimination in other aspects of the criminal legal system or society.

This pattern changes after the reform—white defendants were 1.3–4 p.p. (1.8–5.6%) more likely than minority defendants to be prosecuted, after accounting for racial differences in the re-offence outcomes if prosecuted. Even though the budget reform reduces prosecution rates overall, the reduction is greater for minority defendants than for white defendants, even after conditioning on re-offence outcomes if prosecuted.

Alternative approaches to estimate discrimination here yield estimates outside of our estimated bounds. ‘Selection-on-observable’ estimates that control for age, gender, criminal history, and court fixed effects would estimate white–minority prosecution gaps of 1.2 p.p. before the budget reform and 4.4 p.p. after the reform. This alternative estimate of discrimination in prosecution before the reform is potentially incorrectly signed, and the alternative estimate of discrimination after the reform is biased upward by 10%–238%.

Robustness checks

These findings are robust to changing our empirical definitions of racial groups, prosecution, and re-offence outcomes. First we demonstrate that these patterns are not driven by our definition of the ‘minority’ group. [Figure A11](#) and [Figure A12](#) show that the average re-offence outcomes if prosecuted and discrimination estimates after excluding Asian American & Pacific Islander defendants from the minority group are qualitatively and quantitatively similar to our baseline. Our findings are also robust to a broader definition of prosecution that includes fine-only punishments as prosecution—[Figure A13](#) shows racial gaps that are qualitatively similar to and overlap considerably with the baseline bounds. The racial gaps in prosecution that we estimate remain stable even if we reduce or expand the time horizon used to measure re-offending (see Panel (a) of [Figure A14](#)).

Our results are also robust to weakening key identifying assumptions. The patterns in [Figure 6](#) are not driven by the weak monotonicity assumption imposed in [Figure 4](#). [Figure A15](#) relaxes that assumption and constructs the widest possible bounds by allowing never takers’ outcomes to lie between 0 and 1—i.e., allowing none or all of the never takers to re-offend if prosecuted (Manski, 1989). This exercise yields nearly identical discrimination estimates to our baseline.

Finally, [Figure A16](#) reports estimates of discrimination when we simultaneously adjust for potential re-offence outcomes and baseline covariates (age, gender criminal history, and court fixed effects). We continue to find patterns similar to our baseline estimates, implying that covariates in this context provide little additional information not already captured by potential outcomes. These results also suggest that the unwarranted disparities we estimate are not mediated by the covariates we have access to.

3.3 Understanding drivers of the racial gap in prosecution after the reform

Our results so far document that in King County: 1) there was little evidence of discrimination in prosecution before the budget reform and that 2) even though the reform reduced overall prosecution rates, white defendants were more likely to be prosecuted than minority defendants who would experience identical re-offence outcomes if prosecuted. Next, we investigate potential factors that could be driving this result.

Since the budget reform posed significant strain on resources, the relatively higher prosecution rate for white defendants (even conditional on prosecuted outcome) could stem from cases involving minority defendants needing more resources to prosecute. Discrimination in pre-prosecution stages of the criminal legal system, such as policing, could lead cases involving minority defendants to be backed up by weaker evidence (Goncalves and Mello, 2021; Owens and Ba, 2021; Jordan, 2024). Since cases with weak evidence would require greater resources to prosecute successfully, such cases may have been less likely to be pursued after the budget reform, when resources became scarce. Instead, prosecutors might have shifted resources to cases that they were more likely to win. As mentioned in [Section 3.1](#), the King County Prosecutors’ Office expressed concerns at the time about not being able to prosecute resource-intensive and time-consuming cases.

We investigate this explanation by using our approach to estimate discrimination in two sub-

samples that vary in terms of average case quality. We classify offence types into two bins based on the share of charges that ended up being successfully punished in King County, using pre-reform data from September 2004 to September 2010. The logic is that we should see a high conversion rate of charges into punishments for offences where the average case is typically backed up by high quality evidence. We refer to such offences as “high quality” cases, which include drug, driving under the influence, property, prostitution, and weapons violations. On the other hand, arrests for traffic, violent, and ‘other’ offences have a relatively low share of charges that are punished, and we refer to these types of cases as “low quality”.⁴¹ As Table 2 shows, before the budget reform, cases involving minority defendants were more likely to be lower “quality” than cases involving white defendants.

While other factors may also vary between these two categories (e.g., resource intensity), if the budget cut forced prosecutors to re-allocate resources based on case quality, and minority defendants’ cases tended to be lower quality, we should expect to see a more positive post-budget cut gap within the “low quality” subset.

Table 2: Distribution of case “quality” by defendant racial group

	Share of cases that are:	
	High quality	Low quality
White	0.24	0.76
Minority	0.20	0.80

Note: Sample includes all criminal cases disposed in the King County District Courts between October 2008 and September 2010 (i.e., in the analysis sample but before the budget reform). ‘High quality’/‘low quality’ offences are those with an above/below median share of charges that result in any punishment in this time period.

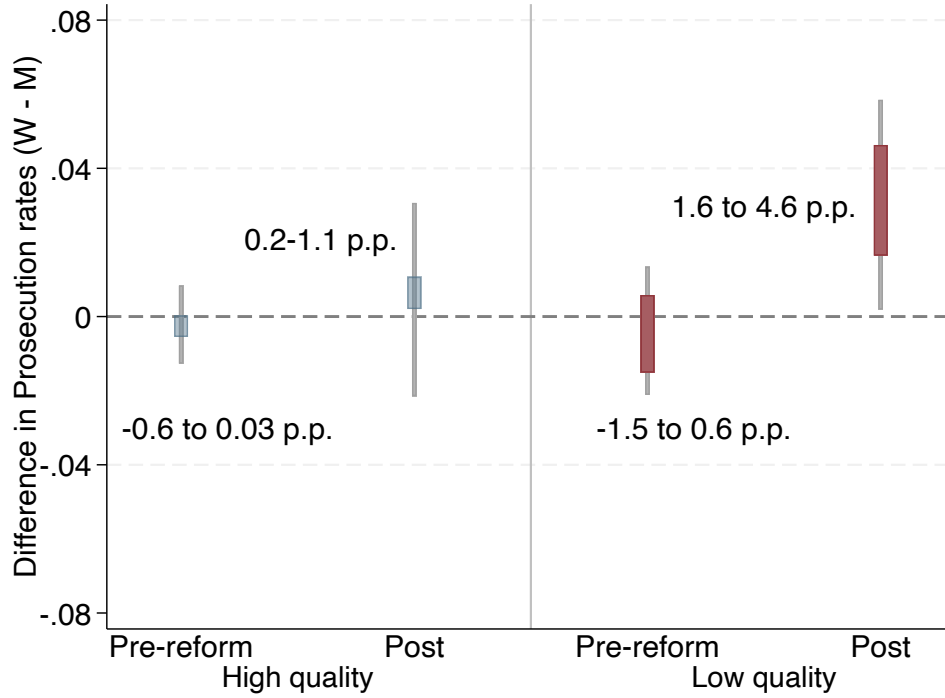
Figure 7 displays our discrimination estimates separately for “high quality” and “low quality” cases. We see that our findings of discrimination after the reform are driven by the “low quality” cases. In this subsample, white defendants after the reform were 1.6–4.6 p.p. more likely than minority defendants to be prosecuted, after accounting for racial differences in the re-offence outcomes if prosecuted. In contrast, we see a post-period gap between 0.2 to 1.1 p.p. among the “low quality” subset of cases, which is much smaller and lies outside of the post-period “high quality” bounds. However, we cannot reject that these bounds overlap due to the reduction in power from splitting our data into two subsamples.⁴²

While this exercise only provides suggestive evidence, the divergence in discrimination patterns between the “high quality” and “low quality” subsamples and the patterns in Table 2 suggest that

⁴¹One potential reason why quality differs across these offence categories is the objectivity of evidence. For example, driving under the influence cases might be backed up by blood alcohol content tests, which are relatively objective and verifiable, while cases involving violent offences may be more likely to rely on subjective witness statements.

⁴²An alternative exercise is to condition on whether the case was successfully sentenced to any punishment. However, such gaps would be biased by the exclusion of potential re-offence outcomes. In contrast, the exercise in Figure 7 conditions on re-offence outcomes as well as an imperfect measure of case quality. Nevertheless, Figure A17 conditions on the potential punishment outcome if a case was prosecuted. Racial gaps from this exercise are similar to the baseline analysis in Figure 6, and we cannot reject that the bounds between the two cases overlap.

Figure 7: Racial prosecution gap, by proxy for case quality



Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on prosecuted outcomes, $Y_i(1)$, using the approach described in Section 2.2. $Y_i(1)$ is whether an individual re-offends one year after disposition, if prosecuted. ‘High quality’/‘low quality’ offences are those with an above/below median share of charges that result in any punishment using pre-reform data. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

minority defendants’ cases might be lower quality, and that such cases are dropped due to the budget reform.⁴³ Taken seriously, this would imply that prosecutors were pursuing most cases before the reform, and passing through any pre-existing disparities from prior stages in the criminal legal system (Harrington and Shaffer, 2024). However, in shifting their focus to high quality cases, prosecutors may have offset some of the pre-existing disparities. While this behavior is consistent with prior work showing how prosecutors can use discretion to attenuate prior discrimination (Harrington and Shaffer, 2023; Jordan, 2024), it contrasts with how other agents in the criminal legal system react when fiscally-constrained—e.g., when budget deficits bind, police alter their search behavior in ways that increase rather than reduce racial disparities (Makowsky, Stratmann, and Tabarrok, 2019).

⁴³Some cases in the data are dismissed without charges ever being recorded and we consider these cases as “low quality” in our baseline classification. If we exclude such cases from the exercise, we see similar shares of “low quality” cases for white defendants (76%) and minority defendants (79%) as in Table 2 and we obtain similar but less precise conclusions as in Figure 7 (see Figure A18).

3.4 Conditioning on the outcome if dismissed, $Y_i(0)$

We now consider an alternate definition of racial discrimination in prosecution that conditions on the outcome if **dismissed**. This definition can be interpreted as an estimate of discrimination that holds fixed a notion of the baseline “risk” that an arrested individual might re-offend. As discussed in Appendix B.1 (Equation 4), estimating this quantity first requires bounding outcomes if dismissed for always takers (who are always prosecuted). Given that always takers are approximately 80% of the population in this setting (see Figure 4), we should expect the bounds on average dismissed outcomes and discrimination to be wider here.

Again, given the time-varying nature of the potential outcomes, we must purge the time trends in potential re-offence outcomes if dismissed ($Y_{it}(0)$) in King County using trends in $Y_{it}(0)$ in adjacent counties (see Appendix B.6 for details). We make a similar parallel trends assumption as we previously did, but now for the re-offence outcomes if **dismissed**. As described in A3, the assumption requires parallel trends to hold between compliers and never takers (not always takers). We use this assumption to account for time trends in the average dismissed outcomes for compliers and never takers, and construct bounds for always takers’ dismissed outcomes.⁴⁴

A3 Without the reform, re-offence outcomes if dismissed, $Y_{it}(0)$, would trend similarly for never takers & compliers and is independent of county.

Using the time trend adjustment, Figure A19 presents estimates of the average re-offence rates if all individuals were dismissed, separately by race and time period. We find suggestive evidence that minority defendants are more likely than white defendants to commit a new offence if dismissed, but we cannot reject that the bounds for the two racial groups overlap. For example, our estimates suggest that before the reform, 4.3%–15.6% of white defendants would commit a new offence if all were dismissed, while 6.4%–19.7% of minority defendants would commit a new offence if dismissed.

We use the bounds on the race-specific average outcomes if dismissed to compute racial gaps in prosecution, conditional on the outcome if dismissed. Figure A20 shows that similar to our baseline results, we cannot reject that there is no discrimination in prosecution prior to the reform, and that white defendants are 0.9–8.2 p.p. (1.3–11.6%) more likely to be prosecuted after the reform. As with our baseline results, the findings here are also robust to altering the time horizon used to measure re-offending (see Panel (b) of Figure A14) and simultaneously controlling for baseline covariates (see Panel (b) of Figure A16).

We also redo the exercise estimating racial gaps in prosecution separately for potentially “high quality” and “low quality” cases, this time conditioning on re-offence outcomes if dismissed. The patterns are qualitatively similar to what we see when conditioning on re-offence outcomes if prosecuted, but the bounds are wider and less precise (see Figure A21).

⁴⁴Similar to the previous validation exercises, we test for differential pre-trends in the outcomes of dismissed individuals across county and covariate subgroups. Figure A26 shows no evidence of differential pre-trends in re-offence outcomes if dismissed across counties but within various demographic subgroups. Figure A27 finds limited evidence of differential pre-trends in re-offence outcomes if dismissed within counties but across subgroups. Almost all estimates are statistically indistinguishable from zero, which we interpret as evidence that A3 is not grossly violated.

3.5 Summary

Our analysis presents the first evidence of racial discrimination in misdemeanor prosecution that directly accounts for meaningful unobservable differences across groups. We find that after a budget reform that limited prosecutor capacity and reduced overall prosecution rates, white defendants were more likely to be prosecuted than minority defendants with similar potential re-offence outcomes. Digging deeper, our evidence suggests that prosecutors seemed to be dismissing low quality and potentially resource-intensive cases, which were more prevalent among minority defendants. As a result, this behavior may have attenuated discrimination from prior stages of the criminal legal system that contributed to the case quality gaps in the first place.

4 Conclusion

This paper shows how to use a natural experiment that generates a binary instrumental variable (IV) to estimate bounds or point estimates of discrimination conditional on potential outcomes or treatment effects. We do this by combining the shifts in treatment rates and outcomes induced by the binary IV with behavioral assumptions on the relationship between selection into treatment and average potential outcomes from the marginal treatment effects literature. We also discuss how to apply our approach when a natural experiment generates a conditionally-random binary IV, as in the case of regression discontinuity or difference-in-difference (DiD) designs. In doing so, we add to the DiD-IV literature by showing how to use selection into being an always taker, complier, or never taker to estimate average potential outcomes with DiD variation.

Our approach does not require random assignment to decision-makers, or even that the key decision-makers are observed in the data, in contrast to existing discrimination estimation methods. Thus, our approach allows researchers to study discrimination in high-stakes settings where individuals are typically not randomly assigned to the decision-makers—for example, managers’ worker promotion decisions or teachers’ student suspension decisions.

We implement our approach to provide novel evidence on racial discrimination in misdemeanor prosecution, the most common form of contact with the criminal courts. We use a DiD-IV strategy generated by an unexpected budget cut that affected the King County Prosecutors’ Office but did not affect adjacent counties. Adjusting for racial differences in potential re-offence outcomes, we find no evidence of discrimination in prosecution before the budget cut. After the cut, white defendants were more likely to be prosecuted than minority defendants with identical potential re-offence outcomes, even though overall prosecution rates fell. We find suggestive evidence of prosecutors responding to fiscal constraints by focusing on easy cases and offsetting disparities from prior stages of the criminal legal system.

While our analysis conditions on a single potential outcome at a time, future methodological steps might incorporate insights from work estimating average population outcomes while accounting for two dimensions of unobservable heterogeneity (Dutz et al., 2021). Such an approach could quantify discrimination between individuals similar on more than one dimension, e.g., discrimina-

tion in prosecution conditional on potential re-offence and employment outcomes. Fruitful next steps for the empirical analysis of discrimination in prosecution might further assess the mechanisms underlying how discrimination is affected by changes in resource constraints. In particular, data that can track individual events within each court case would allow us to better understand which prosecutorial actions amplify versus offset any pre-existing disparities.

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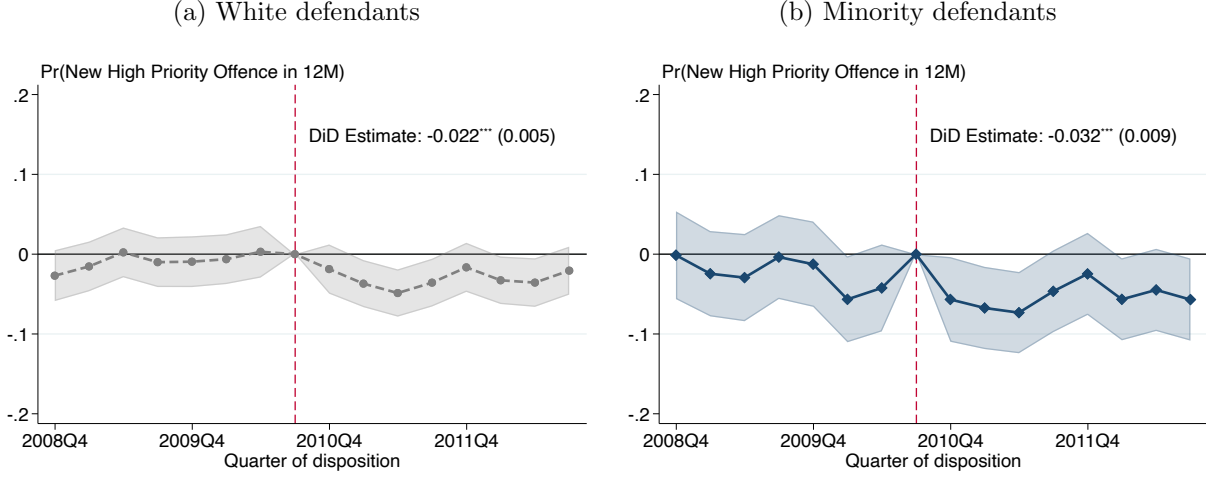
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Appendix A Additional results: Misdemeanor prosecution

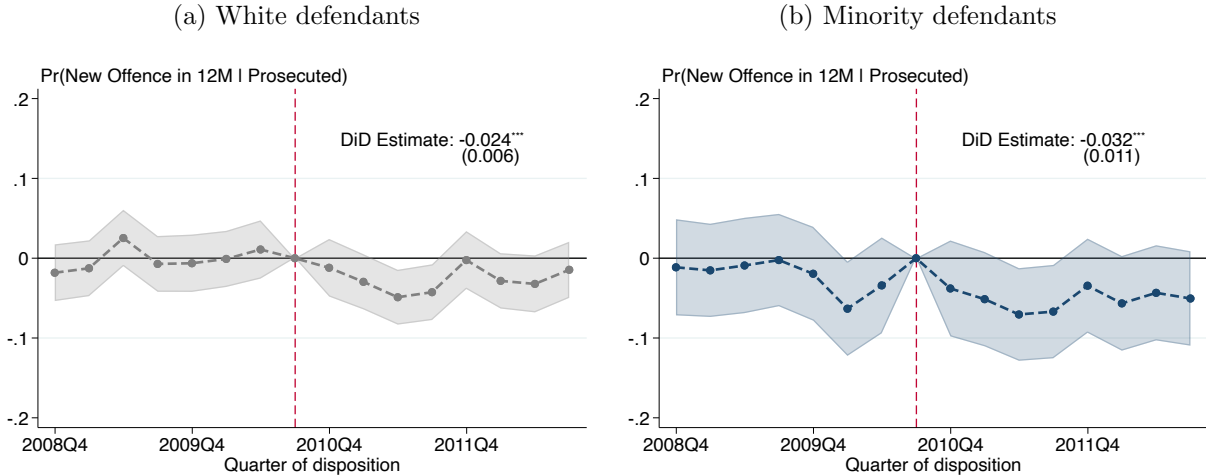
A.1 Robustness checks

Figure A1: Impact of King County budget reform on ‘high-priority’ re-offence within one year



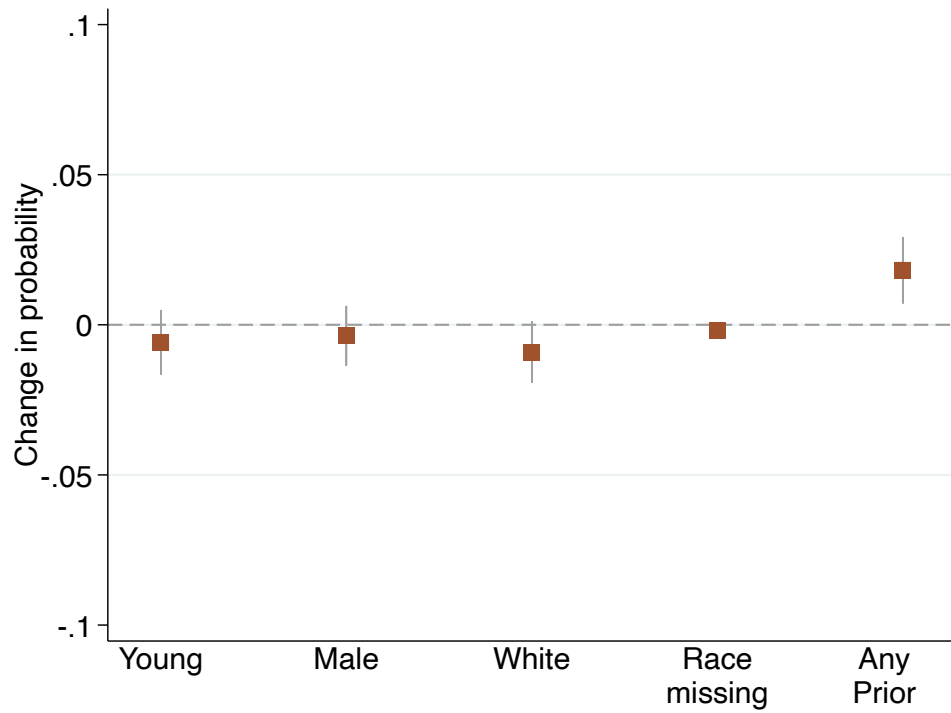
Note: Each Panel presents event study estimates investigating the impact of the King County budget reform. The re-offence outcome includes new ‘high-priority’ offences anywhere in Washington State. ‘High-priority’ offences are those that are not associated with charges that were commonly dismissed in the 2 quarters after the budget reform. Sample includes all misdemeanor defendants, as described in Table 1. ‘DiD Estimate’ pools the coefficients on relative time indicators and estimates $Y_{igt} = \alpha + \delta_1 \mathbb{I}[\text{King County}] + \delta_2 Post_i + \beta^{DD} \mathbb{I}[\text{King County}] \times Post_i + \epsilon_{igt}$, where $Post_i = 1$ if the case is disposed of on or after September 28, 2010, when the budget reform was announced. 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

Figure A2: Impact of King County budget reform on re-offence within one year, only prosecuted defendants



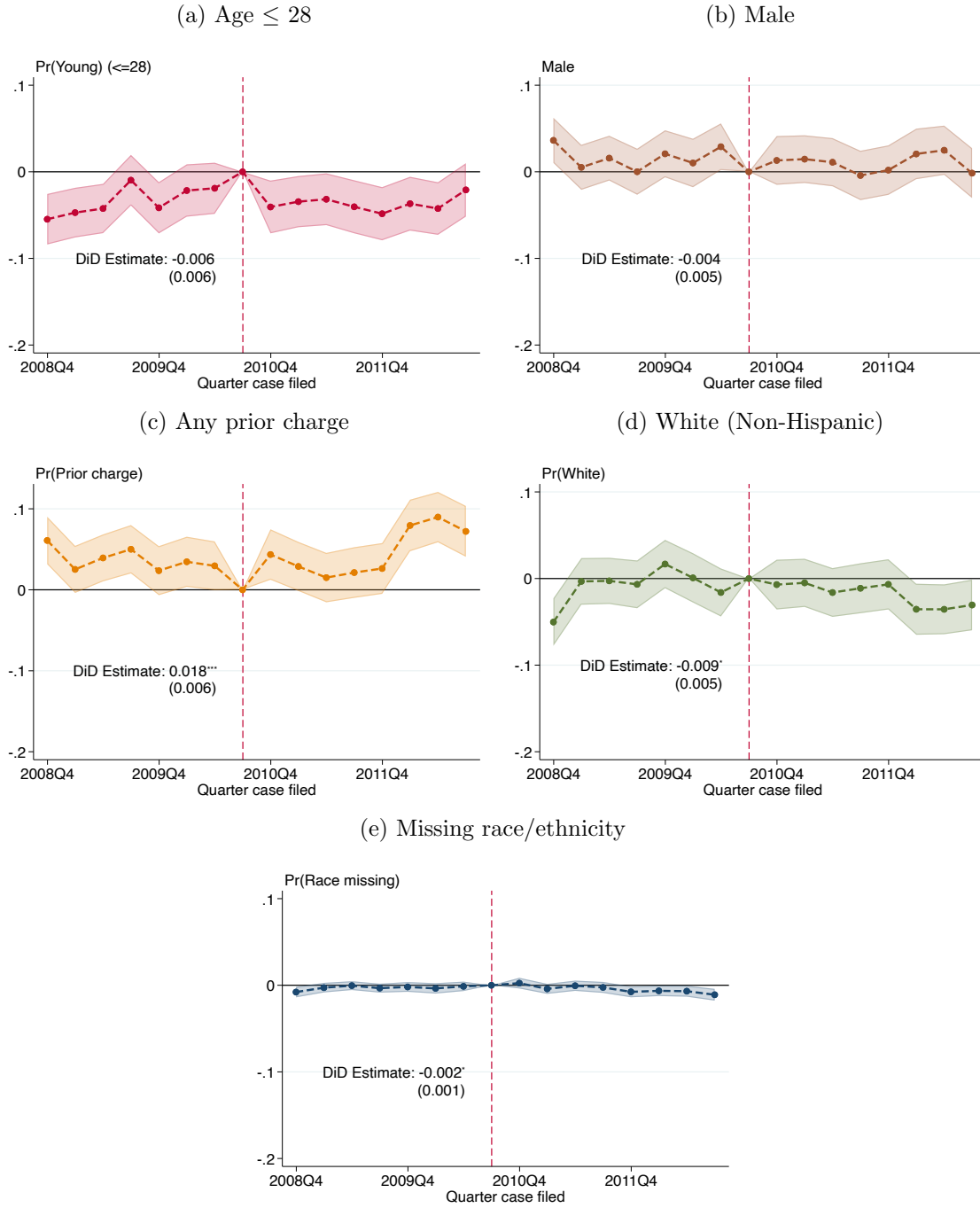
Note: Each Panel is identical to Figure 3, except the sample only includes prosecuted defendants.

Figure A3: Testing for changes in observable characteristics of cases filed



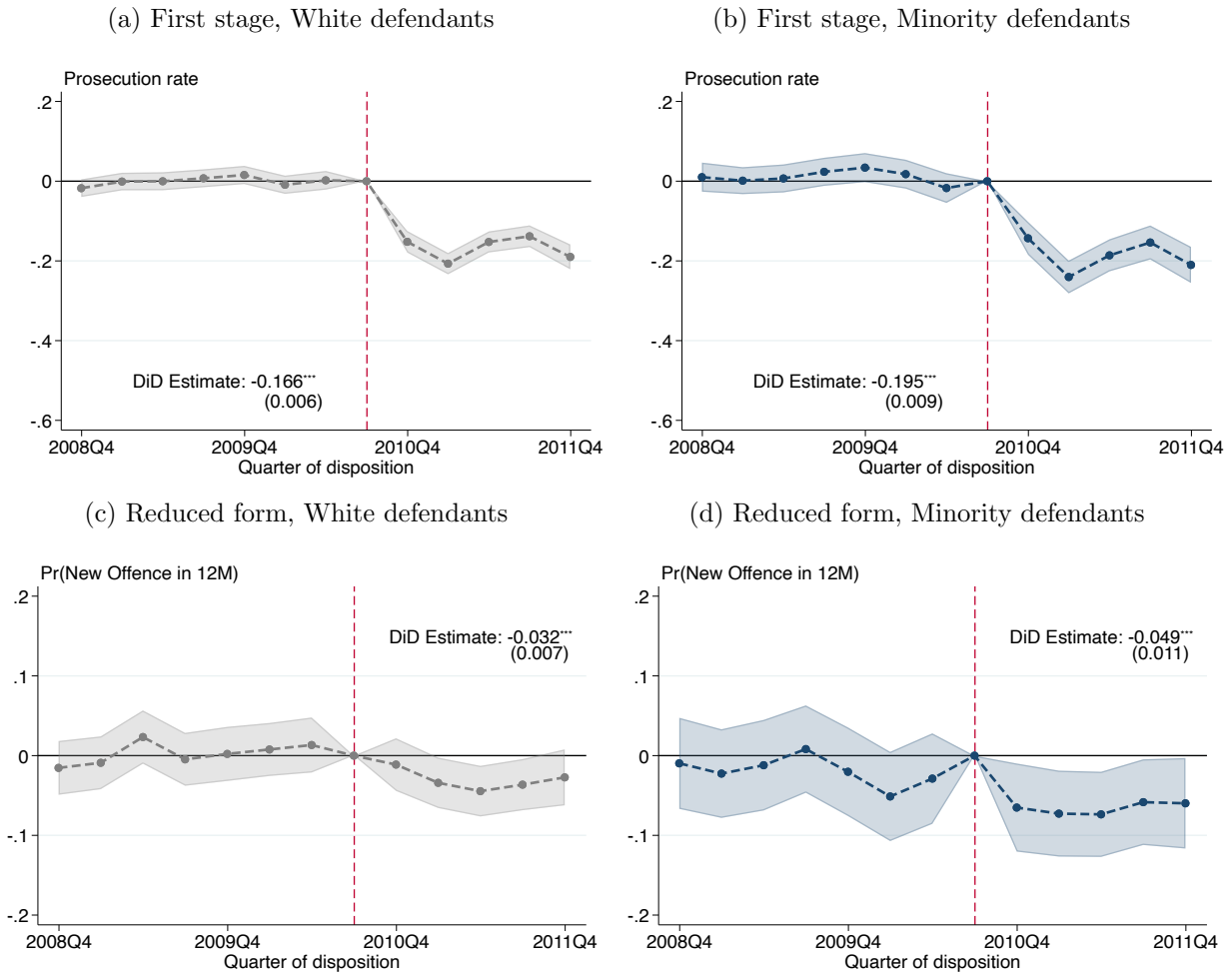
Note: Each square is β^{DD} from $X_{igt} = \alpha + \delta_1 \mathbb{I}[\text{King County}] + \delta_2 Post_i + \beta^{DD} \mathbb{I}[\text{King County}] \times Post_i + \epsilon_{igt}$, where $Post_i = 1$ if the case is filed on or after September 28, 2010, when the budget reform was announced. X_{igt} is the relevant baseline characteristic. ‘Young’ defendants are those who ≤ 28 years old at disposition and ‘Any Prior’ is an indicator for whether an individual has ever been previously charged with an offence in Washington State. 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

Figure A4: Impact of budget reform on caseload characteristics



Note: Each Panel presents event study estimates investigating the impact of the King County budget reform. ‘DiD estimate’ is β^{DD} from $X_{igt} = \alpha + \delta_1 \mathbb{I}[\text{King County}] + \delta_2 Post_t + \beta^{DD} \mathbb{I}[\text{King County}] \times Post_t + \epsilon_{igt}$, where $Post_t = 1$ if the case is filed on or after September 28, 2010, when the budget reform was announced. X_{igt} denotes a baseline characteristic. ‘Young’ defendants are those who ≤ 28 years old at disposition and ‘Any Prior’ is an indicator for whether an individual has been previously charged with an offence in Washington. 95% confidence intervals constructed with heteroscedasticity-robust standard errors.

Figure A5: Robustness of first stage and reduced form to excluding post-SPD investigation period



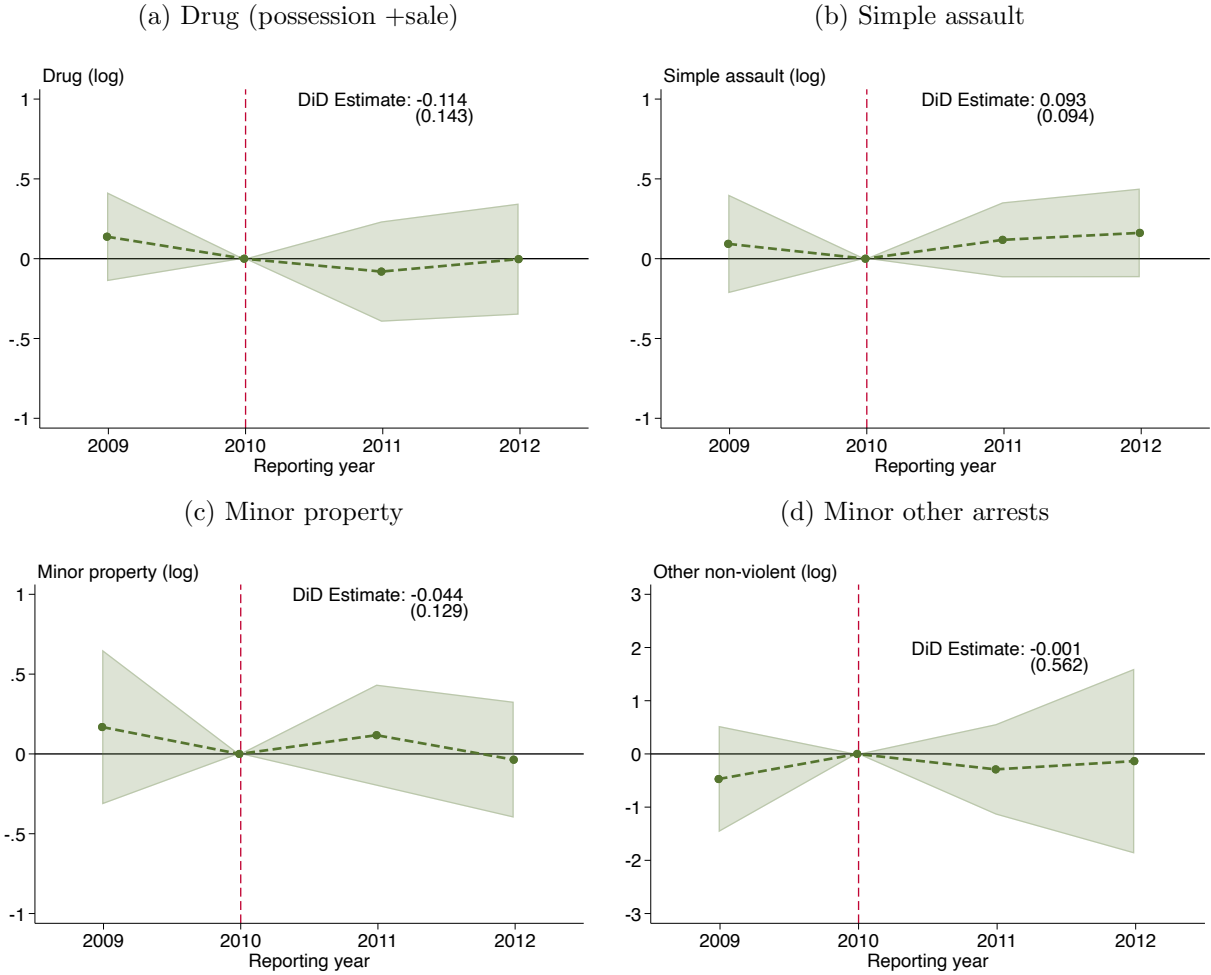
Note: This reproduces Figure 2 (Panels (a) and (b)) and Figure 3 (Panels (c) and (d)), excluding cases disposed after December 11, 2011.

Figure A6: Impact of budget reform on police employment



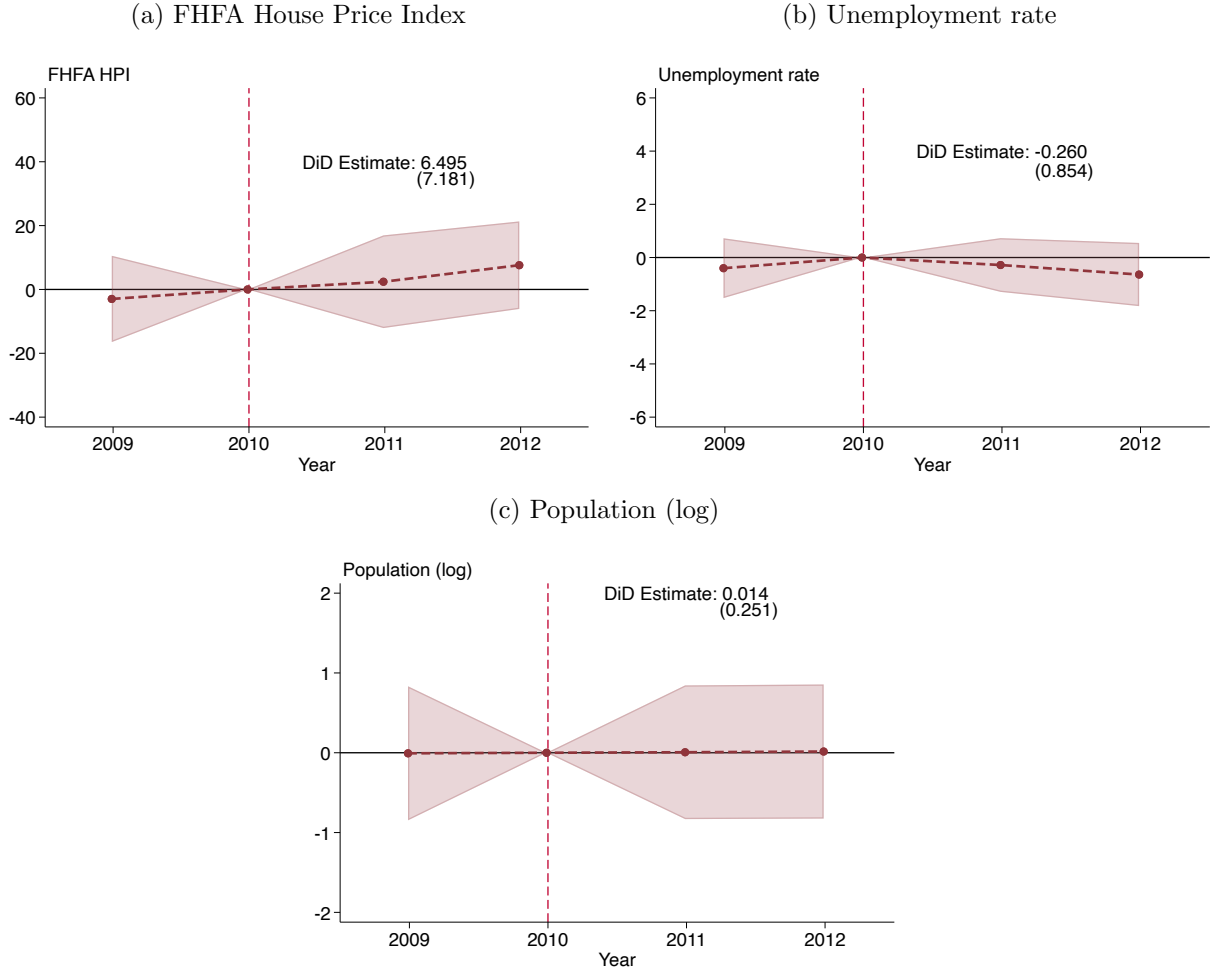
Note: Estimates generated using annual average employment from the Law Enforcement Officers Killed and Assaulted (LEOKA) data (Kaplan, 2023). This specification is similar to those used to estimate the first stage and reduced form except for the inclusion of originating agency (ORI) fixed effects and usage of the average pre-reform county-level population as weights. ORIs in areas with an annual population average less than 1,000 or with zero employment counts throughout the sample are excluded. 95% confidence intervals are constructed using standard errors clustered at the ORI level.

Figure A7: Impact of budget reform on arrests



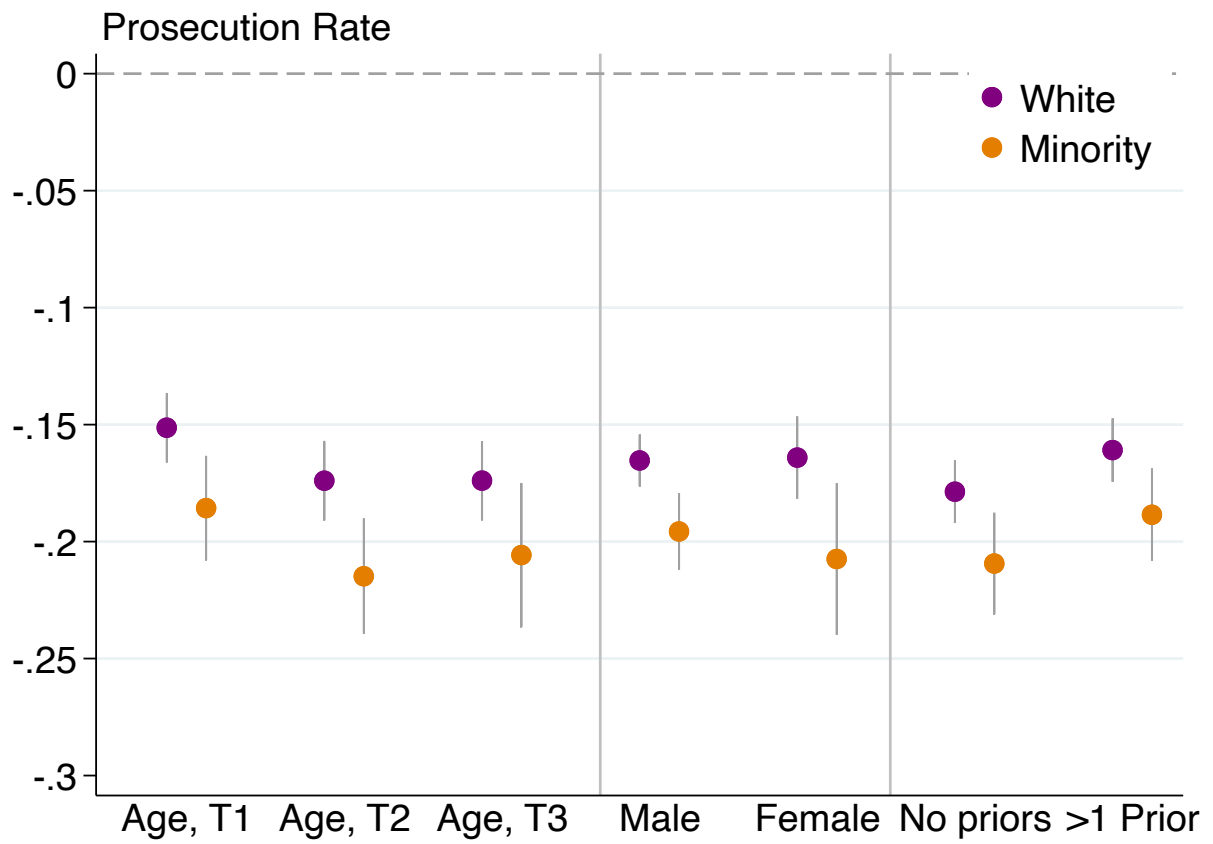
Note: Estimates generated using annual number of arrests from Uniform Crime Report (UCR) data (Kaplan, 2023). ‘Minor property’ arrests include arrests for stolen property, fraud, forgery, and theft. This specification is similar to those used to estimate the first stage and reduced form except for the inclusion of originating agency (ORI) fixed effects and usage of the average pre-reform county-level population as weights. ORIs in areas with an annual population average less than 1,000, with limited time coverage between 2008-13 or with zero arrest counts in a given year are excluded. 95% confidence intervals are constructed using standard errors clustered at the ORI level.

Figure A8: Impact of budget reform on non-crime economic factors



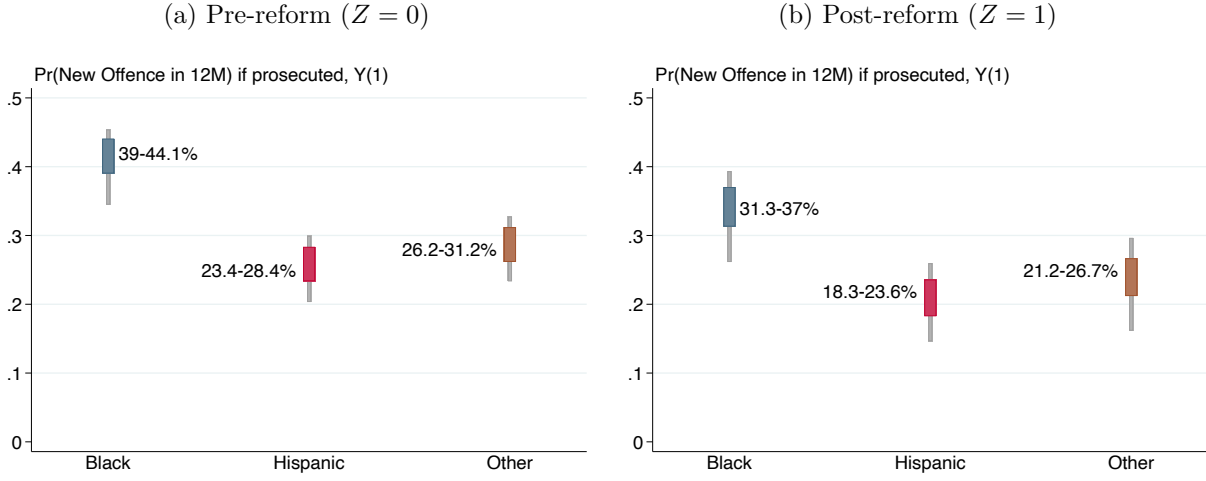
Note: The specification used here is similar to those used to estimate the first stage and reduced form except for the usage of the average pre-reform county-level population as weights in Panels (a) and (b). County-level data on house prices are from the Federal Housing Finance Agency (FHFA), unemployment rates from Local Area Unemployment Statistics and population counts are from the Census Bureau. 95% confidence intervals are constructed using heteroskedasticity-robust standard errors.

Figure A9: Testing for defiers: First stage by subgroup



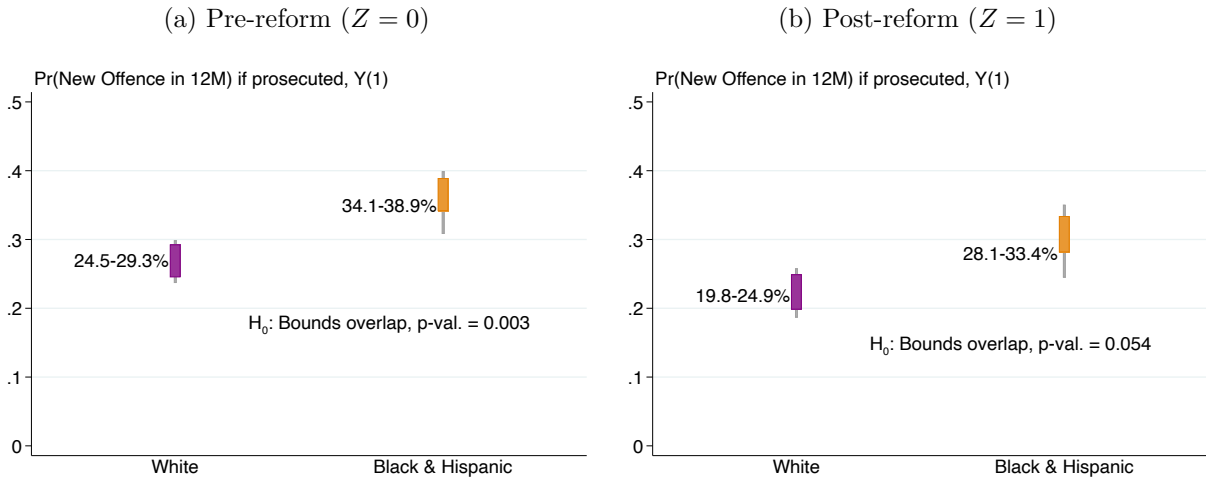
Note: Each Panel presents DiD estimates as reported in [Figure 2](#), but for different covariate subgroups. Age at disposition is split into terciles, represented by T1–T3.

Figure A10: Average re-offence outcomes if prosecuted, disaggregated by minority subgroup



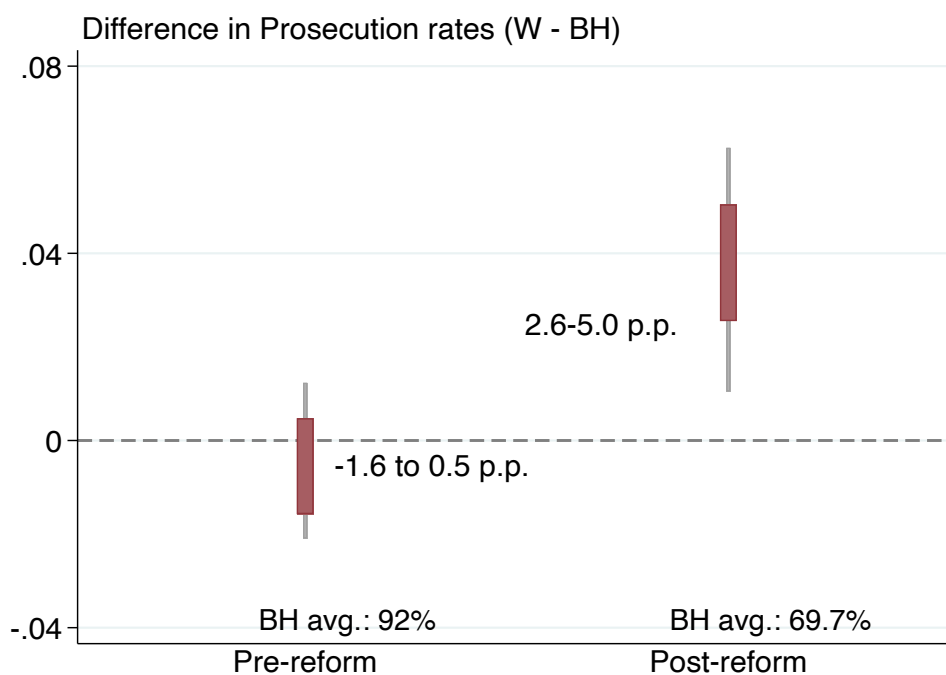
Note: This figure presents bounds on the average treated outcome obtained using the approach described in Section 2.2, separately by time period and subgroups within minority defendants. The treatment is prosecution and the treated outcome, $Y_i(1)$, is whether an individual re-offends one year after disposition, if prosecuted. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure A11: Average re-offence outcomes if prosecuted: White vs. Black/Hispanic defendants



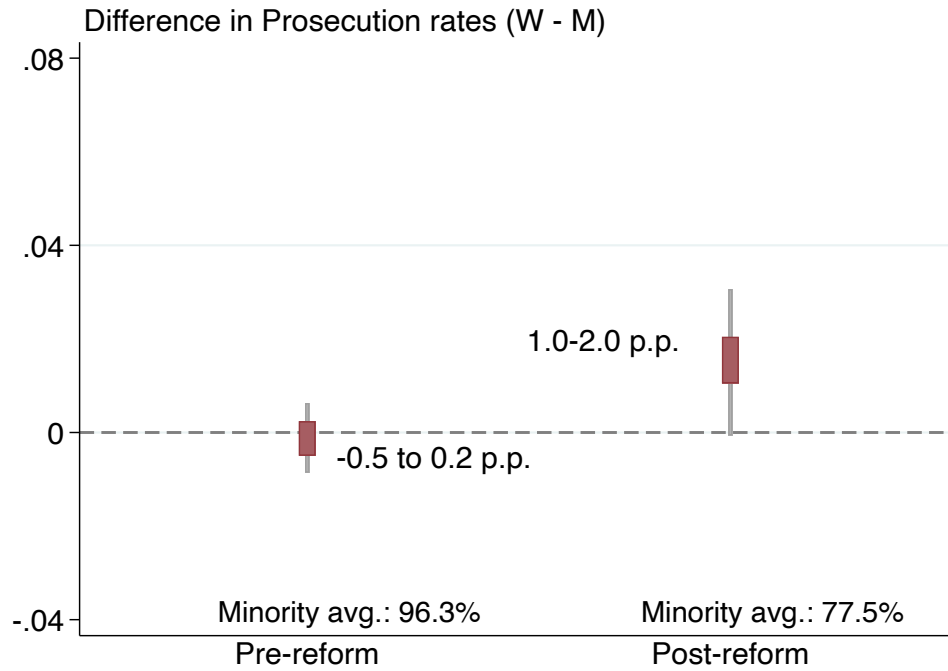
Note: This figure presents bounds on the average treated outcome obtained using the approach described in Section 2.2, separately by race and time period. The treatment is prosecution and the treated outcome, $Y_i(1)$, is whether an individual re-offends one year after disposition, if prosecuted. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap. The p-value is from a formal bootstrapped test of whether the identified sets overlap, described in Appendix B.7.

Figure A12: Racial prosecution gap conditional on prosecuted outcome: White vs. Black/Hispanic



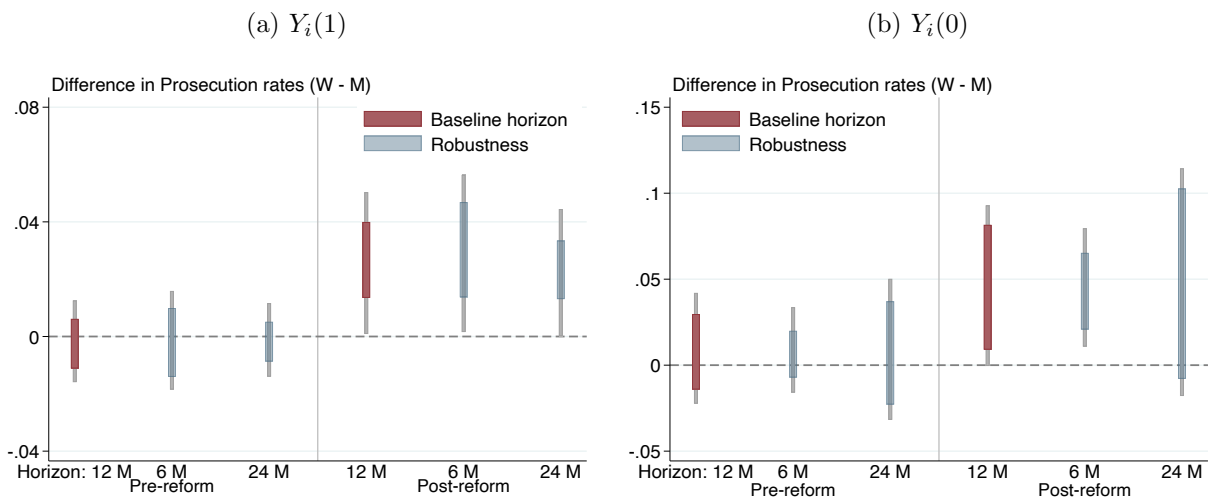
Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on prosecuted outcomes, $Y_i(1)$, using the approach described in Section 2.2. $Y_i(1)$ is whether an individual re-offends one year after disposition, if prosecuted. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure A13: Racial prosecution gap cond. on prosecuted outcome: Broad prosecution definition



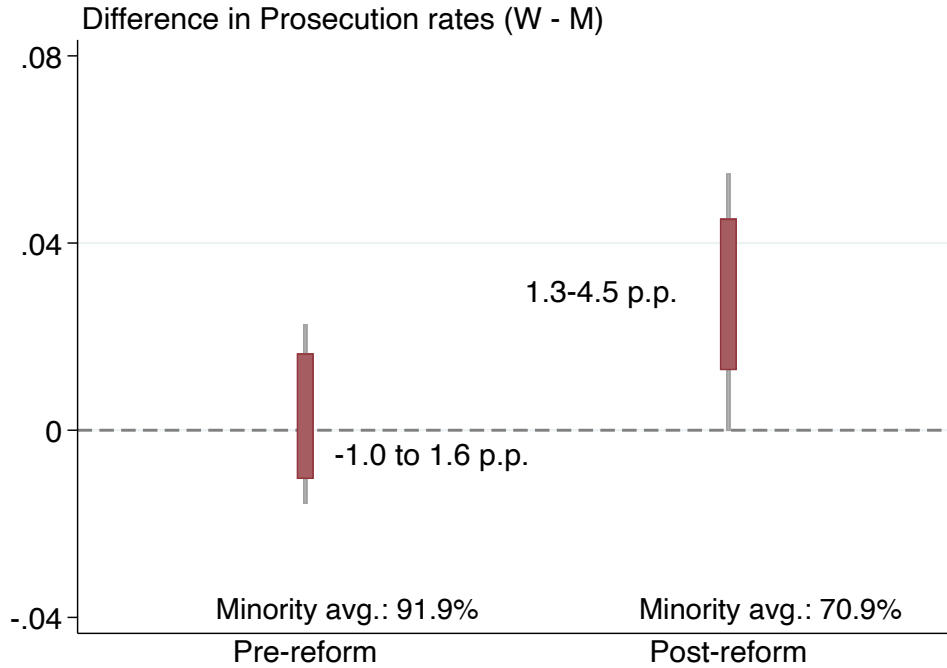
Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on prosecuted outcomes, $Y_i(1)$, using the approach described in Section 2.2. $Y_i(1)$ is whether an individual re-offends one year after disposition, if prosecuted. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure A14: Racial prosecution gap conditional on potential outcomes: Variation by outcome horizon



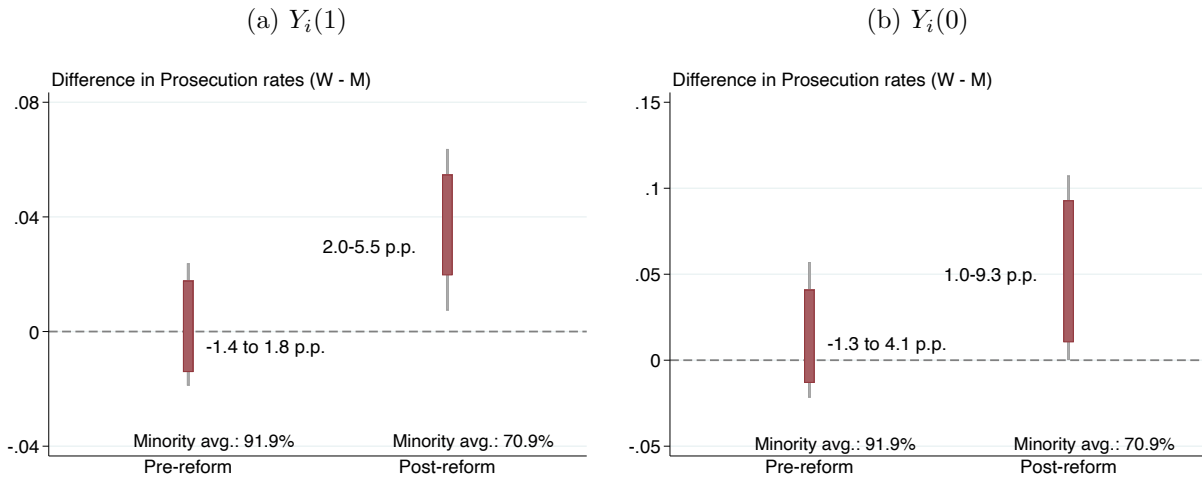
Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on potential outcomes, $Y_i(\cdot)$. $Y_i(\cdot)$ is whether an individual re-offends within the amount of time labelled on the x-axis after disposition, if prosecuted (Panel (a)) or if dismissed (Panel (b)). Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure A15: Racial prosecution gap conditional on prosecuted outcome: Worst case bounds



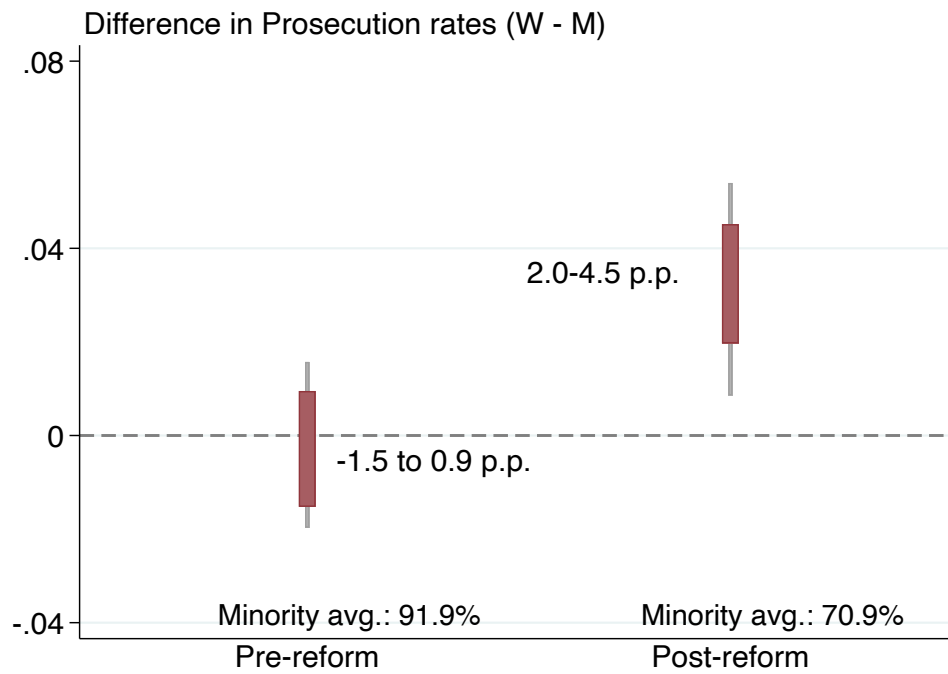
Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on prosecuted outcomes, $Y_i(1)$, assuming that never takers' outcomes are bounded between 0 and 1. $Y_i(1)$ is whether an individual re-offends one year after disposition, if prosecuted. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure A16: Racial prosecution gap, conditional on potential outcomes and baseline covariates



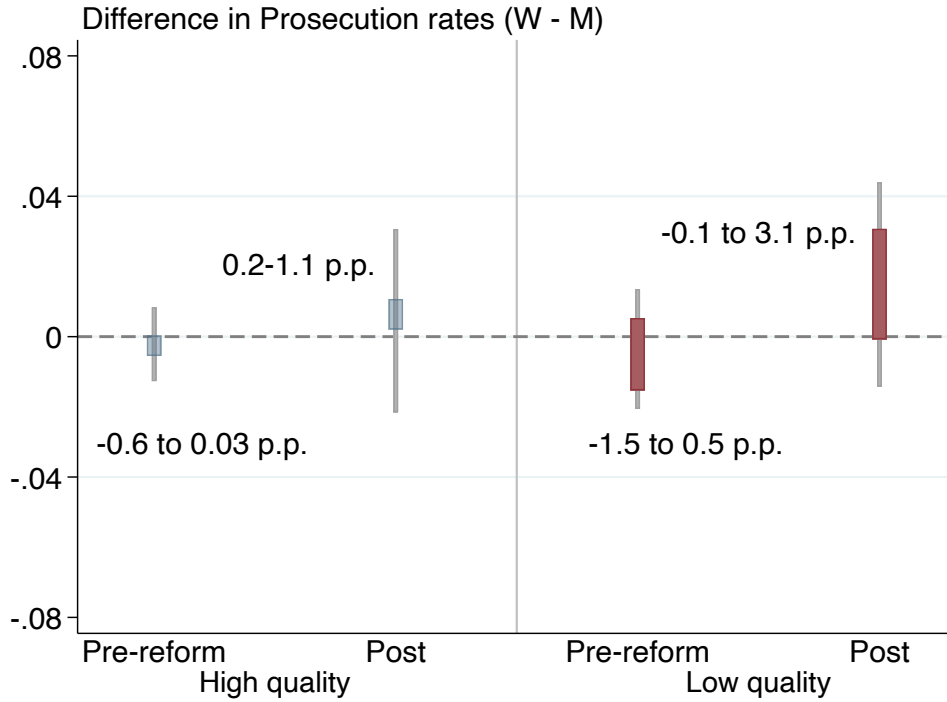
Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on potential outcomes, $Y_i(\cdot)$, and baseline covariates. $Y_i(\cdot)$ is whether an individual re-offends one year after disposition if prosecuted (Panel (a)) or if dismissed (Panel (b)). Baseline covariates include age, gender, criminal history (convictions and charges), and court fixed effects. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure A17: Racial prosecution gap conditional on receiving any punishment if prosecuted



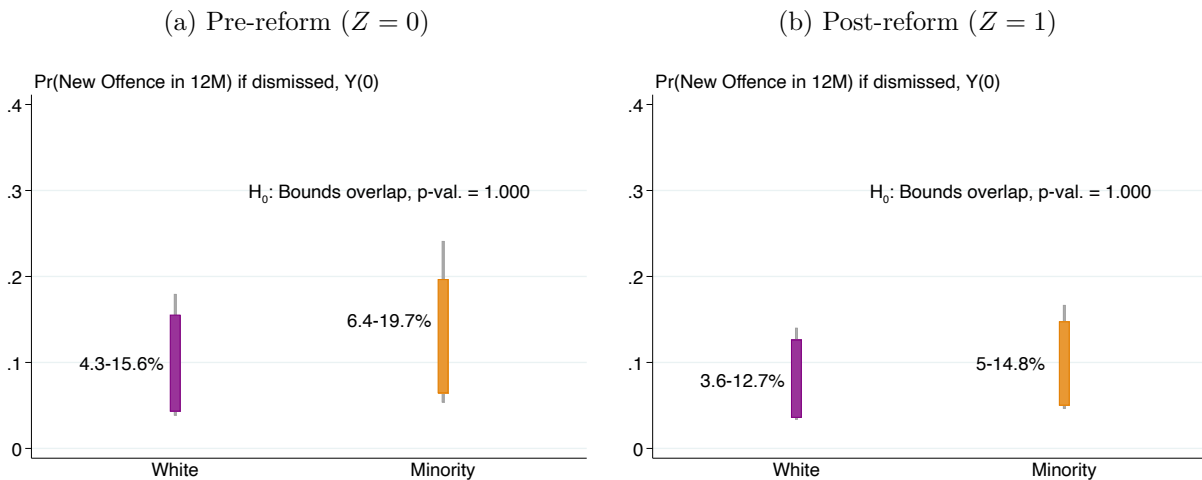
Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on case outcome if prosecuted, $Y_i(1)$, using the approach described in Section 2.2. $Y_i(1)$ is whether an individual is sentenced to any non-fine punishment, if prosecuted. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure A18: Racial prosecution gap, by proxy for case quality: excluding cases with missing charges



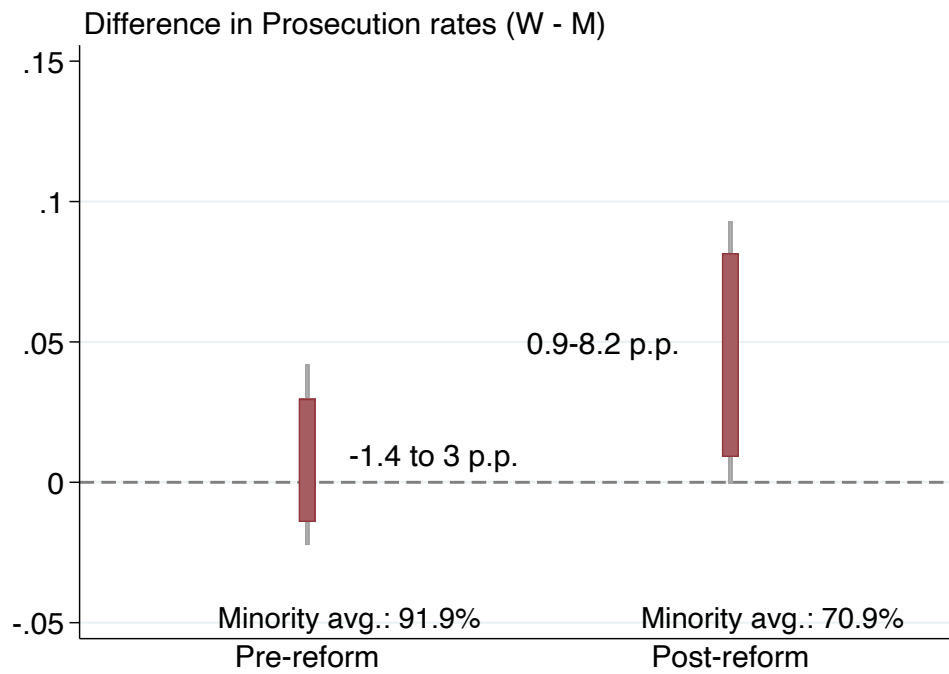
Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on prosecuted outcomes, $Y_i(1)$, using the approach described in Section 2.2. $Y_i(1)$ is whether an individual re-offends one year after disposition, if prosecuted. ‘High quality’/‘Low quality’ offences are those with an above/below median share of charges that result in any punishment using pre-reform data, excluding cases where no charges are listed. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure A19: Average re-offence outcomes if dismissed, $E[Y_{it}(0)|R_i = r]$



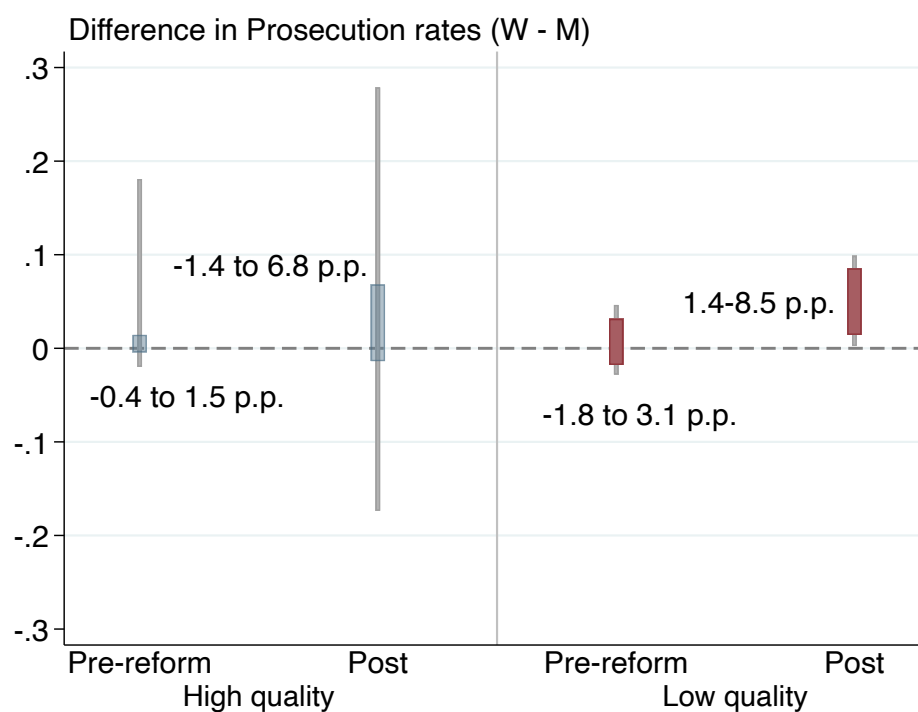
Note: This figure presents bounds on the average untreated outcome obtained using the approach described in Section 2.2, separately by race and time period. The treatment is prosecution and the untreated outcome, $Y_i(0)$, is whether an individual re-offends one year after disposition, if dismissed. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure A20: Racial prosecution gap conditional on dismissed outcome



Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on dismissed outcomes, $Y_i(0)$, using the approach described in Section 2.2. $Y_i(0)$ is whether an individual re-offends one year after disposition, if dismissed. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

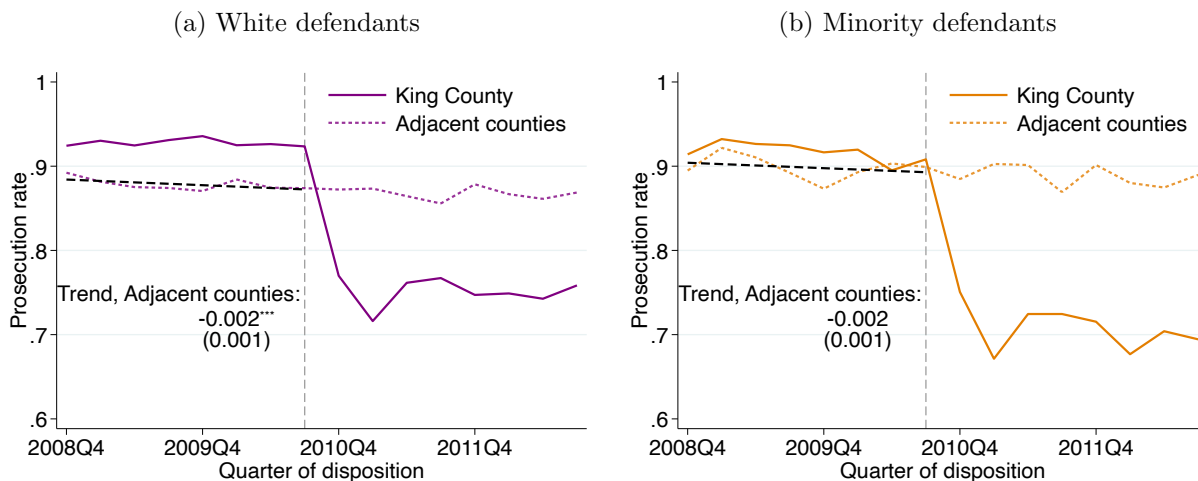
Figure A21: Racial prosecution gap, by proxy for case quality: conditional on dismissed outcome



Note: This figure presents bounds on the average difference in prosecution rates in each time period, conditional on dismissed outcomes, $Y_i(0)$, using the approach described in Section 2.2. $Y_i(0)$ is whether an individual re-offends one year after disposition, if dismissed. ‘High quality’/‘Low quality’ offences are those with an above/below median share of charges that result in any punishment using pre-reform data. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

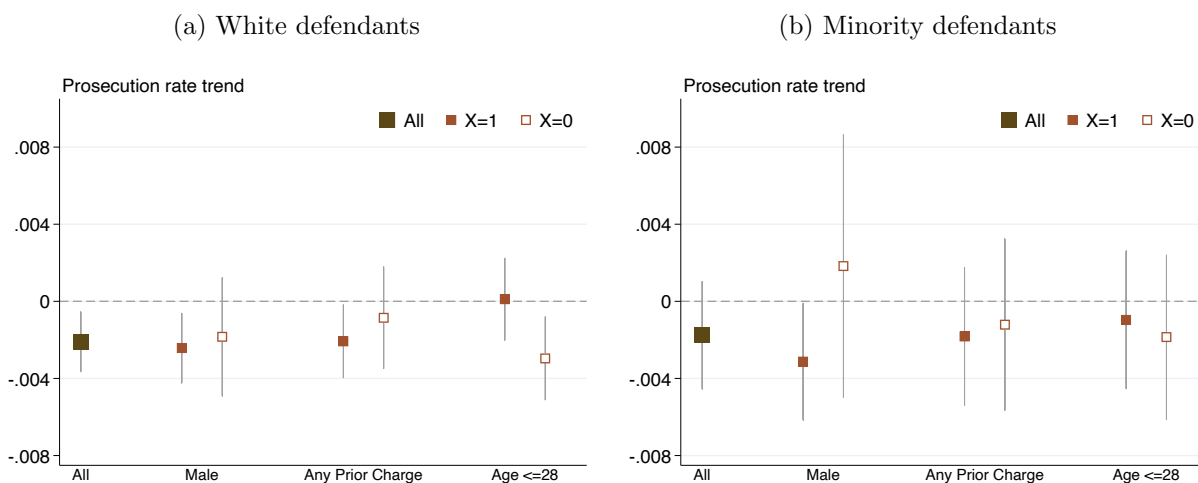
A.2 Empirically validating DiD adjustment assumptions

Figure A22: Testing trends in prosecution rates in adjacent counties



Note: The displayed coefficients are from estimating a linear regression of prosecution on a linear quarterly trend using pre-period data in the counties adjacent to King County. Standard errors on pre-period trend estimates are heteroscedasticity-robust.

Figure A23: Prosecution trends in adjacent counties, by covariate subgroup



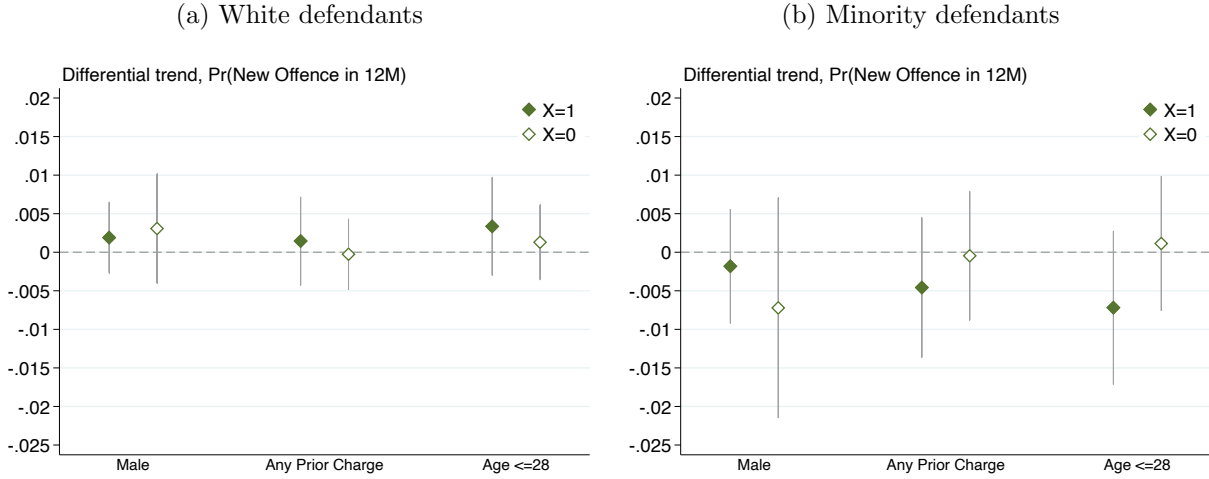
Note: Each coefficient is from estimating a linear regression of prosecution on a linear quarterly trend using pre-period data on individuals from a given subgroup (denoted by X) in the counties adjacent to King County. The estimate labelled 'All' reproduces the overall trend estimate from [Figure A22](#). 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

Figures [A24–A27](#) estimate the following regression; [Figure A24](#) and [Figure A26](#) test hypotheses 1) and 2), while [Figure A25](#) and [Figure A27](#) test hypotheses 3) and 4):

$$Y_{itg} = \beta_1 t + \beta_2 X_i + \beta_3 \text{King County} + \delta_1 X_i \times t + \delta_2 X_i \times \text{King County} + \delta_3 t \times \text{King County} + \delta_4 X_i \times t \times \text{King County} + \varepsilon_{igt}$$

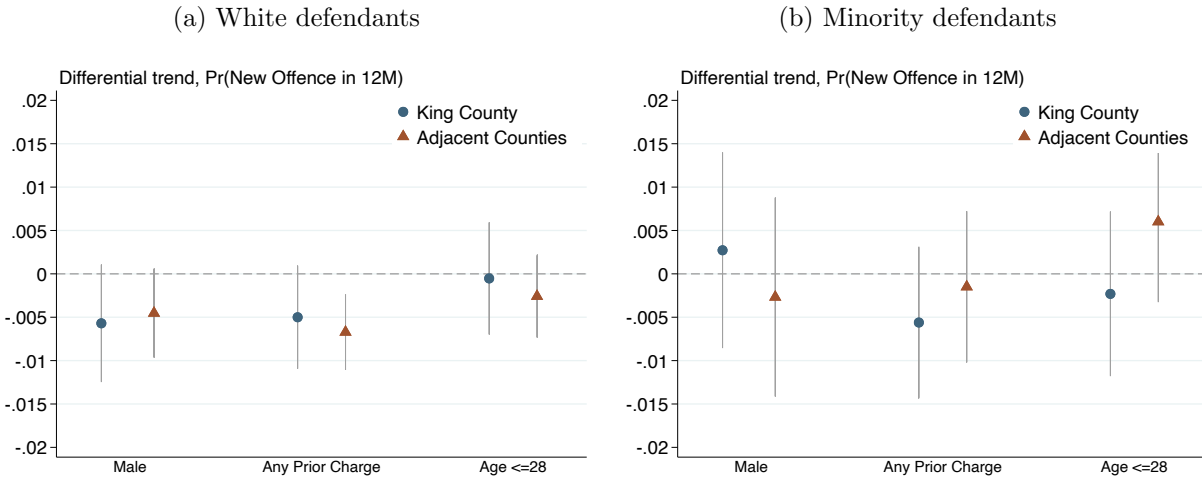
- 1) $X = 0 : H_0 : \delta_3 = 0$
- 2) $X = 1 : H_0 : \delta_3 + \delta_4 = 0$
- 3) King County $H_0 : \delta_1 + \delta_4 = 0$
- 4) Adjacent $H_0 : \delta_1 = 0$

Figure A24: Testing for differential trends in group-specific treated outcomes across counties



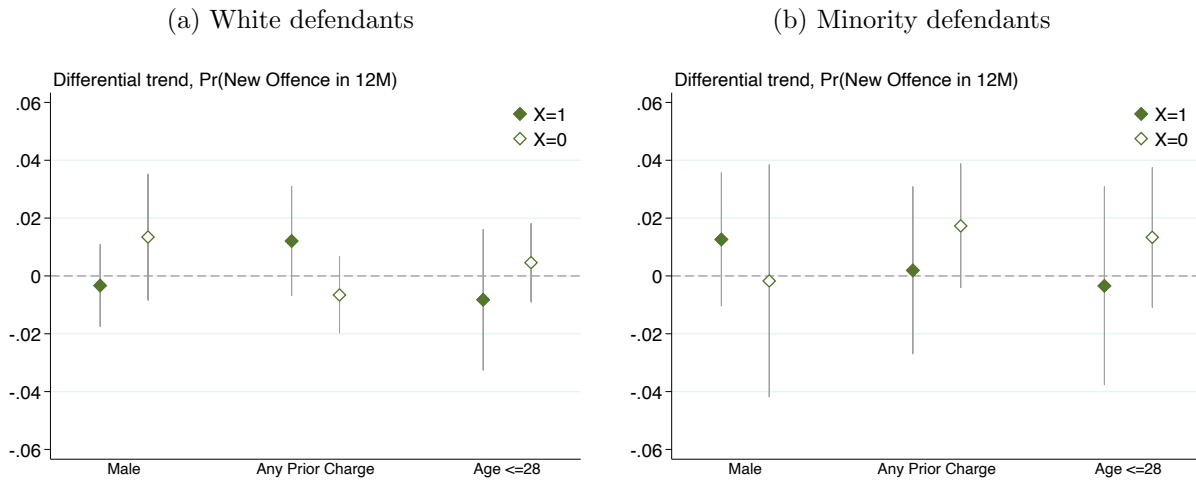
Note: Each coefficient is an estimate of the difference in pre-period trends in treated outcomes (outcomes if prosecuted) across counties, for a given covariate value. The sample only includes individuals who are prosecuted prior to the budget reform. For example, the first solid green diamond in Panel a) represents the difference in trends in treated outcomes for white male defendants between King County and other counties. 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

Figure A25: Testing for differential trends in treated outcomes between groups, within each county



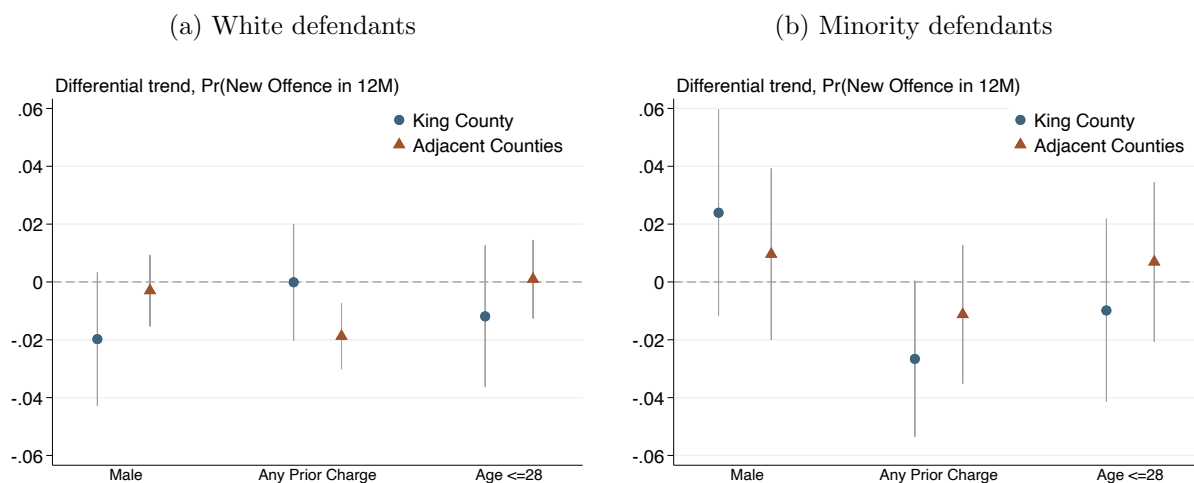
Note: Each coefficient is an estimate of the difference in pre-period trends in treated outcomes (outcomes if prosecuted) across covariate groups, within a given county. The sample only includes individuals who are prosecuted prior to the budget reform. For example, the first blue circle in Panel a) represents the difference in trends in treated outcomes between white male and female defendants in King County. 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

Figure A26: Testing for differential trends in group-specific untreated outcomes across counties



Note: Each coefficient is an estimate of the difference in pre-period trends in untreated outcomes (outcomes if not prosecuted) across counties, for a given covariate value. The sample only includes individuals who are dismissed prior to the budget reform. For example, the first green diamond in Panel a) represents the difference in trends in untreated outcomes for white male defendants between King County and other counties. 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

Figure A27: Testing for differential trends in untreated outcomes between groups, within each county



Note: Each coefficient is an estimate of the difference in pre-period trends in untreated outcomes (outcomes if not prosecuted) across covariate groups, within a given County. The sample only includes individuals who are dismissed prior to the budget reform. For example, the first blue circle in Panel a) represents the difference in trends in untreated outcomes between white male and female defendants in King County. 95% confidence intervals are constructed using heteroscedasticity-robust standard errors.

Appendix B Methodological Details

B.1 Discrimination conditional on other functions of potential outcomes

We have discussed estimating discrimination conditional on **treated** potential outcomes. However, it might be more appropriate in certain contexts to measure discrimination as differential treatment among individuals with the same **untreated** potential outcomes, or even the same **treatment effect**. Each of these may also map to different normative notions of fairness. For example, say we wanted to study discrimination in the decision to nominate students for advanced educational programs. Conditioning on the treated/untreated potential outcomes and the treatment effect in this example would provide an understanding of group differences in educational program nominations between individuals who would: i) do equally well in the program, ii) do equally well without the program, and iii) have equal gains from the program. We next discuss estimating these other discrimination estimands, and the additional assumptions that may be required, again assuming that potential outcomes are binary for simplicity.

Conditioning on untreated potential outcomes, $Y_i(0)$

Differential treatment among individuals with the same untreated potential outcome is a function of period-, race-, and potential outcome-specific treatment rates, as shown in Equation 4. This diverges from the treatment rates conditional on treated potential outcomes from Equation 2 in two ways. 1) The denominator is now the average outcome that would be realized if **no one was treated**. 2) The first term in the numerator, the average **untreated** outcome among those who were **treated**, is no longer directly observed in the data, since always takers are always treated.

$$E[D_i|Z = z, R_i = r, Y_i(0) = 1] = \frac{E[Y_i(0) = 1|Z = z, R_i = r, D_i = 1] \times E[D_i|Z = z, R_i = r]}{E[Y_i(0) = 1|R_i = r]} \quad (4)$$

Assumptions on the relationship between treatment propensity and average **untreated** potential outcomes, analogous to those described for treated outcomes, deal with both instances of divergence. These assumptions bound or point-identify the average outcomes if no one were treated, $E[Y_i(0) = y|R_i = r]$. Extrapolating $E[Y_i(0) = y|R_i = r]$ involves extrapolating the average untreated outcomes of always takers (since they are always treated) using estimates of the untreated outcomes of compliers and never takers. Note that always takers' untreated outcomes are a component of $E[Y_i(0) = y|Z = z, R_i = r, D_i = 1]$. Plugging in bounds/point estimates from each of these steps into Equation 4 yields bounds/point estimates for $E[D_i|Z = z, R_i = r, Y_i(0) = y]$. These treatment rates would then be aggregated up, in a way analogous to Equation 2, to estimate group differences in treatment among those who would have identical outcomes if not treated.

Alternatively, since treatment is binary here, the **prosecution rate** conditional on outcome if dismissed is equivalent to subtracting the **dismissal rate** for those who would have that outcome if dismissed from 1. To see this, consider Equation 5, which shows how the share of individuals **not**

treated, with a given untreated outcome (e.g., the dismissal rate for people with a specific outcome if dismissed) can be used to quantify the share of **treated** individuals with a specific untreated outcome, abstracting away from the period-specific notation Z . The first line subtracts the share of individuals not treated with a given untreated outcome from 1, where $\bar{D} = E[D_i|R_i = r]$, and where $E[D_i = 0|R_i = r, Y_i(0) = y]$ is rewritten in the same way as described in Equation 1. The third line uses the fact that the average untreated outcome observed if everyone were untreated is a weighted average of the untreated outcomes for the treated individuals and the untreated individuals: $E[Y_i(0) = y|R_i = r] = E[Y_i(0) = y|R_i = r, D_i = 1]\bar{D} + E[Y_i(0) = y|R_i = r, D_i = 0](1 - \bar{D})$. This recovers the share of treated individuals with a specific untreated outcome, described in Equation 4.

$$\begin{aligned}
1 - E[D_i = 0|R_i = r, Y_i(0) = y] &= 1 - \frac{E[Y_i(0) = y|R_i = r, D_i = 0] \times (1 - \bar{D})}{E[Y_i(0) = y|R_i = r]} \\
&= \frac{E[Y_i(0) = y|R_i = r] - E[Y_i(0) = y|R_i = r, D_i = 0] \times (1 - \bar{D})}{E[Y_i(0) = y|R_i = r]} \\
&= \frac{E[Y_i(0) = y|R_i = r, D_i = 1] \times \bar{D}}{E[Y_i(0) = y|R_i = r]} \\
&= E[D_i|R_i = r, Y_i(0) = y]
\end{aligned} \tag{5}$$

Given this mapping, if one was interested in estimating discrimination conditional on untreated outcomes, an attractive natural experiment is one that generates a small share of always takers. While this discussion has focused on a simple binary IV, Appendix B.6 discusses the assumptions required to estimate discrimination conditional on untreated potential outcomes with DiD variation.

Conditioning on treatment effects, $Y_i(1) - Y_i(0)$

Equation 6 describes treatment rates that condition directly on the treatment effect (τ_i). This object departs from Equation 2 in two ways. 1) The denominator is the share of individuals in the population who would realize a given treatment effect value. 2) The first term in the numerator is the share of treated individuals who would realize a given treatment effect value. We cannot estimate these quantities in the same way that we have done when conditioning on potential outcomes because τ_i is never observed for any individual and can take multiple values. For example, even if both $Y_i(1)$ and $Y_i(0)$ are binary, $\tau_i \in \{-1, 0, 1\}$. Thus even though the IV assumptions identify the local average treatment effect for compliers, we will not know the prevalence of a given value of τ_i .

$$\begin{aligned}
E[D_i|Z = z, R_i = r, \tau_i = y] &= \frac{E[\tau_i = y|Z = z, R_i = r, D_i = 1] \times E[D_i|Z = z, R_i = r]}{E[\tau_i = y|R_i = r]} \\
\tau_i &\equiv Y_i(1) - Y_i(0)
\end{aligned} \tag{6}$$

Conditioning on the treatment effect requires restricting the support of the treatment effect. If we assume τ_i is binary, say due to a theoretical or empirical justification, we can overcome the two challenges above. For example, we may be in a context where the treatment (prosecution) is unlikely to reduce future re-offending, i.e., there are no deterrence effects. We then might assume $\tau_i \in \{0, 1\}$, which implies that the average treatment effect (ATE), $E[\tau_i | R_i = r]$, coincides with the denominator of Equation 6. This solves the first challenge. Similarly, $\tau_i \in \{0, 1\}$ implies that $E[\tau_i = y | Z = z, R_i = r, D_i = 1]$ is the average treatment effect on the treated (ATT). Recall that we directly observe average treated outcomes for the treated in the data and that the discussion regarding conditioning on untreated outcomes provided an estimate of the average untreated outcomes for the treated. The difference between these two averages is the ATT. Under the restriction that $\tau_i \in \{0, 1\}$, this identifies the proportion of treated individuals with a treatment effect of $\tau_i = 1$: $E[\tau_i = 1 | Z = z, R_i = r, D_i = 1]$.

Plugging in bounds/point estimates from each of these steps into Equation 6 yields bounds/point estimates for $E[D_i | Z = z, R_i = r, Y_i(0) = y]$. These treatment effect-specific treatment rates would then be aggregated up in a way analogous to Equation 2, to estimate group differences in treatment among those who would have the same treatment effect, assuming binary treatment effects.

Finally, note that bounding the treatment effect-specific treatment rates requires bounding a non-linear function of point identified and partially identified objects. To see how we might compute the bounds in such a case, consider the following. The equation below reproduces Equation 6, where $\tau_i \equiv Y_i(1) - Y_i(0)$. The right hand side is a function of two partially identified objects (first term in numerator and denominator) and a point estimate (second term in numerator). The two partially identified objects are dependent—the denominator is a function of the first term in the numerator.

$$E[D_i | Z = z, R_i = r, \tau_i = y] = \frac{E[\tau_i = y | Z = z, R_i = r, D_i = 1] \times E[D_i | Z = z, R_i = r]}{E[\tau_i = y | R_i = r]}$$

We rewrite the treatment rate conditional on the treatment effect, omitting group conditioning for brevity and considering the case where the policy reform has taken effect, i.e. $Z = 1$. Here, we assume that $\tau_i \in \{0, 1\}$, because this assumption is required to identify this type of treatment rate, as described above.

$$\begin{aligned} E[D_i | Z = 1, \tau_i = 1] &= \frac{E[\tau_i | Z = z, D_i = 1] \times E[D_i | Z = z]}{E[\tau_i]} \\ &= \frac{(p_A E[\tau_i | A] + p_C E[\tau_i | C]) / (p_A + p_C)}{p_A E[\tau_i | A] + p_C E[\tau_i | C] + p_N E[\tau_i | N]} \times E[D_i | Z = 1] \end{aligned} \tag{7}$$

The average treatment effect on the treated is a function of the bounds on the treatment effect for always takers and the point estimated treatment effect for compliers. The denominator ($E[\tau_i]$) is a function of two partially identified objects, $E[\tau_i | A]$ and $E[\tau_i | N]$, since we never observe untreated outcomes for always takers or treated outcomes for never takers.

This implies that the treatment rate conditional on the treatment effect is of the form $y = f(p, q) = \frac{p \times r}{p + q}$, where r is a point estimate and the other quantities are bounds. We compute bounds for y by holding one partially identified object fixed at a time, evaluating the function at the extremes of the other partially identified object. That is, we take the minimum and maximum values of y over the following cases: $f(\underline{p}, \underline{q})$, $f(\underline{p}, \bar{q})$, $f(\bar{p}, \underline{q})$, $f(\bar{p}, \bar{q})$.

B.2 Comparing change in Δ_z to observed change in disparity

Denote the change in discrimination due to a reform by $\Delta_{\text{change}} = \Delta_1 - \Delta_0$, following the definition of Δ_z in Definition 2. Consider a case with binary potential outcomes. The observational analogue to quantifying how an intervention affected group differences in treatment might be to compare the change in observed treatment rates for each group by estimating the regressions below and constructing the gap defined in Definition 3.

$$\begin{aligned} D_i &= \alpha_w + \beta_w Z_i + \varepsilon_{iw}, \text{ if } R_i = w \\ D_i &= \alpha_m + \beta_m Z_i + \varepsilon_{ib}, \text{ if } R_i = m \end{aligned}$$

Definition 3. Observed change in group treatment gaps due to policy reform (Δ_{obs})

$$\Delta_{\text{obs}} \equiv \beta_w - \beta_m, \text{ where } \beta_r = E[D_i|Z = 1, R_i = r] - E[D_i|Z = 0, R_i = r] \quad (8)$$

We are interested in whether the observed change in treatment responses coincides with the change in discrimination among individuals with identical treated potential outcomes. First, note that each expectation in Definition 3 is a weighted average of treatment rates for individuals with each potential outcome, where the weights are the prevalence of each binary value of the treated outcome. Equation 9 shows this, where $\pi_{zry} \equiv E[Y_i(1) = y|Z = z, R_i = r]$. The third and fourth lines follow from the fact that $Y_i(1) \perp Z$.

$$\begin{aligned} E[D_i|Z = z, R_i = r] &= E[Y_i(1) = 1|Z = z, R_i = r]E[D_i|Z = z, R_i = r, Y_i(1) = 1] \\ &\quad + E[Y_i(1) = 0|Z = z, R_i = r]E[D_i|Z = z, R_i = r, Y_i(1) = 0] \\ &= E[Y_i(1) = 1|R_i = r]E[D_i|Z = z, R_i = r, Y_i(1) = 1] \\ &\quad + E[Y_i(1) = 0|R_i = r]E[D_i|Z = z, R_i = r, Y_i(1) = 0] \\ \implies E[D_i|Z = z, R_i = r] &= Pr(Y_i(1) = 1|R_i = r)\pi_{zr1} + (1 - Pr(Y_i(1) = 1|R_i = r))\pi_{zr0} \end{aligned} \quad (9)$$

Plugging this into Definition 3, we can rewrite the observed change (Δ_{obs}) in terms of these group- and outcome-specific treatment averages, where $\mu_r \equiv Pr(Y_i(1) = 1|R_i = r)$:

$$\begin{aligned} \Delta_{\text{obs}} &= [(\mu_w \pi_{1w1} + (1 - \mu_w) \pi_{1w0}) - (\mu_w \pi_{0w1} + (1 - \mu_w) \pi_{0w0})] \\ &\quad - [(\mu_m \pi_{1m1} + (1 - \mu_m) \pi_{1m0}) - (\mu_m \pi_{0m1} + (1 - \mu_m) \pi_{0m0})] \end{aligned} \quad (10)$$

Similarly, Equation 11 rewrites the change in discrimination as a function of group- and outcome-specific treatment averages:

$$\begin{aligned}
\Delta_{\text{change}} &= Pr(Y_i(1) = 1) \underbrace{[\Delta_{11} - \Delta_{01}]}_{\text{Change, } Y_i(1) = 1} + (1 - Pr(Y_i(1) = 1)) \underbrace{[\Delta_{10} - \Delta_{00}]}_{\text{Change, } Y_i(1) = 0} \\
&= \underbrace{[Pr(Y_i(1) = 1)(\pi_{1w1} - \pi_{1m1}) + (1 - Pr(Y_i(1) = 1))(\pi_{1w0} - \pi_{1m0})]}_{\text{Gap when } Z=1} \\
&\quad - \underbrace{[Pr(Y_i(1) = 1)(\pi_{0w1} - \pi_{0m1}) + (1 - Pr(Y_i(1) = 1))(\pi_{0w0} - \pi_{0m0})]}_{\text{Gap when } Z=0}
\end{aligned} \tag{11}$$

Comparing Equation 10 and Equation 11, there are only 2 cases in which $\Delta_{\text{change}} = \Delta_{\text{obs}}$:

Case 1. $Y_i(1)$ is similar across groups: $Cov(Y_i(1), R_i) = 0 \implies Pr(Y_i(1) = 1 | R_i = r) = Pr(Y_i(1) = 1) \forall r \in R_i$

Case 2. Group-specific policy-induced responses are constant across $Y_i(1)$: $\pi_{1r1} - \pi_{0r1} = \pi_{1r0} - \pi_{0r0} = \theta_r$

The following shows how $\Delta_{\text{change}} = \Delta_{\text{obs}}$ under the described conditions. We will consider each case in turn.

Case 1

Consider that groups are similar in terms of $Y_i(1)$. Then, substituting $Pr(Y_i(1) = 1)$ for each $Pr(Y_i(1) = 1 | R_i = r)$ in Equation 10, we have:

$$\begin{aligned}
\Delta_{\text{obs}} &= [(Pr(Y_i(1) = 1)\pi_{1w1} + (1 - Pr(Y_i(1) = 1))\pi_{1w0}) - (Pr(Y_i(1) = 1)\pi_{0w1} + (1 - Pr(Y_i(1) = 1))\pi_{0w0})] \\
&\quad - [(Pr(Y_i(1) = 1)\pi_{1m1} + (1 - Pr(Y_i(1) = 1))\pi_{1m0}) - (Pr(Y_i(1) = 1)\pi_{0m1} + (1 - Pr(Y_i(1) = 1))\pi_{0m0})] \\
&= Pr(Y_i(1) = 1)[(\pi_{1w1} - \pi_{0w1}) - (\pi_{1m1} - \pi_{0m1})] + (1 - Pr(Y_i(1) = 1))[(\pi_{1w0} - \pi_{0w0}) - (\pi_{1m0} - \pi_{0m0})] \\
&= \Delta_{\text{change}}
\end{aligned}$$

Case 2

Here, groups differ in terms of $Y_i(1)$ but the policy-induced treatment response for each group is the same, regardless of $Y_i(1)$: $\pi_{1r1} - \pi_{0r1} = \pi_{1r0} - \pi_{0r0} = \theta_r$. Reorganizing Equation 10 to group the policy-induced treatment response terms by race and potential outcome level:

$$\begin{aligned}
\Delta_{\text{obs}} &= [Pr(Y_i(1) = 1|R_i = w) (\pi_{1w1} - \pi_{0w1}) + (1 - Pr(Y_i(1) = 1|R_i = w)) (\pi_{1w0} - \pi_{0w0})] \\
&\quad - [Pr(Y_i(1) = 1|R_i = m) (\pi_{1m1} - \pi_{0m1}) + (1 - Pr(Y_i(1) = 1|R_i = m)) (\pi_{1m0} - \pi_{0m0})] \\
&= \underbrace{\theta_w - \theta_m}_{\text{Gap in potential outcome-invariant responses}}
\end{aligned}$$

Substituting θ_r for $\pi_{1ry} - \pi_{0ry}$ in Equation 11, we see that $\Delta_{\text{change}} = \Delta_{\text{obs}}$.

$$\begin{aligned}
\Delta &= Pr(Y_i(1) = 1)[\theta_w - \theta_m] + (1 - Pr(Y_i(1) = 1))[\theta_w - \theta_m] \\
&= \theta_w - \theta_m = \Delta_{\text{obs}}
\end{aligned}$$

B.3 Point identifying discrimination

This section discusses how to implement our discrimination estimation approach to obtain point estimates instead of bounds. We first sketch a simplified selection model commonly used to understand treatment non-compliance and treatment effect heterogeneity (Vytlacil, 2002). Individuals are treated if the benefit of treatment, $p_i(Y_i(1), Y_i(0), Z)$, outweighs the cost, where Z is a binary instrument that shifts the benefit of treatment. $p_i(Y_i(1), Y_i(0), Z)$ can also be interpreted as an individual's treatment propensity, and can be normalized such that $p_i(Y_i(1), Y_i(0), Z) \in [0, 1]$. Let $u_i \in U[0, 1]$ be a unidimensional measure summarizing possibly multiple factors that determine an individual's cost of treatment. Given the IV assumptions listed in Section 2.1, $Y_i(D_i)$ and u_i are unaffected by Z . Consolidating this notation, individual i is treated if $D_i = \mathbb{I}[p_i(Y_i(1), Y_i(0), Z) \geq u_i]$.

Returning to the discussion of always takers (A), compliers (C), and never takers (N) in Section 2.2, always takers have lower cost of treatment than compliers, who in turn have lower cost of treatment than never takers, $u_A \leq u_C \leq u_N$ (Angrist, Imbens, and Rubin, 1996).⁴⁵ Assuming that the relationship between treatment propensity and treated/untreated potential outcomes is linear, Equation 12 describes the expression for each marginal treatment response function, where $\bar{Y}_{(\cdot)}$ represents the average treated/untreated outcome for always takers, compliers, or never takers (Brinch, Mogstad, and Wiswall, 2017; Mogstad, Santos, and Torgovitsky, 2018; Kowalski, 2023b). The remaining steps to estimate discrimination follow Section 2.2, except that we estimate average treated or untreated potential outcomes by integrating the marginal treated or untreated functions ($MTO(p)$ or $MUO(p)$) over the full support of the treatment propensity. As a result, we obtain point estimates for the average treated/untreated outcomes and discrimination estimands.

⁴⁵Vytlacil (2002) demonstrates how latent index selection models coincide with the local average treatment effect framework (Imbens and Angrist, 1994).

$$\begin{aligned}
MTO(p) &\equiv E[Y_i(1)|p = u_i] = \bar{Y}_{T,A} - \frac{p_A}{p_C} (\bar{Y}_{T,AC} - \bar{Y}_{T,A}) + \frac{2}{p_C} (\bar{Y}_{T,AC} - \bar{Y}_{T,A}) \times p \\
MUO(p) &\equiv E[Y_i(0)|p = u_i] = \frac{(2 - p_N)\bar{Y}_{U,NC} - (1 + p_A)\bar{Y}_{U,N}}{p_C} + \frac{2}{p_C} (\bar{Y}_{U,N} - \bar{Y}_{U,NC}) \times p \\
MTE(p) &\equiv E[Y_i(1) - Y_i(0)|p = u_i] = MTO(p) - MUO(p)
\end{aligned} \tag{12}$$

Empirical exercise: Racial discrimination in incarceration

We briefly demonstrate point identifying discrimination using the context of racial discrimination in incarceration decisions and publicly-available case-level records from Bexar County Criminal District (felony) Courts. We use a large reform meant to reduce overcrowding in Texas jails (SB 1067 in 1994). The goal of the reform was to reduce the burden on correctional facilities by limiting the incarceration rates for low-level offenders. The reform created a new category of felony: the state jail felony (SB 1067 Article 1, Subchapter C, §12.35) which reduced the punishment associated with a wide range of common offences, including many property and drug crimes. These provisions only applied to offences committed on or after September 1, 1994.⁴⁶

Figure B1 validates this natural experiment separately for white and minority defendants. Panels a) and b) show that the reform resulted in a 6 p.p. (7%) increase in non-incarceration among white defendants and a 14 p.p. (20%) increase among minority defendants. Panels c) and d) demonstrate that future involvement with the criminal legal system falls by 3.1 p.p. for white defendants (-32.6%) and increases by 3.8 p.p. for minority defendants (29%), although the former estimate is imprecise. Panels e) and f) show that a summary measure of the baseline characteristics of defendants is smooth around this date cut-off.

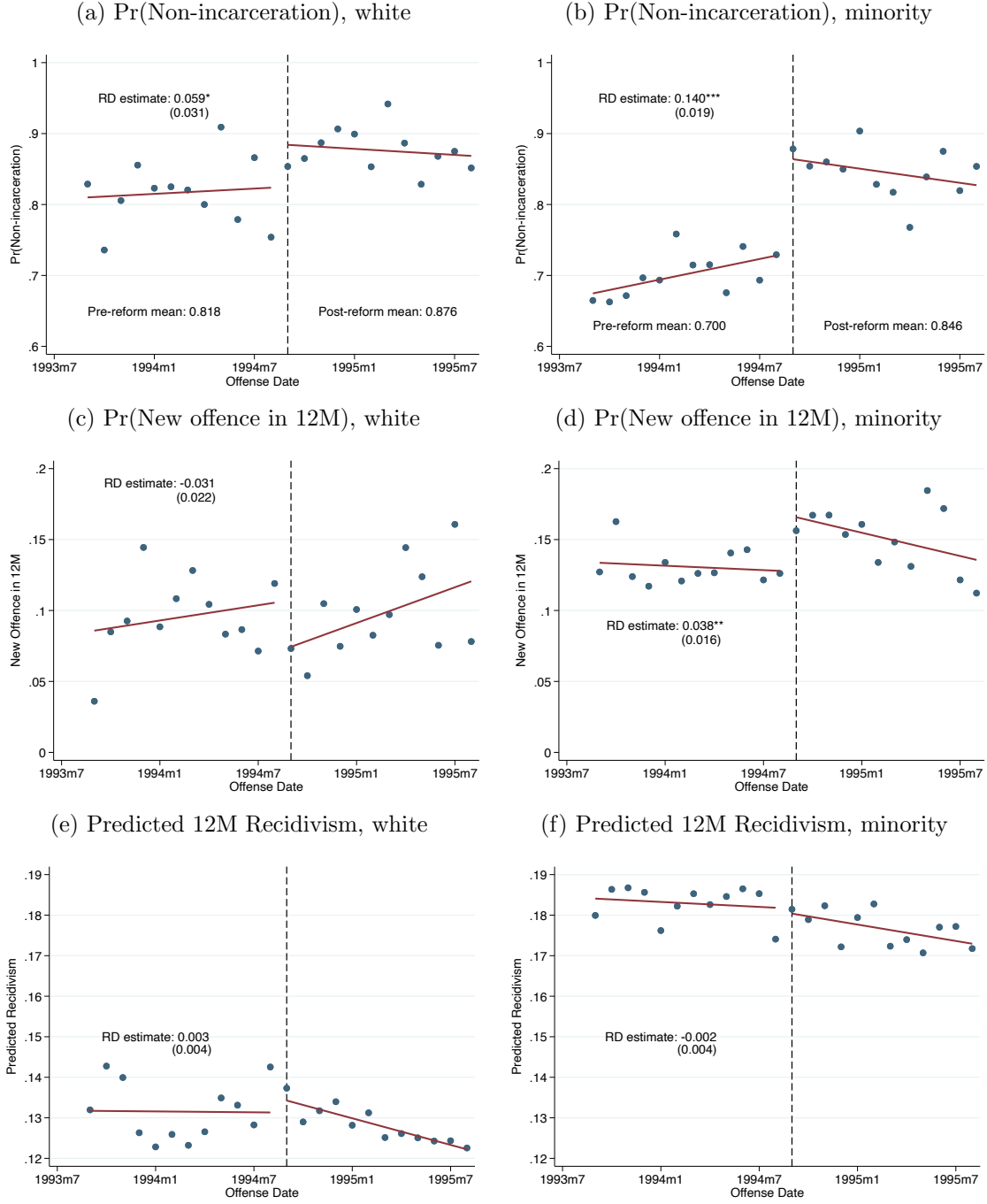
Given that the reform is a valid natural experiment, we apply it to estimate the marginal treatment response functions as described above. We define treatment as $D_i = 1$ if an individual is not incarcerated (referred to as ‘released’ henceforth) and $D_i = 0$ if incarcerated. If an individual is released, we observe their treated re-offence outcome $Y_i(1)$. Here, we define $Y_i(1) = 1$ if an individual commits a new offence in Bexar County in the 12 months after they are released. Finally, while we assessed the validity of the natural experiment using regression discontinuity techniques, we parametrize Z as a binary instrument for simplicity: $Z = 1$ if an individual committed an offence after September 1, 1994.⁴⁷

Figure B2 plots the estimated race-specific marginal treated outcome functions, $MTO_r(p)$ and Table B1 integrates these functions to estimate the average outcome that would be realized if everyone were released. The point estimates suggest large differences in underlying potential outcomes. 9.7% of white defendants would re-offend if released, while 17.7% of minority defendants would, although these difference are not significant.

⁴⁶Mueller-Smith and Schnepel (2021) use data from Harris County, TX to study a related aspect of the same legislation, which changed the incentives to offer deferred adjudication.

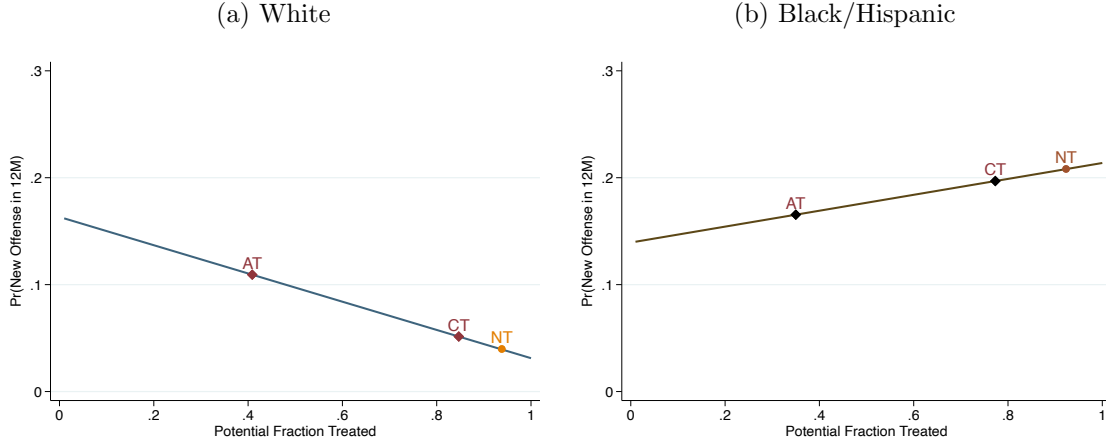
⁴⁷We alternatively could have implemented this using RD variation as in Appendix B.4. Given the lack of trends in treatment and in re-offence outcomes here, we proceed with this binary implementation for simplicity.

Figure B1: Validating the legislative reform



Note: Each Panel presents RD estimates from regressions of the form $Y_i = \alpha + \beta \mathbf{1}(T_i > t) + \delta_1 T_i + \delta_2 \mathbf{1}(T_i > t) \times T_i + \varepsilon_i$, where T_i denotes the running variable, and t denotes the cut-off date of September 1, 1994. Sample includes all felony defendants who committed an offence in a year around the cut-off date. The lines of best fit are estimated on the monthly averages, represented by the blue dots. Incarceration is defined as serving an incarceration sentence. Predicted recidivism is computed by estimating $\mathbf{1}(\text{New Offence})_i = \alpha + \beta X_i + \nu_i$ using pre-reform data, and excluding the RD sample. These coefficients are then used to predict the probability of re-offending for the RD sample. X includes: indicators for race, offence type, felony category, gender, age, criminal history, and neighbourhood characteristics.

Figure B2: Average re-offence extrapolation by race



Note: This figure displays the marginal treated outcome curves ($MTO(p)$) (Mogstad, Santos, and Torgovitsky, 2018) for non-incarceration. The treatment is non-incarceration, or ‘release’, and the outcome is whether an individual commits a new offence within 1 year, if not incarcerated. Each $MTO(p)$ is identified by assuming a linear relationship between potential outcomes of always takers (‘AT’), compliers (‘CT’), and never takers (‘NT’) and their treatment propensities. Lower values of the x-axis denote individuals who are more likely to be released.

Table B1: Average re-offence estimates (p.p.)

	Average (1)	White (2)	Minority (3)
μ	0.155	0.097	0.177
95% CI	[0.121,0.188]	[0.011,0.180]	[0.143,0.211]

Note: This table presents estimates of the average treated outcome obtained using the approach described in Section 2.2, separately by race. The treatment is non-incarceration and the treated outcome is whether an individual re-offends within one year after disposition. Confidence intervals are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Following Equation 2, Table B2 displays point estimates of racial discrimination in non-incarceration decisions that condition on re-offence outcomes if not incarcerated. Prior to the reform, white individuals were 12.3pp more likely to be released than minority individuals. The reform significantly narrowed this disparity—after the reform, release rates were 3.4pp higher for White defendants, a 8.9pp reduction.

Table B2: Estimated disparities (p.p.)

	Pre ($Z = 0$) (1)	Post ($Z = 1$) (2)	Change (3)
Δ	0.123	0.034	-0.089
95% CI	[0.084,0.447]	[0.003,0.276]	[-0.170,-0.050]

Note: This figure presents the average disparities in each time period, conditional on treated potential outcomes, using the approach described in Section 2.2. The treatment is non-incarceration and the treated outcome, denoted by $Y_i(1)$, is whether an individual re-offends within one year after disposition. Confidence intervals are bootstrapped using 1,000 replications and a Bayesian bootstrap.

B.4 Measuring discrimination with a regression discontinuity (RD) design

We briefly illustrate how to use a regression discontinuity (RD) design to measure discrimination conditional on potential outcomes. We do this in the context of studying socio-economic discrimination in the decision to promote Michigan public school students to the next grade.⁴⁸ The variation is generated by legislation that intended to improve the reading skills of 3rd graders in public schools. One component of the bill, also known as the ‘Read by Grade 3’ (RBG3) law, introduced a formulaic rule to determine when a student should be retained instead of being promoted. This component of the bill stipulated that 3rd graders scoring below 1253 (approximately the 5th percentile) on the standardized reading test (English Language Arts Michigan Student Test of Educational Progress, or ELA M-STEP) were to be retained while the rest were to be promoted to 4th grade. The promotion component of the policy came into effect in the 2020-21 school year but was ultimately repealed due to its unpopularity (Donahue, 2023; Povich, 2023). As a result, formula-based promotion and retention decisions affected students who were in 3rd grade during the 2020-21 and 2021-22 school years. As shown in prior work studying the impacts of the RBG3 policy, the formula induced shifts in promotion decisions and student outcomes at the cut-off (Westall et al., 2022a,b; Berne et al., 2023; Westall, Utter, and Strunk, 2023). We use these shifts to quantify whether there are differences by socio-economic status (SES) in the underlying potential outcome of promotion and adjust for them when measuring discrimination.

We use the same student-level data as used in prior work studying the RBG3 policy. Our analysis sample includes all first-time 3rd graders who scored within 15 points of the ELA M-STEP cut-off during the two years that the formula-based retention rule was active, which is the baseline sample in Berne et al. (2023). Students in our sample are more likely to be economically-disadvantaged, demonstrate limited English proficiency, and participate in special education programming than the average 3rd grader in Michigan’s public schools.⁴⁹ We refer interested readers to the prior work for more details on these students’ characteristics. For brevity, we refer to economically-disadvantaged students as ‘low SES’ and the rest of the students as ‘high SES’.

Assessing the first stage & identifying assumptions

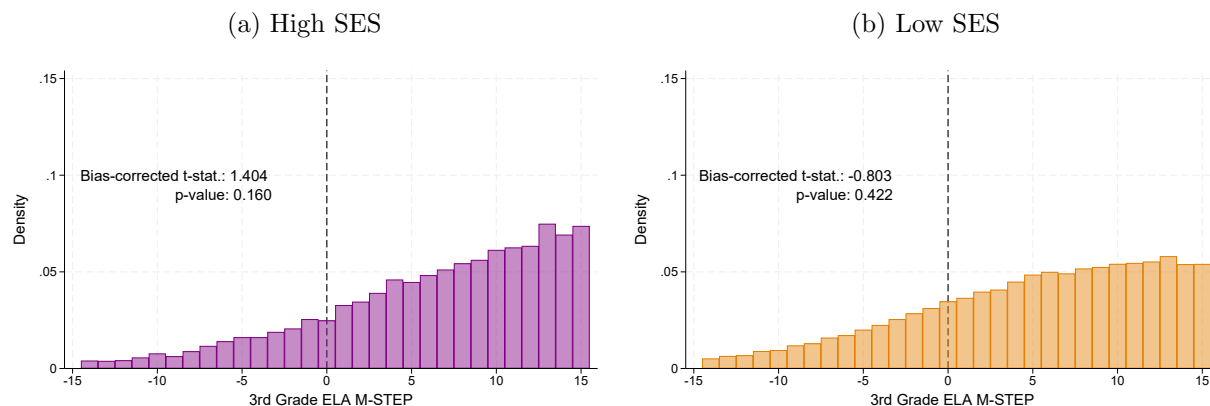
While this variation has been validated in aggregate by prior work, we assess the variation separately by SES group since we implement our approach within these groups. [Figure B3](#) finds no evidence of manipulation in the running variable, using the Cattaneo, Jansson, and Ma (2018) density test. [Figure B4](#) presents a summary test of whether 4th grade ELA performance, predicted

⁴⁸This research used data structured and maintained by the MERI-Michigan Education Data Center (MEDC). MEDC data are modified for analysis purposes using rules governed by MEDC and are not identical to those data collected and maintained by the Michigan Department of Education and/or Michigan’s Center for Educational Performance and Information. This research was funded by a grant R305H1900004 through the U.S. Department of Education’s Institute of Education Sciences, which is a collaboration between the University of Michigan and researchers from the Education Policy Innovation Collaborative (EPIC) at Michigan State University’s College of Education. Any opinions, findings, conclusions, or recommendations expressed in this material are those of the authors and do not reflect the views of any other entity.

⁴⁹A student is designated as economically-disadvantaged if the student: was eligible for free/reduced-price lunch, received SNAP/TANF, was homeless, was a migrant, or was in foster care.

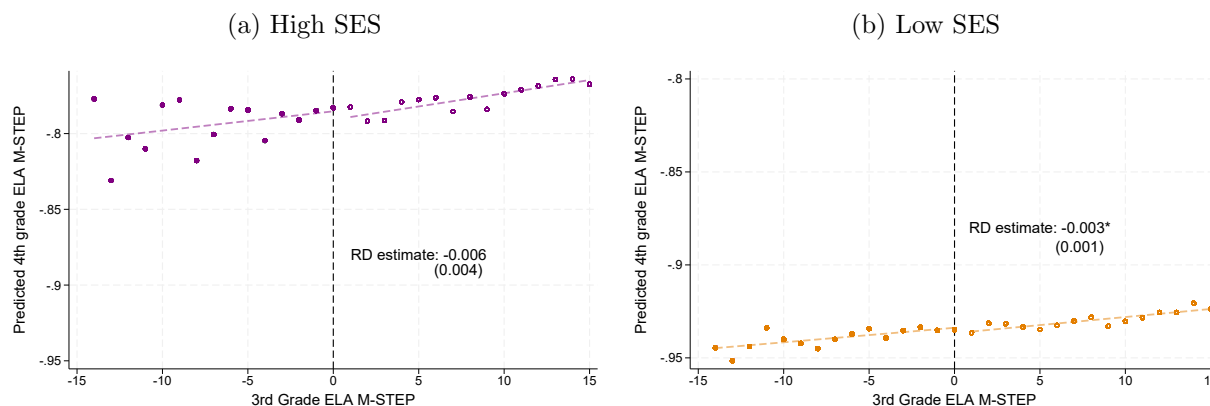
using baseline covariates, is smooth around the cut-off. While the covariates would predict that low SES students above the cut-off would score lower than low SES students below the cut-off, this estimate is marginally significant and economically small—the estimated discontinuity represents 0.005 of the standard deviation of the 4th grade ELA test score.

Figure B3: Smoothness of test score distribution density around cut-off



Note: These figures present results of the Cattaneo, Jansson, and Ma (2018) density test for the smoothness of the running variable. The x-axis represents the running variable, the 3rd grade ELA M-STEP, re-centred by the cut-off of 1252. The sample includes first-time 3rd graders during academic years 2020-21 and 2021-22 who scored within 15 points of the ELA M-STEP cut-off.

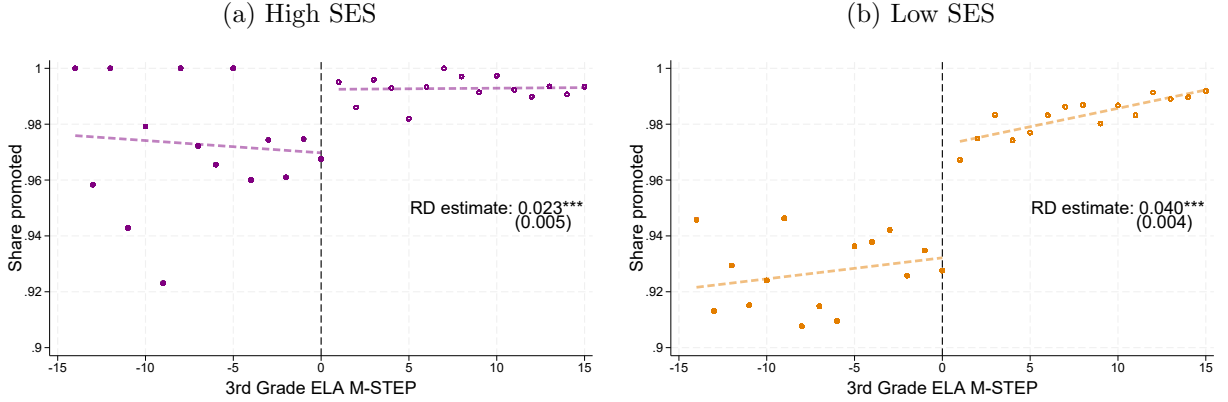
Figure B4: Smoothness of predicted 4th grade ELA M-STEP around test score cut-off



Note: These figures present RD estimates investigating the impact of the RBG3 test score-based promotion policy on the predicted ELA M-STEP score taken in the following school year (regardless of actual promotion status), using demographics, Limited English Proficiency and special education status, whether the student was previously retained, whether the student is new to the district, and school fixed effects. ‘RD estimate’ presents β from $X_i = \alpha + \beta \mathbb{I}(\text{Score}_i > 1252) + \delta_1 \text{Score}_i + \delta_2 \mathbb{I}(\text{Score}_i > 1252) \times \text{Score}_i + \varepsilon_i$. Standard errors are clustered at the level of the running variable. The x-axis represents the running variable, the 3rd grade ELA M-STEP, re-centred by the cut-off of 1252. The sample includes first-time 3rd graders during academic years 2020-21 and 2021-22 who scored within 15 points of the ELA M-STEP cut-off.

Figure B5 plots the relationship between the 3rd grade ELA M-STEP and the probability of being promoted, separately for high and low SES students. High SES students just above the cut-off are 2.3 p.p. (2.4% of the average promotion rate within 3 points of the cut-off) more likely to be

Figure B5: Effect of test score cut-off on promotion rates



Note: Each Panel presents RD estimates investigating the impact of the RBG3 test score-based promotion policy on promotion rates, using a local linear specification. The x-axis represents the running variable, the 3rd grade ELA M-STEP, re-centred by the cut-off of 1252. The sample includes first-time 3rd graders during academic years 2020-21 and 2021-22 who scored within 15 points of the ELA M-STEP cut-off. ‘RD estimate’ presents β from $\text{Promoted}_i = \alpha + \beta \mathbb{I}(\text{Score}_i > 1252) + \delta_1 \text{Score}_i + \delta_2 \mathbb{I}(\text{Score}_i > 1252) \times \text{Score}_i + \varepsilon_i$. Standard errors are clustered at the level of the running variable.

promoted than high SES students below the cut-off. Low SES students just above the cut-off are 4.0 p.p. (4.3% of the average promotion rate within 3 points of the cut-off) more likely to be promoted than their counterparts below the cut-off. While the RBG3 law stipulated a formulaic approach to promotion and retention policy, these figures suggest that a large amount of discretion was still used in making promotion decisions, consistent with prior work studying the law’s implementation (Westall et al., 2022a,b). Promotion rates do not jump from zero below the cut-off to one above the cut-off due to a section of the law that exempted certain students below the cut-off from being retained.⁵⁰ However, despite how common these exemptions seem to be, the test score cut-off still caused a modest but meaningful share of students who would have otherwise been retained to be promoted.

Estimating discrimination in grade promotion using the test score cut-off

Mapping the objects from the empirical discussion to the potential outcomes framework in Section 2.2, students are treated if they are promoted to 4th grade for the upcoming school year ($D_i = 1$) and untreated if retained in 3rd grade. Students are either high or low SES, denoted by $R_i = r \in \{h, l\}$. The treated potential outcome, $Y_i(1)$, is how well a student would perform in 4th grade standardized tests if promoted to the 4th grade for the upcoming school year. We construct our empirical analog of $Y_i(1)$ using students’ test scores in the next school year, if they are promoted.⁵¹ We do not observe how well students who are retained do in the 4th grade, since they are still in the 3rd grade in the following year—as a result, we do not observe $Y_i(0)$. We

⁵⁰Common exemptions include students who: are English language learners, have disabilities, and whose parents submit an exemption request (Westall et al., 2022b).

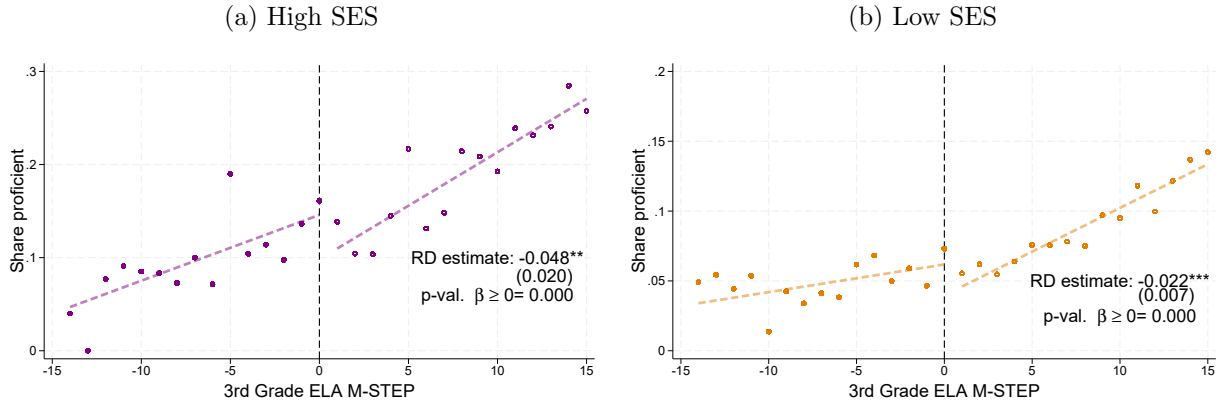
⁵¹This definition of $Y_i(1)$ maps well to the underlying behavior of teachers and other educational staff. In fact, educators were concerned that some students were advancing to 4th grade without the skills to cope and the reform was intended to improve skills that educators viewed as key inputs to students’ success in later grades (French, 2019; Povich, 2023).

define $Y_i(1) = 1$ if a student demonstrates “any proficiency” in both the Math and ELA M-STEP tests in the 4th grade, i.e., if they receive a score of at least Level 2 (out of 4) according to the Department of Education guidelines (Michigan Department of Education, 2023).⁵² We consider this binary outcome to be a proxy for whether a promoted student was ready for the 4th grade. Hence $Y_i(1) = 1$ if a student is ready for 4th grade and $Y_i(1) = 0$ if a student is not ready. As discussed in Section 2.3, the instrument, Z , is an indicator for being above or below the RD cut-off.

Estimating shifts in the outcome if promoted

As before, we use quasi-experimental variation in promoted outcomes to estimate how average promoted outcomes vary across always takers, compliers, and never takers. We use that information to bound the average test score outcomes if everyone in the analysis sample were to be promoted. Figure B6 plots the relationship between the 3rd grade ELA M-STEP score and 4th grade outcomes. We limit the sample here to only students who were promoted, since we are trying to understand how promoted outcomes vary around the cut-off. Marginal high SES students who are promoted due to the RBG3 policy are 4.8 p.p. (35.9% of the average share proficient within 3 points of the cut-off) less likely to demonstrate ‘any proficiency’ in 4th grade than promoted students below the cut-off. Marginal low SES students are 2.2 p.p. (36.7% of the average share proficient within 3 points of the cut-off) less likely to demonstrate ‘any proficiency’ in the 4th grade than promoted students below the cut-off.

Figure B6: Impact of test score cut-off on 4th grade outcomes (Only promoted students)



Note: Each Panel presents RD estimates investigating the impact of the RBG3 test score-based promotion policy on 4th grade proficiency rates, using a local linear specification. ‘Share proficient’ represents the share of individuals who demonstrated any proficiency on both the Math and ELA M-STEP in 4th grade, as defined by Michigan Department of Education guidelines. The x-axis represents the running variable, the 3rd grade ELA M-STEP, re-centred by the cut-off of 1252. The sample includes first-time 3rd graders during academic years 2020-21 and 2021-22 who scored within 15 points of the ELA M-STEP cut-off, and were promoted to 4th grade. ‘RD estimate’ presents β from $\text{Any Proficiency}_i = \alpha + \beta \mathbb{I}(\text{Score}_i > 1252) + \delta_1 \text{Score}_i + \delta_2 \mathbb{I}(\text{Score}_i > 1252) \times \text{Score}_i + \varepsilon_i$. The p -value in the second line is a one-sided test of whether the ‘RD estimate’ is weakly positive. Standard errors are clustered at the level of the running variable.

Following Section 2.3, we can use the intercepts of the local linear lines of best fit at the cut-off to

⁵²We focus on a binary proficiency measure since our sample consists of students with 3rd grade ELA M-STEP scores around the 5th percentile of the score distribution. As a result, most of the variation in outcomes is between Levels 1 and 2.

estimate the proportions of always takers, compliers, and never takers as well as the average treated outcomes for always takers and compliers. As described in Section 2.2, this is the information we need to bound average treated outcomes for never takers and subsequently bound the average promoted outcomes if everyone **at the cut-off** from both SES groups were to be promoted.

Bounding the socio-economic group-specific average outcomes if promoted

Before presenting our SES-specific estimates of average outcomes if everyone were promoted, we first address the fact that the estimated discontinuities in outcomes if promoted are large relative to the first stage. For example, taken at face value, the estimates for high SES students suggest that a 2.3 p.p. increase in the probability of being promoted causes a 4.8 p.p. reduction in the probability of demonstrating any proficiency in 4th grade. The large magnitude of the discontinuity in outcomes could be due to noise or treatments other than promotion shifting around the cut-off.⁵³

We deal with this issue by taking the sign of the discontinuity seriously but being agnostic about the magnitude. This approach assumes that marginally promoted students at the cut-off are weakly less likely to be ready for 4th grade than students who would have been promoted anyway. This is consistent with the literature finding that some 3rd grade students might benefit from retention in terms of academics (Jacob and Lefgren, 2004; Hwang and Koedel, 2022). The p -values presented in each Panel of Figure B6 suggest that our estimate of the sign of the relationship is credible—in both cases we reject the null that the relationship is weakly positive.⁵⁴

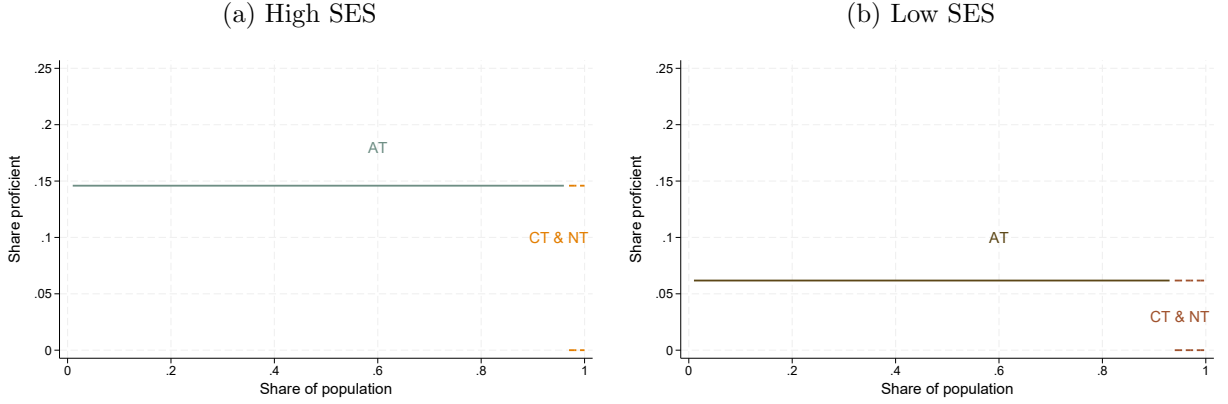
We use this estimated **sign** of the discontinuity, rather than the **magnitude**, to adjust the bounding approach described in Section 2.2. Figure B6 implies that, for both SES groups, always takers for promotion are more likely to demonstrate ‘any proficiency’ in 4th grade than compliers. The approach described in Sections 2.2 and 2.3 would involve: i) backing out the proficiency rate if promoted for compliers and ii) assuming never takers’ proficiency rate if promoted lies between zero and the compliers’ proficiency rate if promoted. We modify this to assume that the average proficiency rate for both **compliers and never takers** together is bounded above by that of always takers, and below by zero. Under this adjustment, we still assume weak monotonicity of the relationship between compliance groups’ treatment propensity and the average treated outcomes, using the estimated direction of the relationship between treated outcomes for always takers and compliers. Figure B7 displays the resulting estimates of average outcomes if promoted by compliance group and SES. Compliers and never takers, the portion of the population whose treated outcomes we bound, make up 3% and 6.8% of the population of high and low SES students respectively.

With these adjustments, Figure B8 shows that there are indeed large SES differences in the underlying 4th grade outcomes that would be realized if all students in the analysis sample were

⁵³The policy mandated that students below the cut-off receive additional interventions and encouraged (but did not mandate) this for students above the cut-off. Berne et al. (2023) find evidence that some school districts provided the optional additional supports.

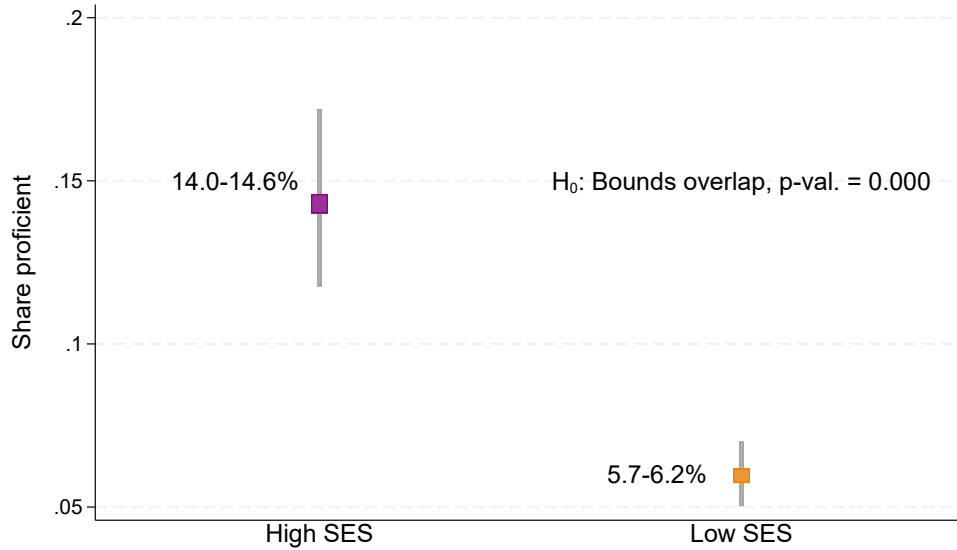
⁵⁴We also conduct a placebo exercise that re-estimates the RD specification in Figure B6, using every other possible test score in our analysis sample as the cut-off, while keeping the bandwidth the same. Only 1.2% and 3.5% of placebo estimates for the high and low SES sample respectively are larger than the ones we observe, providing additional evidence validating the estimated discontinuity in treated outcomes.

Figure B7: Average outcomes if promoted ($Y_i(1)$) by compliance group



Note: This figure shows the average treated outcomes for always takers ('A'), compliers ('C'), and never takers ('N'). The treatment is promotion and the treated outcome, $Y_i(1)$, is whether a student demonstrated any proficiency on both the Math and ELA M-STEP in 4th grade, as defined by Michigan Department of Education guidelines. The bounds for the treated outcomes for never takers and compliers come from the assumption of weak monotonicity of average treated outcomes across compliance groups, and that $Y_i(1) \in \{0, 1\}$.

Figure B8: Average outcomes if promoted, $E[Y_i(1)|R_i = r]$



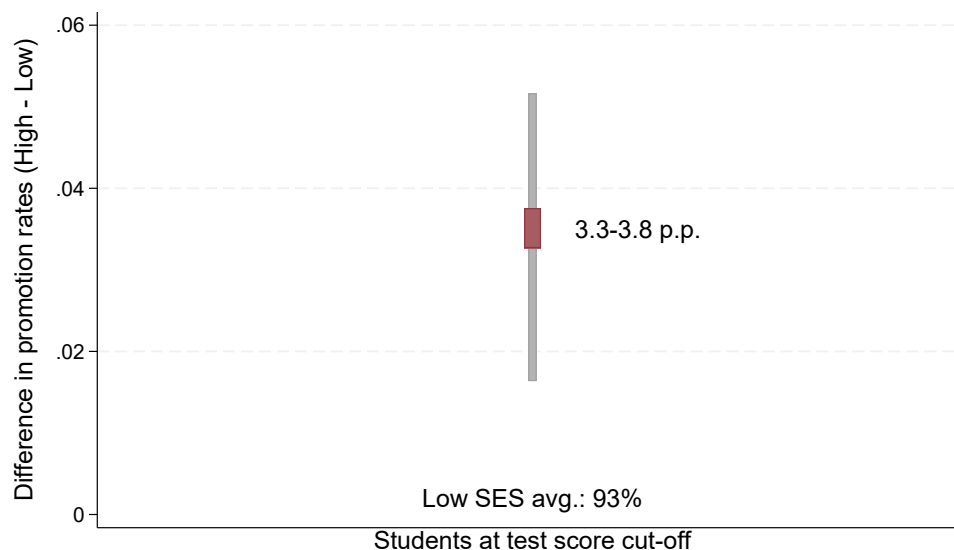
Note: This figure presents bounds on the average treated outcome obtained using the approach described in Section 2.2. The treatment is promotion and the treated outcome, $Y_i(1)$, is whether a student demonstrated any proficiency on both the Math and ELA M-STEP in 4th grade, as defined by Michigan Department of Education guidelines. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap. The p -value is from a formal bootstrapped test of whether the identified sets overlap, described in Appendix B.7.

promoted. 14.0–14.6% of high SES students would demonstrate ‘any proficiency’ if promoted to 4th grade, while this is true for only 5.7–6.2% of low SES students. Using a bootstrapped inference procedure described in Appendix B.7, we reject the null that these bounds overlap. Given that there are meaningful cross-SES differences in the underlying readiness for 4th grade, observed promotion

gaps by SES are likely contaminated by differences in this unobservable factor.

Estimating the socio-economic promotion gap

Figure B9: SES promotion gap conditional on promoted outcome



Note: This figure presents bounds on the average difference in promotion rates conditional on treated potential outcomes, using the approach described in Section 2.2. The treatment is promotion and the treated outcome, denoted by $Y_i(1)$, is whether a student demonstrated any proficiency on both the Math and ELA M-STEP in 4th grade, as defined by Michigan Department of Education guidelines. Confidence intervals are for the true parameter and are bootstrapped using 1,000 replications and a Bayesian bootstrap.

Figure B9 presents our estimates of the SES gap in promotion rates for students who are at the test score cut-off. Our results indicate that high SES students are 3.3–3.8 p.p. (3.5–4.1%) more likely to be promoted relative to low SES students, even after accounting for SES differences in how prepared students are for the 4th grade.⁵⁵ Without our approach, an alternative strategy might be to measure promotion disparities that control for observable characteristics. ‘Selection-on-observables’ approaches that control for gender, special education status, English language learner status, race, school district characteristics, and neighborhood share with at least a BA would estimate high–low SES promotion gaps of 2.8 p.p., which is 15–26% smaller than the one that we find.

Our analysis suggests that despite the intended formulaic nature of the RBG3 policy rules, the discretion that decision-makers exercised resulted in unwarranted disparities in promotion decisions by SES. These results make clear that the promotion disparities documented in recent work on the

⁵⁵The results are robust to weakening the assumption that despite any shifts in multiple treatments, the estimated sign of the discontinuity in promoted outcomes is still correct. We also estimate bounds for discrimination by allowing promoted outcomes for compliers and never takers to lie between 0 and 1. The resulting estimates imply that high SES students are 3.3–6.0 p.p. more likely to be promoted than low SES students. These patterns are in line with our baseline estimates, suggesting that estimates in Figure B9 are not driven by the assumption on the sign of the discontinuity in promoted outcomes.

RBG3 law are not solely driven by differences in underlying unobservables (Westall et al., 2022a,b; Westall, Utter, and Strunk, 2023).

B.5 Extrapolating discrimination away from a regression discontinuity cut-off

Here we discuss the assumptions needed to apply the average potential outcome estimates derived from information at the cut-off to adjust treatment rates away from the cut-off. Let the analysis sample include the following values of the running variable (e.g., a test score): $s \in [\underline{s}, \bar{s}]$ and let s^* be the cut-off. Let D_i denote the treatment decision (e.g., whether student i is promoted) and Z denote whether a student is above or below the test-score cut-off. Consider that we want to condition on the treated potential outcome, $Y_i(1)$. Finally, any assumptions discussed below will need to hold by subgroup, although we omit subgroup notation for brevity.

Assume that the following assumptions hold:

RD1: $D_i \perp s$ and $E[D_i|Z = 1, s^*] - E[D_i|Z = 0, s^*] = E[D_i|Z = 1, s] - E[D_i|Z = 0, s] \forall s \in [\underline{s}, \bar{s}]$

RD2: $E[Y_i(1)|s = s^*] = E[Y_i(1)|s < s^*] = E[Y_i(1)|s > s^*]$

To understand the plausibility of these assumptions, consider them in the context of the student grade promotion application in Appendix B.4. RD1 states that within the analysis window, the test score does not influence promotion decisions by itself—only the cut-off does. RD1 also implies that the size of the first-stage would be the same in counterfactuals where the test score cut-off was elsewhere in the window. This ensures that the proportions of always takers, compliers, and never takers identified at the cut-off are applicable to the wider window. RD2 assumes that the average promoted outcomes that would be realized if all students **at the cut-off** were promoted is equal to the average promoted outcomes if students elsewhere in the window were all promoted. In the simplest case, this would be satisfied if $Y_i(1)$ did not vary across s .

While RD1 may hold in some settings, RD2 is a strong assumption that may not be satisfied in many contexts, including student grade promotion. To see this, note that we can split $E[Y_i(1)|s = s^*]$ into average treated outcomes for always takers, compliers, and never takers at the cut-off s^* . Let $p_{(\cdot)}$ denote the proportion of always takers (A), compliers (C), or never takers (N). Consider the first part of the equality in RD2, which compares students at the cut-off to those below. RD2 implies that the equality in Equation 13 must hold.

$$\begin{aligned}
E[Y_i(1)|s = s^*] &= E[Y_i(1)|s < s^*] \\
\implies p_A E[Y_i(1)|A, s = s^*] + p_C E[Y_i(1)|C, s = s^*] + p_N E[Y_i(1)|N, s = s^*] &= \\
p_A E[Y_i(1)|A, s < s^*] + p_C E[Y_i(1)|C, s < s^*] + p_N E[Y_i(1)|N, s < s^*] & \\
\implies p_C (E[Y_i(1)|C, s = s^*] - E[Y_i(1)|C, s < s^*]) + p_N (E[Y_i(1)|N, s = s^*] - E[Y_i(1)|N, s < s^*]) &= \\
= p_A (E[Y_i(1)|A, s < s^*] - E[Y_i(1)|A, s = s^*]) &
\end{aligned} \tag{13}$$

We can map the implications of [Equation 13](#) to how promoted outcomes vary with the running variable in the data. In [Figure B6](#), all individuals below the cut-off are always takers for promotion (they are being promoted despite being below the cut-off). Here, always takers below the cut-off clearly have lower treated outcomes than those at the cut-off, implying that always takers **below the cut-off** are less likely to be prepared for 4th grade than always takers **at the cut-off**. Mathematically, this means that the right hand side of the final expression of [Equation 13](#) is negative. For this assumption to hold with equality, the left hand side needs to be sufficiently negative to offset this. However, that would imply that compliers and never takers **at the cut-off** are **less prepared** for 4th grade than compliers and never takers **below the cut-off**, which is at odds with reasonable models of selection into promotion. Comparing the average treated outcomes at the cut-off to that above the cut-off yields a similar conclusion.

B.6 Identifying average potential outcomes with difference-in-difference designs

This section describes conditions under which we can estimate average potential outcomes in DiD settings with individual-level treatment non-compliance and heterogeneity in potential outcomes, focusing on non-parametrically bounding these quantities rather than obtaining point estimates. While ‘treatment’ in a typical DiD might be at an aggregate level, e.g., county-level, we consider a setting where individuals in either county may be treated both before or after some policy. The estimation of causal effects is confounded by the effects of time, which are correlated with the policy adoption. We use changes in the unaffected control county to purge these effects of time. Our approach is similar to “time-corrected” Wald approach to estimate the local average treatment effect (LATE) from recent work (De Chaisemartin and D’Haultfœuille, 2018). However, we require stronger assumptions to identify average potential outcomes for those who are not compliers.

We start with the following notation:

- $T \in \{0, 1\}$: Denotes periods before (pre) and after (post) a policy
- $G \in \{0, 1\}$: 1 if the county adopts the policy in $T = 1$, 0 if not.
- $Z \equiv T \times G \in \{0, 1\}$: This is the binary instrument

- $D_i(g, z) \in \{0, 1\}$: Whether an **individual** takes up treatment or not.
- $Y_{it}(d, g)$: Potential outcomes for person i , given the time period, treatment status, and the county they are in.
- Refer to always takers, compliers and never takers as “compliance groups”

We make the following assumptions, many of which are common in IV implementations.

Assumption 1. First stage: $Pr(D_i(g, Z = 1)) > Pr(D_i(g, Z = 0)) \forall g$

Assumption 2. Independence and exclusion: $(Y_{it}(1, g), Y_{it}(0, g), D_i(g, 1), D_i(g, 0)) \perp Z|g$.

This implies that within each county, the instrument is random and only affects outcomes via changes in treatment status (Imbens and Angrist, 1994). This allows for a) potential outcomes to differ across counties and b) time-varying factors to directly affect potential outcomes.

Assumption 3. No spillovers: The potential outcomes of individual i are unrelated to the treatment status of other individuals (Angrist, Imbens, and Rubin, 1996).

Assumption 4. IV monotonicity: $D_i(g, 1) \geq D_i(g, 0) \forall g$

This allows the instrument to (weakly) shift individuals in only one direction across treatment contrasts and **does not allow secular trends to change treatment status of individuals**. Together, this means that only the following shifts between treatment contrasts are permitted:

1. $D_i(g, z) = 1 \forall z$: These are always takers in group g
2. $D_i(g, z) = 0 \forall z$: These are never takers in group g
3. $D_i(g, 1) = 1$ and $D_i(g, 0) = 0$: These are compliers in group g , shifted by the instrument

Finally, we make an additional assumption that is in the spirit of parallel trends assumptions, but is not typically made in IV settings or DiD estimation:⁵⁶

Assumption 5. Parallel trends in potential outcomes: This assumes that i) the average change in treated outcomes is the same for always takers and compliers and is independent of county and ii) the average change in untreated outcomes is the same for never takers and compliers and is independent of county. This restricts the effects of time on potential outcomes to be constant across subsets of compliance groups, but not all of them, and does not force the effects of time to be identical across individuals.⁵⁷

⁵⁶This is similar to the assumption underlying the “time-corrected” Wald estimand in De Chaisemartin and D’Haultfœuille (2018). There, the treated (untreated) potential outcomes for those treated (not treated) in the pre-period are the same across group. Their assumption is enough to identify the LATE, but does not allow us to identify the average potential outcomes of each compliance group separately. It pins down time trends in a) treated outcomes for always takers and b) an average of untreated outcomes for both never takers and compliers. This does not pin down time trends for compliers specifically without further assumptions.

⁵⁷This assumption is stronger when using our approach to obtain point estimates of average potential outcomes and discrimination rather than bounds. Obtaining point estimates requires assuming that the outcomes of always takers, compliers, and never takers are linearly related. Hence, if time trends in potential outcomes are assumed to be equal for two of the compliance groups, the assumption must extend to the remaining compliance group as well.

$$E[Y_{i1}(1, g) - Y_0(1, g)|g, \text{Always taker}] = E[Y_{i1}(1, g) - Y_0(1, g)|g, \text{Complier}] \text{ and } \perp g$$

$$E[Y_{i1}(0, g) - Y_0(0, g)|g, \text{Never taker}] = E[Y_{i1}(0, g) - Y_0(0, g)|g, \text{Complier}] \text{ and } \perp g$$

We now show how, under these assumptions, we can identify the proportions and average treated/untreated outcomes of always takers (A), never takers (N), and compliers (C) in the $G = 1$ county.

Given Assumption 4, there are no shifts in treatment status due to secular changes. Hence we have that the proportion of always takers (p_A), compliers (p_C) and never takers (p_N) in $G = 1$ are directly observed in the data for $G = 1$.

$$p_A = E[D_i|G = 1, T = 0]$$

$$p_N = 1 - (E[D_i|G = 1, T = 1]) \tag{14}$$

$$p_C = 1 - (p_A + p_N)$$

However, since common time trends can affect potential outcomes, the treated and untreated potential outcomes for each of these groups is not directly observed. To see this, recall that in settings where a binary instrument Z increases treatment take-up, the outcomes of individuals who are treated when $Z = 0$ identifies the treated outcomes for always takers. Equation 15 shows that if we try to estimate the treated outcomes for always takers in $G = 1$ using pre-period data (since $Z = 0 \ \& \ G = 1 \implies T = 0$), we only recover treated outcomes for always takers in the pre-period. The difference between this and the treated outcomes for always takers in the post-period is the trend in treated potential outcomes, θ_1 (second line of Equation 15).

$$E[Y_i|D_i = 1, G = 1, Z = 0] = E[Y_i|D_i = 1, G = 1, T = 0] = E[Y_{i0}(1, 1)|A, G = 1]$$

$$\underbrace{E[Y_{i1}(1, 1)|A, G = 1]}_{\text{Unobserved}} = \underbrace{E[Y_{i0}(1, 1)|A, G = 1]}_{\text{Observed}} + \underbrace{\theta_1}_{\text{Unobserved}} \tag{15}$$

Equation 16 makes the same point for the untreated outcomes for never takers. In settings where a binary instrument Z increases treatment take-up, the outcomes of individuals who are not treated when $Z = 1$ identifies the untreated outcomes for never takers. Here, the outcomes of individuals who are not treated in the post-period only identifies the untreated outcomes for never takers in the post-period. Similarly, the difference between this and the untreated outcomes for never takers in the pre-period is the trend in untreated potential outcomes, θ_0 (second line of Equation 16).

$$\begin{aligned}
E[Y_i|D_i = 0, G = 1, Z = 1] &= E[Y_i|D_i = 0, G = 1, T = 1] = E[Y_{i1}(0, 1)|N, G = 1] \\
\underbrace{E[Y_{i0}(0, 1)|N, G = 1]}_{\text{Unobserved}} &= \underbrace{E[Y_{i1}(0, 1)|N, G = 1]}_{\text{Observed}} - \underbrace{\theta_0}_{\text{Unobserved}}
\end{aligned} \tag{16}$$

Assumption 5 lets us identify the time trends in treated and untreated outcomes in $G = 1$ (θ_1 & θ_0) using the change over time in $G = 0$. Starting with treated outcomes, note that the only individuals who would be treated in the control county are always takers. The average change in treated outcomes in $G = 0$ identifies the time trend for always takers in $G = 1$, since Assumption 5 states that the trend in treated potential outcomes for always takers is identical across counties and is equal to the trend for compliers (see Equation 17).

$$\begin{aligned}
\theta_1 &= E[Y_{i1}(1, 1) - Y_{i0}(1, 1)|G = 1, A] = E[Y_{i1}(1, 1) - Y_{i0}(1, 1)|G = 1, C] \\
&= E[Y_{i1}(1, 0) - Y_{i0}(1, 0)|G = 0]
\end{aligned} \tag{17}$$

Similarly, the individuals who are untreated in the control county, $G = 0$, consist of compliers and never takers. From Assumption 5, the change in untreated outcomes for never takers and compliers are identical to each other, which means the average change in untreated outcomes in $G = 0$ equals the average change in untreated outcomes for never takers in $G = 0$. Additionally, Assumption 5 states that the time trend in untreated outcomes for never takers is identical across counties, which allows us to identify the time trend in untreated outcomes for never takers in $G = 1$ (see Equation 18).

$$\begin{aligned}
\theta_0 &= E[Y_{i1}(0, 1) - Y_{i0}(0, 1)|G = 1, N] = E[Y_{i1}(0, 1) - Y_{i0}(0, 1)|G = 1, C] \\
&= E[Y_{i1}(0, 0) - Y_{i0}(0, 0)|G = 0]
\end{aligned} \tag{18}$$

Equation 19 restates this by combining this with Equations 15 and 16, to show how the time trend in the treated outcomes of always takers and untreated potential outcomes of never takers in $G = 1$ can be identified using the aggregate changes in treated and untreated potential outcomes in the control county, $G = 0$.

$$\begin{aligned}
E[Y_{i1}(1, 1)|A, G = 1] &= E[Y_{i0}(1, 1)|A, G = 1] + \theta_1 \\
&= E[Y_{i0}(1, 1)|A, G = 1] + E[Y_{i1}(1, 0) - Y_{i0}(1, 0)|G = 0] \\
E[Y_{i0}(0, 1)|N, G = 1] &= E[Y_{i1}(0, 1)|N, G = 1] - \theta_0 \\
&= E[Y_{i1}(0, 1)|N, G = 1] - E[Y_{i1}(0, 0) - Y_{i0}(0, 0)|G = 0]
\end{aligned} \tag{19}$$

We now have the treated outcomes for always takers and the untreated outcomes for never takers from $G = 1$ in both periods. We can use this information with other observed moments in the data to estimate the treated and untreated outcomes for compliers (Imbens and Rubin, 1997).

Starting with treated outcomes, the first line of Equation 20 notes that the observed outcomes among treated individuals in period $T = 1$ is a weighted average of treated outcomes for always takers and compliers in $T = 1$. Rearranging this expression, the second line shows that the treated outcomes for compliers in $T = 1$, $E[Y_{i1}(1,1)|C, G = 1]$, is a function of moments that we can estimate. $E[Y_{i1}(1,1)|A, G = 1]$ is obtained from Equation 19, $E[Y_i|D_i = 1, G = 1, T = 1]$ is a sample average, and each of the proportions, $p(\cdot)$, is obtained using Equation 14.

$$\begin{aligned} E[Y_i|D_i = 1, G = 1, T = 1] &= \frac{p_A E[Y_{i1}(1,1)|A, G = 1] + p_C E[Y_{i1}(1,1)|C, G = 1]}{p_A + p_C} \\ E[Y_{i1}(1,1)|C, G = 1] &= \frac{(p_A + p_C) E[Y_i|D_i = 1, G = 1, T = 1] - p_A E[Y_{i1}(1,1)|A, G = 1]}{p_C} \end{aligned} \quad (20)$$

We can estimate untreated outcomes for compliers in a similar way. The first line of Equation 21 shows that the observed outcomes among untreated individuals in period $T = 0$ is a weighted average of untreated outcomes for never takers and compliers in $T = 0$. Rearranging this expression, the second line shows that the untreated outcomes for compliers in $T = 0$, $E[Y_0(0,1)|C, G = 1]$, is a function of moments that we can estimate. $E[Y_0(1,1)|N, G = 1]$ is obtained from Equation 19, $E[Y|D_i = 0, G = 1, T = 0]$ is a sample average, and each of the proportions, $p(\cdot)$, is obtained using Equation 14.

$$\begin{aligned} E[Y_i|D_i = 0, G = 1, T = 0] &= \frac{p_N E[Y_{i0}(0,1)|N, G = 1] + p_C E[Y_{i0}(0,1)|C, G = 1]}{p_N + p_C} \\ E[Y_{i0}(0,1)|C, G = 1] &= \frac{(p_N + p_C) E[Y_i|D_i = 0, G = 1, T = 0] - p_N E[Y_{i0}(0,1)|N, G = 1]}{p_C} \end{aligned} \quad (21)$$

As a result, we have identified the following objects:

- Treated outcomes in each period for always takers
- Untreated outcomes in each period for never takers
- Treated outcomes in $T = 1$ for compliers
- Untreated outcomes in $T = 0$ for compliers

We are missing 2 objects: Treated outcomes in $T = 0$ for compliers and untreated outcomes in $T = 1$ for compliers. Assumption 5 allows us to recover this, since it implies the time trend in treated/untreated potential outcomes of each compliance group in $G = 1$ can be identified using

the aggregate changes in treated/untreated potential outcomes in the control county, $G = 0$. We now have average treated and untreated outcomes for each compliance group and in each period.

$$\begin{aligned} E[Y_{i0}(1, 1)|C, G = 1] &= E[Y_{i1}(1, 1)|C, G = 1] - \theta_1 \\ E[Y_{i1}(0, 1)|C, G = 1] &= E[Y_{i0}(0, 1)|C, G = 1] + \theta_0 \end{aligned} \tag{22}$$

B.7 Inference for tests of overlapping bounds

Here we discuss how we test whether the average potential outcome bounds estimated for each group overlap. Let the true parameter of interest for each group be μ_r , where $r \in \{m, w\}$ denotes the group. Let the estimated bounds be $[\mu_{r,L}, \mu_{r,U}]$. The goal is to test whether $[\mu_{m,L}, \mu_{m,U}]$ and $[\mu_{w,L}, \mu_{w,U}]$ overlap.

We construct a set that denotes the difference between the upper bound for one group and the lower bound for another: $\mathbf{M_d} \equiv [\mu_{m,L} - \mu_{w,U}, \mu_{m,U} - \mu_{w,L}] = [\tilde{\mu}_L, \tilde{\mu}_U]$. Note that $0 \in \mathbf{M_d}$ only if the bounds for each group are overlapping. To see this, consider the following three cases.⁵⁸

Case 1. $[\mu_{m,L}, \mu_{m,U}]$ and $[\mu_{w,L}, \mu_{w,U}]$ are disjoint. E.g., $[\mu_{m,L}, \mu_{m,U}] = [0.5, 0.6]$ and $[\mu_{w,L}, \mu_{w,U}] = [0.2, 0.3]$. Then, $\mathbf{M_d} = [0.2, 0.4]$

Case 2. $[\mu_{m,L}, \mu_{m,U}]$ and $[\mu_{w,L}, \mu_{w,U}]$ overlap but one is not a subset of the other. E.g., $[\mu_{m,L}, \mu_{m,U}] = [0.25, 0.6]$ and $[\mu_{w,L}, \mu_{w,U}] = [0.2, 0.3]$. Then, $\mathbf{M_d} = [-0.05, 0.4]$

Case 3. $[\mu_{m,L}, \mu_{m,U}]$ is contained within $[\mu_{w,L}, \mu_{w,U}]$. E.g., $[\mu_{m,L}, \mu_{m,U}] = [0.2, 0.8]$ and $[\mu_{w,L}, \mu_{w,U}] = [0.3, 0.4]$. Then, $\mathbf{M_d} = [-0.2, 0.5]$

Our goal is to test the following null hypothesis: $H_0 : 0 \in \mathbf{M_d}$. We bootstrap the estimation of the bounds using a Bayesian bootstrap (Rubin, 1981). For each bootstrap replication, we construct the interval $\mathbf{M_d} \equiv [\mu_{m,L} - \mu_{w,U}, \mu_{m,U} - \mu_{w,L}] = [\tilde{\mu}_L, \tilde{\mu}_U]$. We then calculate a p -value as the share of the bootstrap replications in which $0 \in \mathbf{M_d}$, i.e., in which the bounds overlap.

⁵⁸There are 3 more cases if you switch m and w , but they yield the same conclusions. These conditions also hold if the intervals themselves contain 0.